END OF WELL REPORT

AZADEGAN OIL FIELD

END OF WELL REPORT

AZNS – 85 (PAD #79), TYPE: IIIF

Drilling Department

MAY 2024

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1. General Information

1.1 Field Introduction

Azadegan South Oil Field is situated in West Karoun, Khuzestan province on a large long – axis anticline with N – S trending with approximate dimensions about 20^{km} width and 60^{km} length near to Iran – Iraq border, It is located in The Northwest of The YADAVARAN Field and in The West of JUFEYR Field. The Average Monthly Temperature is 45$^{°c}$ in summer and Highest Temperature is 52$^{°c}$. The Lowest Temperature is -4.4$^{°c}$ in winter. Maximum Rainfall is 78^{mm} in February, Winter Season.

The trap area of the south Azadegan is limited to 43.2x17.2km. The Azadegan Oil field was discovered in 1999, when Well AZN-1 was drilled vertically down to the Cretaceous Fahliyan Formation with total depth of 4,643m. The results of this well, revealed the reservoir characteristics of the field. Following to the result of the well no. AZN-1, another 5 exploration wells was drilled in different parts of this huge field down to Fahliyan Formation. In the later stage the well No.10 assumed as exploration well for evaluation of deeper formations. This well was drilled in 2008, down to the depth of 5340m (Neyriz Fm.), mainly to evaluate the Jurassic formations.

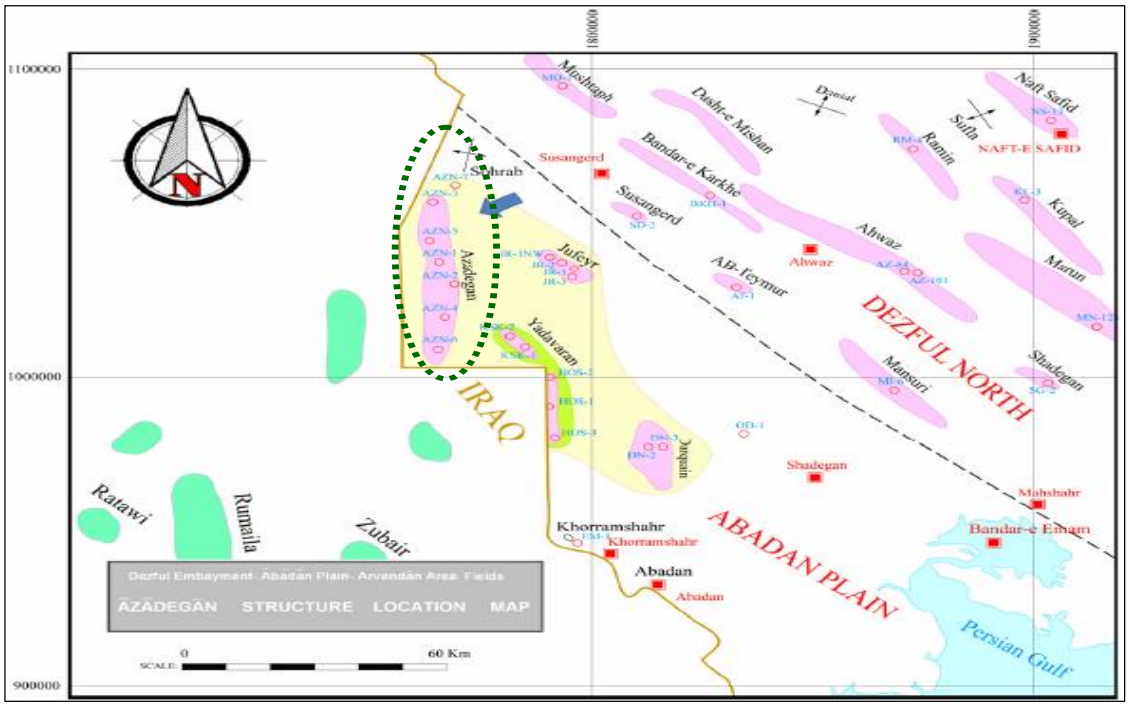


Figure 1-1: AZNS Oilfield Location Map

1.2 AZNS-85 Well Location Map

This well is located on "Pad – 79" of South Azadegan Oil field in Block-III.

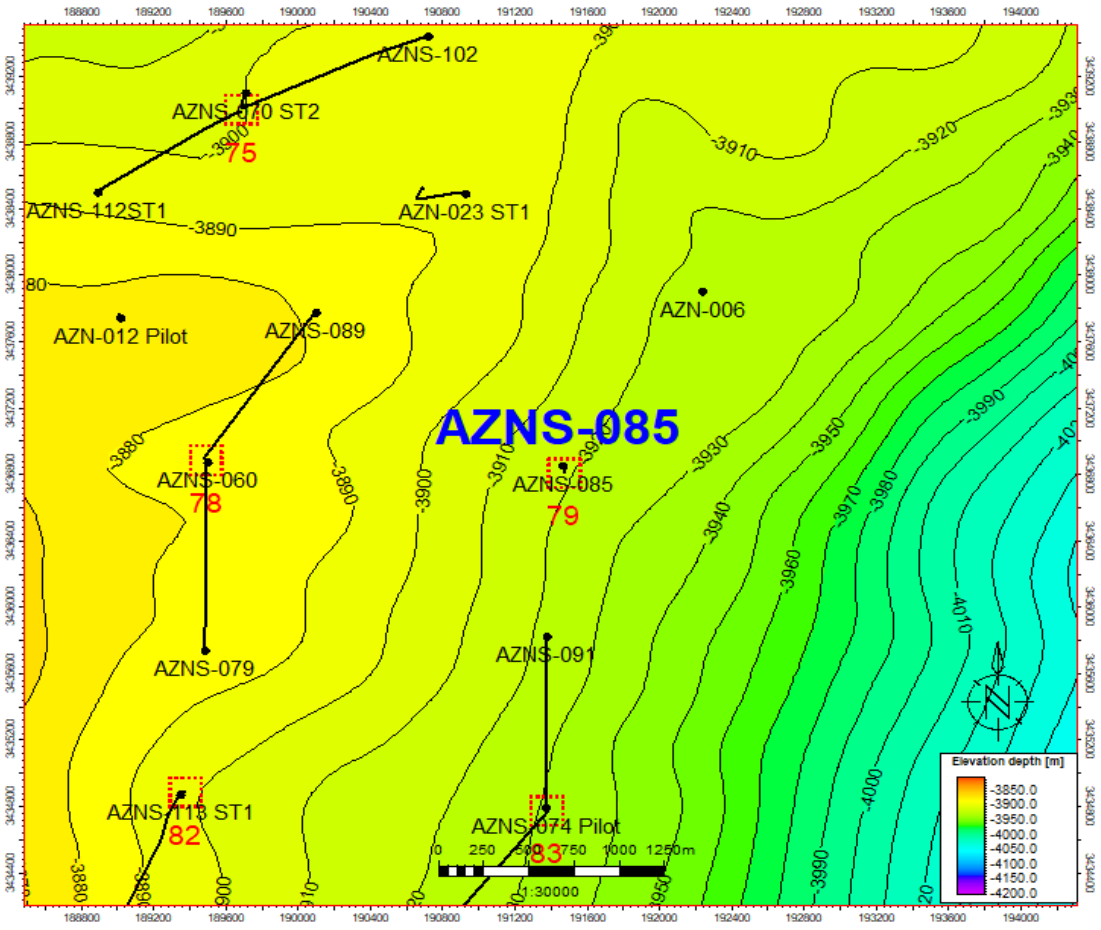


Figure 1-2: AZNS-85 Well Location Map

2. Well History

2.1 Well Informaion

Rig name	201	**Owner**	PEDEC/PPL
Rig type	Land Rig 2000HP	**Contractor**	MSA/OEOC
Well Name	AZNS – 85	**Management Contractor**	MSA
Well Type	Vertical IIIF	**Field**	South Azadegan
Completion Type	4-1/2" * 3-1/2" 10k	**RTE**	12.96
Actual TD	4090^{m}	**GLE**	3.95
Surface Coordinate		N: 3,436,812 – E: 191,473	
Target Coordinate		N: 3,436,854.7 – E: 191,467.9	
Drilling & Suspension		**Completion**	
Spud Date	25/01/2024	**Spud Date**	12/03/2024
Released Date	18/03/2024	**Released Date**	18/03/2024
End of Drilling	-	**End of Drilling**	-
Rig Days Actual/Plan	53 / 42	**Rig Days Actual/Plan**	-

Size (in)	Shoe Depth (mMD)	Specification
20	150	J – 55, 133^{ppf}, BTS
13 – 3/8	1277	N – 80, 72^{ppf}, BTS /NKT, 72^{ppf}, BTS
9 – 5/8	3950	L–80, 53.5^{ppf}, VAM/TN110SS, 47^{ppf}, TPCQ
7	3829 – 4090	VM110SS, 35^{ppf}, N.VAM
tie-back #1	3012 – 3829	L – 80, 29^{ppf}, HSM1
tie-back #2	2907 – 3012	L – 80, 29^{ppf}, VAM
SN Packer	2904 – 2907	L – 80, 29^{ppf}

Table 2-1: AZNS-85 Workover Well Information

2.2 AZNS-85 Well History

The AZNS – 85 well was drilled & suspended by NIDC from date 1393.07.16 to 1395.05.27 as describe below:

- 26" hole size drilled to 152^{m} & 20" casing ran to 150^{m} & cemented as program by 60^{bbl} cement returned.
- 17 – 1/2" hole size drilled to 1277^{m}, while increased Mw from 67 to 85^{pcf}, 13 – 3/8" casing ran & cemented to 1277^{m} by 180^{bb} cement returned.
- 12 – 1/4" hole size drilled to 3367^{m} with 6 – 150^{bph} loss at depth 3237^{m} while pumped LCM Pill & bypass system, took set log, complete loss & pumped LCM Pill several times, decreased Mw from 80 to 77^{pcf}, stuck string & complete loss while hole condition, decreased Mw to 70^{pcf} & continued drilling to 3661^{m}, observed well flow & increased Mw to 78^{pcf}, drilled to 3950^{m}, took log as program, 9 – 5/8" casing ran & cemented to 3950^{m} with no cement returned.
- 8 – 1/2" hole size drilled to 3957^{m}, took CBL/VDL/CCL/GR logs from 9 – 5/8" casing, drilling continued to 4034^{m}, increased Mw to 101^{pcf} due to well flow, drilling continued to TD @4090^{m} while Mw increased to 103^{pcf}, 7" liner ran to 4090^{m}, pressure tested & injected 150^{bbl} cement in 9 – 5/8" casing @3103^{m}, dry test 7" liner lap was Ok, took CBL/VDL/CCL/GR log from 7" liner, set cement plug from 3882 to 3650^{m} & tested 9 – 5/8" casing was not Ok, took CAST log from 3648 to 1200^{m} in 9 – 5/8" casing, observed tear in 9 – 5/8" casing from 3157 to 3140^{m}, ran 7" tie – back liner to 3829^{m}, dry test 7" liner lap was not Ok, ran 7" tie – back liner to 2903^{m}, set cement plug from 2903 to 2763^{m}, suspended the well without completion string & rig released @1395.05.27.

Casing Spec.	20"	J – 55, 133^{ppf}, BTS	0 – 152^{m}
	13-3/8"	N – 80, 72^{ppf}, BTS	0 – 1098^{m}
		NKT, 72^{ppf}, BTS	1098 – 1277^{m}
	9-5/8"	110SS, 47 ^{ppf}, TPCQ	0 – 1844^{m}
		L – 80, 53.5^{ppf}, VAM	1844 – 3950^{m}
	7"	VM110SS, 35^{ppf}, N.VAM	3829 – 4090^{m}
	Tie-back #1	L – 80, 29^{ppf}, HSM1	3012 – 3829^{m}
	Tie-back #2	L – 80, 29^{ppf}, VAM	2907 – 3012^{m}

Table 2-2: AZNS-85 Casing History

Summary of Workover Operation

The AZNS-85 workover well was De-Complete by rig OEOC – 201 from "25 – January – 2024" to "18 – March – 2024" as described in the following:

Workover Summary:

Date	25 – January – 2024
WorkOver Depth	4090^{m}
Formation	Fahliyan
Mud Weight	103 – 106^{pcf}

Table 2-3: AZNS-85 WorkOver Information

P/U & M/U 5" DPs joint by joint & R/B, meanwhile test the THS valves & body, Ok. Changed WETCO THS valves with new ones & pressure test same & P – seal & connections between THS & DSA to 7500^{psi}, Ok. L/D 6 – 3/8" 10k blind flange & block master gate valve (BTM: 11" 10k, Top: 6 – 3/8" 10k, WETCO gray), while work on all nuts & bolts to made loosen due to covered & clogged with rust. Opened all tie down screws of THS. Engaged with tubing hanger, tried to P/U tubing hanger by applying max 40^{klb} overpull, No Succeed. Tried to turn round tubing hanger with max 5$^{klb.ft}$, Obs. stuck same on spool, L/U rig pumps to THS by step, side valve, prepared 400^{psi} pressure while apply max 40^{klb} over-pull step by step, No Succeed. Started heat the THS body & engaged with tubing hanger then applied 40 – 70^{klb} over-pull by jerking up string several times, retrieved tubing hanger successfully. POOH & L/D same.

Install 11" 10k * 13 – 5/8" 10k DSA on THS & related accessories & tested all rams & annular, Ok. Prepared to perform BOP test as NIOC policy with 500^{psi}, observed leakages several times, rectified leakages & performed BOP test successfully. Tried to POOH 11" plug tester, observed abnormal over-pull, applied 70^{klb} over-pull several times to release plug tester without any succeeds. L/D DPs, P/U & M/U fishing jar & work on stuck, retrieved 11" plug tester successfully & L/D same. Continued BOP pressure test successfully.

P/U & M/U 8 – 1/2" used Varel PDC bit & D/O BHA & RIH to 1970^{m} until took weight, tried to wash down to continue without any succeed, started hard R/D from this depth to

2763[m] w/NL-NG, while pumping 30[bbl] Hi-Vis Pill. Took SCR, tag TOC @2763[m], commenced D/O cement to 2907[m] (7" liner lap depth), meanwhile weighting up the in hole mud to 103[pcf]. Performed static flow check, Ok. Commenced POOH D/O BHA to surface.

P/U & M/U 5 – 7/8" new open kingdream B.bit & C/O BHA & started RIH to 2897[m], continued W/D from 2897 to 2904[m], R/D from 2904 to 2965[m], took weight at this depth, tried to pass this point with different parameters, no succeed. Pumped 30[bbl] Hi-Vis Pill while observing pieces of rubbers on shale shakers, D/O plugs, took SCR. Continued R/D from 2966 to 3001[m]. Commenced D/O hard cement with C/O BHA from 3001 to 3069[m]. Soft R/D to 3417[m]. Continued D/O med to hard cement from 3417 to 3444[m], R/D from 3444 to 3665[m]. Commenced POOH to surface & L/D same.

P/U & M/U 9 – 5/8" RTTS packer, RIH & set packer @40[m], N/D BOP stack & N/U 13 – 5/8" BOP & performed pressure test as NIOC policy, Ok. RIH to depth 40[m] & unset RTS packer, POOH to surface & L/D same. P/U & M/U 8 – 1/2" open kingdream B.bit + 9 – 5/8" scraper & RIH to 2900[m] while scraping 2750 to 2900[m] interval several times. POOH to surface & L/D BHA.

Dry Test Operation Summary:

P/U & M/U 9 – 5/8" OEOC dry test tools & RIH to 1935m. Stopped operation due to observe annulus head dropped, then water cushion U-tubed from drill pipes. POOH to surface. Observed crack retainer above drag blocks of packer & 5 separation drags. 9 – 5/8" back-up dry test packer was failed. While waiting to receive 9 – 5/8" new dry test packer, P/U & RIH 5 – 3/16" Rev. Cir. magnet fishing BHA to 3662[m], took weight, RIH with W/D to 3665[m], work on junk sub several times, dropped ball start pumping, pressure increased, work on magnet. POOH fishing magnet & BHA to surface & L/D w/NL-NG. Fish did not retrieve. P/U & M/U 9 – 5/8" OEOC dry test tools & RIH to 2886[m]. L/U dry test lines & manifold, tested them, Ok. Set packer @2884[m], 23[m] above 7" liner lap. Performed dry test, Ok. Unset packer, well condition, static, R/D surface equipment & started POOH to surface & L/D dry test tools, while circulating across wellhead, observed pressure up 340 – 350[psi] in close position.

R/U slickline & ran 2.3" & 2.8" magnet at two stages to catch the fish without any succeed. POOH to surface & L/D slickline equipment. M/U & RIH with 5 – 7/8" junk mill till took weight @3664.5[m], started milling the fish & continued D/O soft to med cement plug to 3804[m], continued D/O hard cement from 3804 to 3879[m] w/NL-NG. Hard

R/D to 4078[m]. POOH milling BHA to surface & L/D. P/U & M/U 11" plug tester & RIH, unset & retrieved wear bushing.

P/U & M/U 9 – 5/8" * 7" tandem scraper with used 5 – 7/8" B.bit & RIH to 4076[m] & pumped 60[bbl] Hi-Vis Pill while scraping SN packer position several times w/NL-NG. Displaced the in hole mud with low solid fresh mud. Pumped 30[bbl] $CaCl_2$ 85[pcf] & spotted in 7" liner as non-solid fluid. Commenced POOH scraping BHA to surface while B/O & L/D BHA & bit.



OEOC - OIL EXPLORATION OPERATION COMPANY

Client : MSA	Well : 85	Client Rep :	
Country : IRAN	Test : DRY TEST	Client Rep :	
Region : khouzestan	Rig : OEOC 201	DRY TEST : E. Liravi	
Field : South Azadegan	Zone :	Memory eng. :	

	Weight	Grade		Job date :			
Casing	53.5		9 5/8"	Set Packer :	2883.95	m	21 FM 0100
Liner	-		-	First Memory :	2888.21	m	
Drill Pipe			5"	Second Memory :	2880.41	m	
Drill Collar	92	IF	6 1/2"			m	

Completion Fluid:	103.0 (pcf)		
Cushion:	Mud 2665m	103pcf	

No.		Length Meter	O.D. Inch		Description
1		0.00	5.000		Bar Drop Device
2	0.00	2752.00	5.000		Drill Pipe
3	2752.00	114.00	6.500		Drill Collar 6 1/2"
4	2866.00	0.40	6.500		REVERS SUB
5	2866.40	9.56	5.000		1 joint Drill Pipe 5"
6	2875.96	0.25	5.000		X-Over 4 1/2" IF Box x 3 1/2" IF Pin
7	2876.21	1.64	5.000		D.C.I.P
8	2877.85	1.56	5.000		Hydrospring
9	2879.41	1.25	5.000		Middle B.T (Second Memory Gauge)
10	2880.66	1.00	5.000		Hydraulic JAR
11	2881.66	0.73	5.000		RH Release Safety Joint
12	2882.39	0.40	5.000		X-Over 3 1/2" FH Box x 3 1/2" IF Pin
13	2882.79	0.40	6.500		X-Over 3 1/2" IF Box x 4 1/2" IF Pin
14	2883.19	0.76	8.187		Above Packer 9 5/8
15	2883.95	1.33	8.250		Middle Packer element
16	2885.28	0.40	5.000		X-Over 3 1/2" EUE Box x 3 1/2" FH Pin
17	2885.68	1.53	5.000		Anchor pipe
18	2887.21	1.26	5.000		Bottom B.T (First Memory Gauge)
					End of tail

Figure 2-1: AZNS-85 Dry Test Sketch

TIME	EVENTS
14 – Feb – 24	
05:50	Connect Memory Gauges battery
08:00	RIH DST tools
15 – Feb – 24	
10:41	Set packer
10:43	Hydro Spring opened (Open the well)
13:43	DCIP closed (Close the well)
19:43	Drop the Bar and start circulating
20:16	Unset packer
21:30	POOH DST String
16 – Feb – 24	
13:20	Disconnect Memory Gauges
	End of job

Table 2-4: AZNS-85 Dry Test Events Table

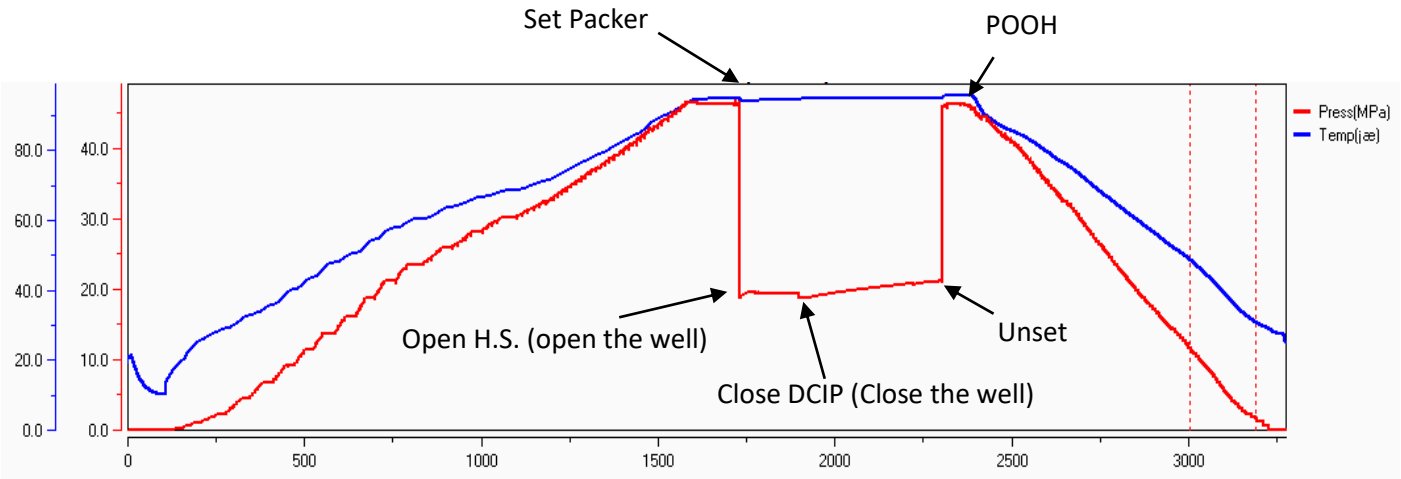


Figure 2-2: AZNS-85 Dry Downhole Test Results graph

Row	Description	Time	Bottom Bt.	Middle Bt.
1	Initial Mud Static Pressure (psi)	10:44	6753	6733
2	Open Well Pressure (psi)	1402/11/26 10:45	2722	2713
		1402/11/26 13:49	2722	2713
3	Close Well Pressure (psi)	1402/11/26 13:50	2723	2714
		1402/11/26 20:15	3065	3059
4	Final Hydrostatic Pressure (psi)	20:31	6739	6708

Table 2-5: AZNS-85 Dry Test Information Table

Wireline Logging Operation Summary:

R/U & M/U logging tools, RIH SBT-VDL-GR-CCL wireline tools as run #1 for 7" liner cement evaluation to 4045^{m}. Observed hold up wireline cable tension, tried to pass the point several times, No success. Started log up from 4045 to 2907^{m}, RIH again to 4045^{m} & tried to pass the choke point several times, No success. POOH to surface, observed debris & junk stuck on DRC arms, caught by GR. Prepared SBT-VDL-GR-CCL wireline tools as run #2 for 9 – 5/8" casing cement evaluation & RIH to 200^{m} till VDL got fail. POOH to surface, used back-up VDL & RIH to 2904^{m} as ReRun #2, log up to surface. RIH back to 1400^{m} & log up to 1280^{m} as repeated ReRun #2. POOH & R/D wireline logging tools.

P/U & M/U used 5 – 7/8" B.bit with new BHA & RIH to 4078^{m}, work on junk sub w/NL-NG. POOH to 2906^{m} & pumped 60^{bbl} Hi-Vis Pill, RIH back to 4087^{m}, spotted 30^{bbl} non-solid fluid as balance in 7" liner lap. Started POOH BHA to surface & L/D same. Observed debris in junk sub. R/U wireline logging tools & tried to perform incomplete ReRun #2 7" liner cement evaluation from 4078 to 4045^{m}. Observed hold up wireline cable tension @3315^{m}, tried to pass this depth several times, No success. POOH wireline

tools & checked for any debris, nothing observed. L/D SBT-VDL-GR-CCL & R/U CBIL-GR-CCL logging tools as run #3 & performed log from 2907^{m} to surface, according to client's request, logging repeated to interval 1350 to 1240^{m}. POOH & L/D wireline logging tools, while observing broken piece of wirelinr centralizer left in hole.

P/U & M/U Wt. bar as dummy run CCL + 2 guns & RIH to 3030^{m}, observed hold up tools & dropped cable tension at this point. Tried several times to pass this depth, No success. POOH & L/D tools. Broken & missed piece of centralizer retrieved by CCL tools. R/D wireline logging tools & equipment.

P/U & M/U 5 – 7/8" junk mill with new C/O BHA & RIH to 4075^{m}, W/D with low flow rate the last Std, spotted 200^{bbl} fresh mud in 7" liner as balance. POOH 5 – 7/8" C/O BHA from 4075^{m} to surface w/NL-NG. No considerable amount of debris in junk subs.

R/U SBT-VDL-GR-CCL wireline tools as ReRun #2 for 7" liner cement evaluation to perform incomplete run #2 & log up from 4076 to 3930^{m}, POOH wireline & L/D tools. R/U CBIL-GR-CCL wireline logging tools as run #4 & log up from 4075 to 2860^{m} & repeated interval from 3010 to 2900^{m}. POOH & L/D wireline tools. P/U & M/U 11" plug tester & set same, performed BOP test as NIOC policy.

Perforation Operation Summary:

R/U perforation tools & 7" shooting nipple, M/U BOP on shooting nipple & pressure test same, Ok. Connect detonator & RIH perforation tools as 1st run to 4061^{m}. Fired the gun & performed perforation from 4057 to 4061^{m}, Obs. Elec. Indication & wireline cable tension indication, Ok. POOH tools to surface, checked the gun visually, all bullets were shot successfully. Connect detonator & RIH perforation tools as 2nd run to 4045^{m}. Fired the gun & performed perforation from 4041 to 4045^{m}, Obs. Elec. Indication & wireline cable tension indication, Ok. POOH tools to surface, checked the gun visually, all bullets were shot successfully. Connect detonator & RIH perforation tools as 3rd run to 4041^{m}. Fired the gun & performed perforation from 4041 to 4036^{m}, Obs. Elec. Indication & wireline cable tension indication, Ok. POOH tools to surface, checked the gun visually, all bullets were shot successfully. Connect detonator & RIH perforation tools as 4th run to 4036^{m}. Fired the gun & performed perforation from 4036 to 4031^{m}, Obs. Elec. Indication & wireline cable tension indication, Ok. POOH tools to surface, checked the gun visually, all bullets were shot successfully, while correlating depth by CCL-GR log in presence of client's witness in all runs. RIH wireline tools to confirm perforation interval to 4065^{m}, logged up & approved perforation interval by client supervisor. POOH

the tools to surface w/NL-NG. R/D wireline tools & equipment. Install mouse hole, P/U & M/U 11" plug tester & RIH to set 11" wear bushing in THS.

P/U & M/U 7 – 5/16" polishing BHA & RIH to 2907^{m}. Tried to make LG profile marked on polish mill nose, polish inside 7" tie-back sleeve several times, Ok. Pass 7" liner lap without rotation & pumped-off, Ok, w/NL-NG. Performed CBU @2097^{m}, while monitoring the well w/NL-NG. Performed flow check @2907^{m}, NL-NG. POOH 7 – 5/16" polish mill & BHA to surface. Obs. LG mark on polish mill nose, Ok, while F/U the well & Circ. CWH, w/NL-NG.

SN Packer Operation Summary:

P/U & M/U 7" SN packer Assy. RIH to 2900^{m} on elevator with controlled speed, while monitoring the well, w/NL-NG. Adjust depth, performed Circ. with 160^{gpm} & 300^{psi}. Recorded P/U weight 265^{klb} & S/O weight 260^{klb}. Cont. RIH 7" SN packer to 2907.5^{m}, Obs. 1st packing indication (SPP increased to 500^{psi}), shut in rig pump & sting in seal nipple inside PBR to 2908^{m}. S/O 50^{klb} weight to set packer, P/U string & checked PKR by 15^{klb} over-pull, Ok. As per PPA supervisor procedure, neutral string, rotated string 27 turn forward & released setting tools, Ok. Checked P/U string weight 265^{klb}, OK. Performed static flow check @2905^{m}, NL-NG. POOH 7" S.N PKR setting tools from 2095 to surface & L/D all equipment.

P/U & M/U 11" plug tester & set the same in THS. Performed BOP top rams pressure test. P/U & M/U 5 – 7/8" C/O BHA with used 5 – 7/8" junk mill & RIH till took weight @4033^{m}. M/U TDS & RIH by W/D to 4065^{m}, reciprocated string & clear perforation interval several times, Ok, w/NL-NG. Performed CBU @4065^{m}, w/NL-NG. POOH to surface & L/D milling BHA.

P/U & M/U 6 – 1/8" PDC bit with C/O BHA + 7" scraper & RIH & pass 7" new liner lap @2904^{m} & SN PKR L.G collar gently, checked same several times, Ok. Scraped inside 7" liner from 2910 to 3070^{m}. Slow Circ. thru string @3100^{m} & wait on receive completion fluid, w/NL-NG. Received completion fluid & prepared same. Pumped 30^{bbl} kill mud as Mw balance, 30^{bbl} Hi-Vis Pill, 30^{bbl} $CaCl_2$ as spacer, then pumped 695^{bbl} $CaBr_2$ as completion fluid. Obs. $CaBr_2$ 104^{pcf} in return on Sh.Shakers, w/NL-NG. POOH 6 – 1/8" BHA from 3100 to surface & L/D BHA.

Completion Operation Summary:

R/U OEOC casing running equipment. P/U & M/U 3 – 1/2" wireline re-entry guide [10.2#, Conn: HSM3, Super 13CR-110KSI] + 3 – 1/2" BTM No-Go Landing [10.2#, Conn: HSM3, Super 13CR-110KSI, Seal bore 2-3/4" Type XN] + 3 – 1/2" TBG [HSM3, L-80, Type-1, 10.2#] + 3 – 1/2" Pup joint [HSM3, L-80, Type-1, 10.2#] + 7" PKR + 3 – 1/2" Anchor Seal [10.2#, Conn: HSM3, Super 13CR-110KSI, 4" Packing OD] + 3 – 1/2" Pup joint [HSM3, L-80, Type-1, 10.2#] + 1Jt 3 – 1/2" TBG [HSM3, L-80, Type-1, 10.2#] + 3 – 1/2" Non-Elastomeric SSD [Super 13CR-110KSI, HSM3 w/ Type X Selective Profile(Profile: 2-3/4", Downward Open)] + 3 – 1/2" Pup joint [HSM3, L-80, Type-1, 10.2#] + 1Jt 3 – 1/2" TBG [HSM3, L-80, Type-1, 10.2#] + 3 – 1/2" Pup joint [HSM3, L-80, Type-1, 10.2#] + 3 – 1/2" Exp. Joint (Super 13CR-110KSI) + 3 – 1/2" Pup joint [HSM3, L-80, Type-1, 10.2#] + 3 Jts 3 – 1/2" TBG [HSM3, L-80, Type-1, 10.2#] + 3 – 1/2" Pup joint [HSM3, L-80, Type-1, 10.2#] + 3 – 1/2" CIV [Super 13CR-110KSI, HSM3, 10.2#, Dual Type & Rupture Disc, w/ Check Valve] + 3 – 1/2" Pup joint [HSM3, L-80, Type-1, 10.2#] + 1Jt 3 – 1/2" TBG [HSM3, L-80, Type-1, 10.2#] + 3 – 1/2" Pup joint [HSM3, L-80, Type-1, 10.2#] + 3 – 1/2" CIV [Super 13CR-110KSI, HSM3, 10.2#, Dual Type & Rupture Disc, w/ Check Valve] + 3 – 1/2" Pup joint [HSM3, L-80, Type-1, 10.2#] & RIH to 75.5^{m}, while monitoring the well, w/NL-NG.

R/U CIL sheave, passed the both CILs thru sheave, then hanged it on derick, threading end of CILs, made them connected to chemical injection mandrel & tested the same one by one w/5000^{psi} – 15min, Ok, while monitoring the well, w/NL-NG. Cont. running completion string by 6 Jts 3 – 1/2" TBG [HSM3, L-80, Type-1, 10.2#] + X-over 4 – 1/2" [HSM3, 18.9#] BOX * 3 – 1/2" [HSM3 , 10.2#] PIN, from 75.5 to 133.5^{m}, while using 3 – 1/2" cross coupling protector as per completion string tally.

R/U slickline equipment & RIH as run #1 with 2.2" S pulling to retrieve prong, POOH slickline to surface, run #2 with 2.75" GS pulling to retrieve plug, POOH to surface, run #3 with 2.7" shifting tools to make sure SSD is closed, Ok. POOH to surface. Run #4 with S pulling to set 2.75" check valve in SSD, POOH to surface, R/D slickline equipment w/NL-NG.

M/U 4 – 1/2" HSM3 Circ. swage on completion string & prepared for pressure test, tried to pressure test surface line of cement unit several times, all got failed. Cont. running completion string by 4 – 1/2" TBG [HSM3, L-80, Type-1, 18.9#] from 133.5 to 741^{m}, while using 4 – 1/2" cross coupling protector as per completion string tally, w/NL-NG. M/U 4 – 1/2" HSM3 Circ. swage on completion string & prepared for pressure test, tried

to test surface line of cement unit several times, all got failed. During pressure test observed return on flow line, B/O & L/D 4 – 1/2" HSM3 Circ. swage, R/U slickline equipment & RIH with SB Pulling as run #1 to retrieve check valve, POOH slickline to surface. RIH with shifting tools as run #2 to make sure SSD is closed, Ok. POOH to surface. RIH with S Pulling as run #3 to set 2.75" check valve in SSD, POOH to surface & R/D slickline equipment. P/U & M/U 4 – 1/2" HSM3 Circ. swage on completion string, installed analog gauge on cement unit line, prepared for pressure test surface line, Ok. Performed pressure test to 7000^{psi} – 15min on completion string using analog gauge, Ok, while monitoring the well, w/ NL-NG.

Cont. running completion string by 4 – 1/2" TBG [HSM3, L-80, Type-1, 18.9#] from 741 to 1756^{m}. M/U 4 – 1/2" HSM3 Circ. swage on completion string & prepared for pressure test, tried to test surface line of cement unit several times, pressure recorder got failed, install analog gauge on cement unit line, prepared for pressure test surface line, Ok. Performed pressure test 7000^{psi} – 15min on completion string, using analog gauge, Ok, while monitoring the well, w/NL-NG. Cont. running completion string by 4 – 1/2" TBG [HSM3, L-80, Type-1, 18.9#] f/ 1756 to 2894.5^{m}.

P/U & M/U 4 – 1/2" Pup Joint [HSM3, L-80, Type-1, 18.9#] + 4 – 1/2" Short Flow Coupling + 4 – 1/2" TRSSV + 4 – 1/2" Long Flow Coupling + 4 – 1/2" Pup Joint [HSM3, L-80, Type-1, 18.9#] & RIH to 2902^{m}, Connect 1/4" CL to 4 – 1/2" TRSSV & performed pressure test with 500 – 8000^{psi}, Ok. Cont. running completion string by 4 – 1/2" TBG [HSM3, L-80, Type-1, 18.9#] f/ 2902 to 2921^{m}. M/U 4 – 1/2" HSM3 Circ. swage on completion string & prepared for pressure test, performed pressure test 7000^{psi} – 15min on string, Ok. Performed TRSCSSV inflow test. Set TRSCSSV in close position, released string pressure through to 1000^{psi}, monitor the pressure for 15min, Ok, then performed equalizing function test of TRSSV, Ok. Cont. running completion string by 4 – 1/2" TBG [HSM3, L-80, Type-1, 18.9#] f/ 2921 to 2949^{m}.

P/U & M/U 4 – 1/2" Pup Joint [HSM3, L-80, Type-1, 18.9#] + Fluted Swage + TBG HGR & RIH to 2952^{m}, pressure up CILs to rupture CIV Rupture Disk by applying 6000^{psi}, Obs. return on completion string, P/U & M/U 4 – 1/2" Landing joint + TBG HGR Running Tool, passed CL & CILs thru Fluted Swage sluts, while monitoring RIH with completion string & TBG HGR till touching THS Bowl, space out 1.5^{m}. R/U slickline tools. Run #1: RIH with 3.437" protection sleeve of TRSSSV to 62^{m} & set the same in its position, POOH to surface. Run #2: RIH with SB Pulling & retrieved 2.75" WFB-2 check valve from SSD, POOH to surface. Run #3: RIH with slickline & set 2.75"

plug & prong in landing nipple, POOH to surface. Run 4#: RIH with shifting tool to confirm SSD is full close, Ok. POOH to surface, R/D slickline tools.

M/U 4 – 1/2" Open all valves of X-Mass Tree, closed BMV, L/U cement unit to kill valve, companion flange & performed pressure test X-Mas tree body & connection between X-Mas Tree & BTM section flange to 8000^{psi} – 15min, Ok. Opened X-Mass Tree Cap & retrieved 2WCV. Installed X-Mas Tree Cap & closed all valves except BMV, Released the OEOC Rig 201 @18 – Mar – 2024, 4:00 a.m.

HSM3 Circ. swage on completion string & prepared for set PKR, pressure test surface lines, Ok. Pressurized completion string pressure to 300^{psi} to make sure Traveling Joint is fully open. Gradually increased pressure to 2700^{psi}, Obs. PKR setting indication & preformed completion string test before shearing Traveling Joint with applying 7000^{psi} stepwise, Ok. Performed PKR pull test by applying 15^{klb} over-pull, Ok, then slack off completion string weight about 20^{klb} to shear Expansion Joint, then land TBG HGR in THS Bowl. M/U 4 – 1/2" HSM3 Circ. swage on completion string, performed Expansion Joint pressure test with 5000^{psi} – 15min, Ok. Connect cement surface line to THS outlet valve & performed pressure test against PKR & hanger seals with 1500^{psi} – 15min, Ok. Fastened all tie-down screws, closed annular BOP on landing joint & performed pressure test against tie-down & TBG HGR with 1000^{psi}, Ok.

R/U slickline tools & RIH to 62^{m} & retrieved 3.437" protection sleeve of TRSSSV. POOH to surface & R/D slickline tools. Back off landing joint on TBG HGR & L/D same. M/U 2WCV & RIH with 2WCV & installed the same in TBG HGR. Closed shear ram & performed pressure test against 2WCV with 7000^{psi} – 15min, Ok. N/D BOP. Installed MM-Seal, N/U BTM section of wellhead including DSA Pack-off head adaptor & 11" * 4 – 1/16" 10k BMV MSP, while passing 1/4" CL & 3/8" CIL thru DSA Pack-off head adaptor. Connected 1/4" CL to CL exit cap. L/U cement unit to BMV & performed pressure test. Installed plug on all CIL & CL ports. N/U 4 – 1/16" 10k X-Mass Tree on top of BTM Master Valve Pack-off.

2.3 AZNS-85 Well Sketch

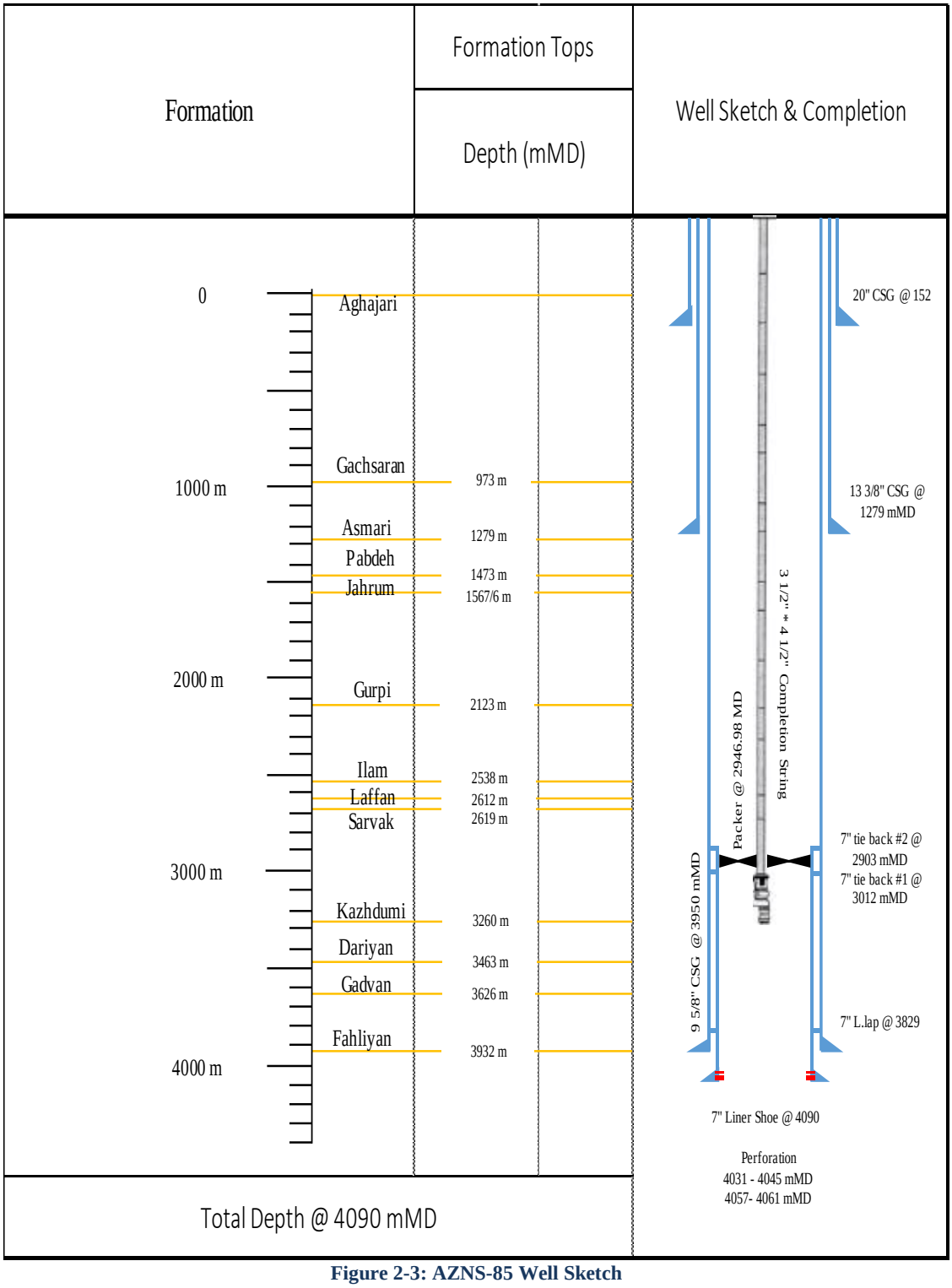


Figure 2-3: AZNS-85 Well Sketch

2.4 Bit Record Table

Bit No	Size	Bit Manufacture	BHA No	Type	IADC Code	Serial No	Jets			CMT	Depth		Metres	Hrs	Krev (for all runs)	WOB (klb)	Surface RPM	Pump Press	GPM		Mud Weight pcf		TFA	IR	OR	DC	Loc	B/S	G/16	OC	R Pulled
							No	x	Size		In	Out																			
1	8-1/2	Varel	1	PDC	M322	6000544	Open			Yes	-	-	240	124.5	-	5 – 15	20 – 50	'	-	-	103	103	-	2	3	CT	A	X	I	WT	BHA
2	5-78	King Dream	2	B.Bit	517	2105637JH	Open			-	-	-	-	-	-	'	'	'	-	-	103	103	-	1	1	WT	A	E	IG	NO	BHA
3	8-1/2	King Dream	3	B.Bit	517	2003235JH	Open			-	-	-	-	-	-	'	'	'	-	-	103	103	-	1	1	WT	A	E	IG	NO	DST
2RR3	5-7/8	King Dream	5	B.Bit	517	2105637JH	Open			-	-	-	-	-	-	'	'	'	-	-	103	103	-	1	1	WT	A	E	I	NO	LOG
4	6-1/8	-	14	PDC	M613	-	-			-	-	-	-	-	-	'	'	'	-	-	103	103	-	-	-	-	-	-	-	-	-

Table 2-6: AZNS-85 Bit Record Table

2.5 Hole Problem History

Not Applicable.

3. Time Analysis Report

3.1 Time vs. Depth Chart

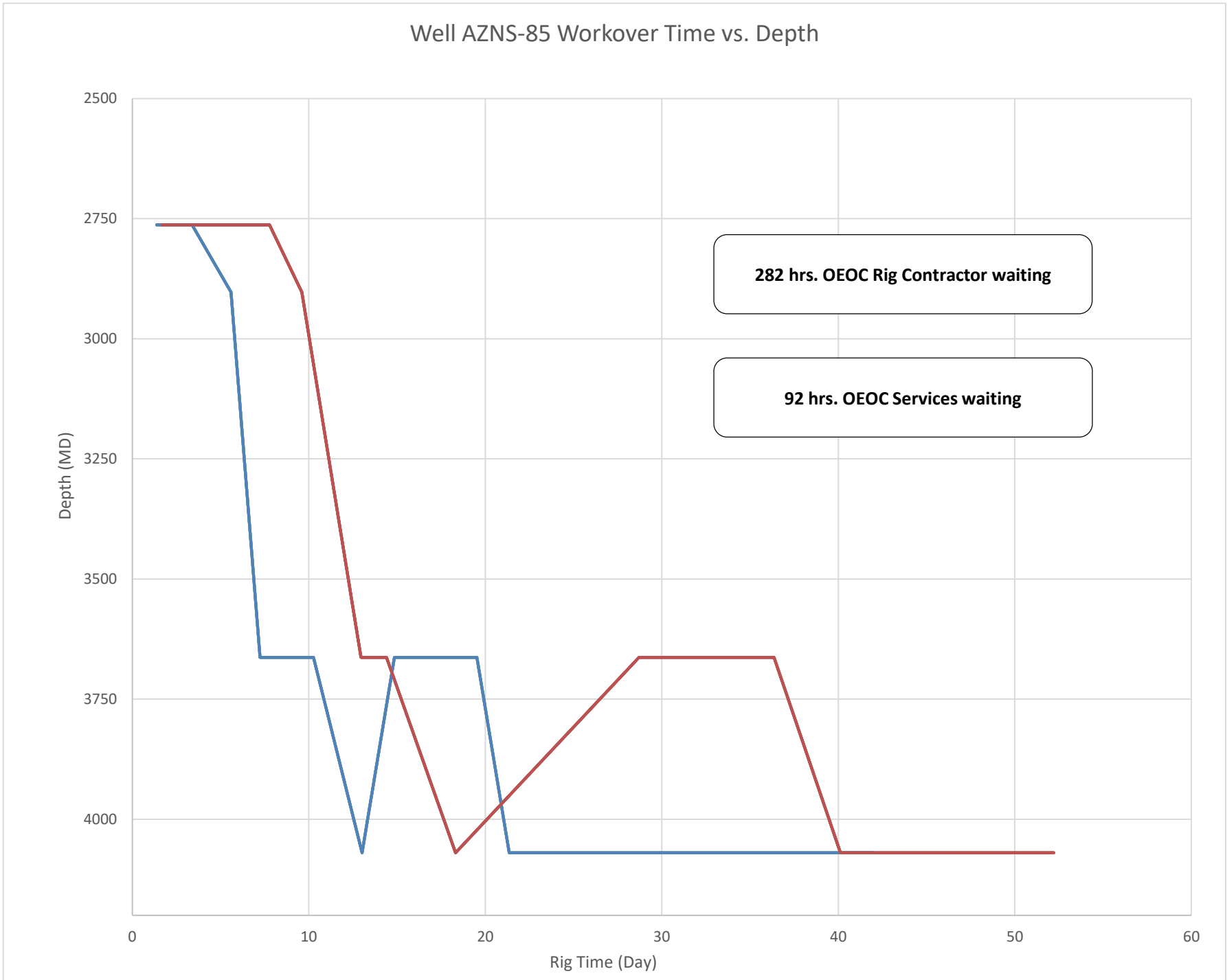


Figure 3-1: AZNS-85 Workover Well Time vs. Depth

3.2 Total Time Break-Down of Operation

WorkOver Section

De-Complete, Work-over and Completion Operation TBD of Well: AZNS-85 Work-over	IADC Code	Depth (meters)	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)	Actual Depth (meter)	Time Duration w/ contingency (Actual Hours)	Time Duration w/ contingency (Actual Days)	Total Duration w/ contingency (Actual Days)	Remark
Work-over														
Held PJSM. Spud the Rig. Stand Back required 5" DPs, 5" HWDPs and 6-1/2" DC.	SPUD	2763	30.0	1.25	1.3	1.4	33.0	1.38	1.38	2763	40.00	1.67	1.67	10 hrs. OEOC waiting due to falling behind of schedule.
Check the wellhead for any damage or leakage. Record well head pressures; check the all annulus pressure. Check the return fluid & Bleed off any annulus pressure to Zero upon client's approval and recheck the well status. Note: If wellhead pressure is not ZERO after bleed off, contractor shall inform client. Remedial plan to be prepared in advance by operators & will be followed after client's confirmation.	Wellhead	2763	6.0	0.25	1.5	0.3	6.6	0.28	1.7	2763		0.00	1.67	
N/D X-Mass Tree	BOP	2763	4	0.17	1.7	0.2	4.4	0.18	1.8	2763	7.00	0.29	1.96	
N/U BOP and pressure test.	BOP	2763	16	0.67	2.3	0.7	17.6	0.73	2.6	2763	18.50	0.77	2.73	
RIH with tubing hanger retrieving tool & engage with TBH. Pull out with TBH to surface.	Trip	2763	4	0.17	2.5	0.2	4.4	0.18	2.8	2763	14.00	0.58	3.31	
Perform wellhead pressure integrity test with cup tester.	Trip	2763	4	0.17	2.7	0.2	4.4	0.18	2.9	2763	17.50	0.73	4.04	
RIH with 8-1/2" D/O BHA to top of cement plug.	Trip	2763	10	0.42	3.1	0.5	11.0	0.46	3.4	2763	89.50	3.73	7.77	14 hrs. OEOC waiting due to falling behind of schedule.
Drill out cement plug to 7" liner lap f/2768m to 2903m (take care of lap@ 2903m), POOH.	D/O	2903	48	2.00	5.1	2.2	52.8	2.20	5.6	2903	44.00	1.83	9.60	
RIH with 5-7/8" clean-out BHA including 9-5/8" scraper , scrape dry test packer setting depths, continue RIH to top of cement @ 3663m & tag, CBU till shakers clean, POOH.	D/O	3663	36	1.50	6.6	1.7	39.6	1.65	7.2	3663	80.50	3.35	12.96	1 hrs. OEOC waiting due to work on TDS.
RIH with 9-5/8" RTTS packer & perform 7" liner lap dry test, POOH. Note: If dry test is not OK, follow contingency plan #1.	Trip	3663	60	2.50	9.1	2.8	66.0	2.75	10.0	3663	10.00	0.42	13.38	
Perform BOP test.	BOP Test	3663	6	0.25	9.3	0.3	6.6	0.28	10.3	3663	24.50	1.02	14.40	
RIH with 5-7/8" D/O BHA (3-1/2" DPs JT by JT), drill-out cement plug inside 7" liner f/3663m to 3879m, continue RIH to 4070m LDC, CBU till shakers clean, POOH to surface.	D/O	4070	60	2.50	11.8	2.8	66.0	2.75	13.0	4070	94.00	3.92	18.31	1 hrs. OEOC waiting due to work on mud pump
RIH with tandem scraper 9-5/8" * 7", scrape inside 9-5/8" casing & 7" liner (dry test packer setting depths), CBU & POOH.	Trip	3663	40	1.67	13.5	1.8	44.0	1.83	14.9	3663	249.50	10.40	28.71	P/U 8-1/2" R.bit w/9-5/8" scraper. 61 hrs. RIH w/ dry test tools. 85 hrs. OEOC Dry Test Dep. Due to dry test PKR failure & Rec. new dry test PKR. 7 hrs. OEOC Dry Test Dep. Due to unable catch the fish. 5 hrs OEOC waiting due to drawworks chain sys. Failure. 7 hrs OEOC waiting due to falling behind schedule. 29 hrs. RIH w/5-7/8" BHA & work on junk & POOH. 10.5 hrs. OEOC waiting due to drawworks failure.
RIH with wireline SBT/VDL/CCL/GR Logging tools to bottom, log up to surface (in two separate runs for 9-5/8" CSG & 7" Liner), POOH.	Log	3663	48	2.00	15.5	2.2	52.8	2.20	17.1	3663	178.00	7.42	36.13	2 hrs SeaLand Co. waiting due to failure in VDL. 47.5 hrs. RIH w/5-7/8" & work on junk. 4 hrs. OEOC due to falling behind of schedule.
RIH with wireline corrosion logging tools to bottom, log up to surface (in two separate runs for 9-5/8" CSG & 7" Liner), POOH.	Log	3663	48	2.00	17.5	2.2	52.8	2.20	19.3	3663		0.00	36.13	
Perform BOP test.	BOP Test	3663	6	0.25	17.8	0.3	6.6	0.28	19.5	3663	5.50	0.23	36.35	
RIH with 7" scraper, scrape inside 7" liner, CBU till shakers clean, displace inside 7" liner with non-solid fllud, POOH to surface.	Completion	4070	40	1.67	19.4	1.8	44.0	1.83	21.4	4070	90.50	3.77	40.13	22 hrs. RIH w/7-5/16" polish mill & POOH. 28 hrs. RIH w/7" SN PKR. 40.5 hrs. RIH w/5-7/8"C/O BHA & POOH.
Run perforation gun assembly & perforate 7" liner in Fah2a (4031m to 4045m) & Fah3a (4057m to 4061m) tentatively, POOH to surface.	Completion	4070	60	2.50	21.9	2.8	66.0	2.75	24.1	4070	38.50	1.60	41.73	
RIH with CCL log to confirm perforation intervals.	Log	4070	12	0.50	22.4	0.6	13.2	0.55	24.7	4070		0.00	41.73	
Perform BOP test.	BOP Test	4070	6	0.25	22.7	0.3	6.6	0.28	24.9	4070	3.00	0.13	41.85	

Table 3-1: AZNS-85 WorkOver Time Breakdown

Completion Section

RIH with 7" scraper to scrape completion packer area & displace the well with non-solid as completion fluid, POOH to surface & L/D DP's & DC's.	Completion	4070	60	2.50	25.2	2.8	66.0	2.75	27.7	4070	100.00	4.17	46.02	RIH w/6-1/8" PDC bit C/O BHA & 7" scraper. 3.5 hrs. wait on bad weather. 18 hrs. wait on Rec. Comp. fluid.
Run 10K completion string as per program.	Completion	4070	96	4.00	29.2	4.4	105.6	4.40	32.1	4070	122.50	5.10	51.13	16 hrs. Spad Energy Co. waiting due to cement unit assets failure.
Set TWCV inside TBH, N/D BOP & install 10K XMT, pressure test same.	Completion	4070	36	1.50	30.7	1.7	39.6	1.65	33.7	4070	10.00	0.42	51.54	
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, Release the Rig.	Completion	4070	5	0.21	30.9	0.2	5.5	0.23	34.0	4070	12.00	0.50	52.04	
Retrieve BPV, RIH with SL, set protection sleeve, retrieve Plug & Prong from NO-GO landing nipple, POOH.	Completion	4070	60	2.50	33.4	2.8	66.0	2.75	36.7	4070		0.00	52.04	
RIH with CT to bottom, displace the well with diesel, POOH.	Completion	4070	12	0.50	33.9	0.6	13.2	0.55	37.3	4070		0.00	52.04	
R/U & RIH with SL, retrieve protection sleeve, POOH & R/D SL.	Completion	4070	6	0.25	34.1	0.3	6.6	0.28	37.5	4070		0.00	52.04	
Flow the well as pre-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for 72 hrs.	Completion	4070	72	3.00	37.1	3.3	79.2	3.30	40.8	4070		0.00	52.04	
R/U & RIH with SL, set protection sleeve, POOH & R/D SL.	Completion	4070	Rig Less	0.00	37.1	0.0	0.0	0.00	40.8	4070		0.00	52.04	
RIH with CT & perform acid wash as per program, POOH to surface.	Completion	4070	Rig Less	0.00	37.1	0.0	0.0	0.00	40.8	4070		0.00	52.04	
R/U & RIH with SL, retrieve protection sleeve, POOH & R/D SL.	Completion	4070	Rig Less	0.00	37.1	0.0	0.0	0.00	40.8	4070		0.00	52.04	
Flow the well as post-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for minimum 96 hrs. Note: The time is only for flowing duration (R/U, R/D time & preparation is excluded. Note: Acidizing & post-acid clean-up maybe perform as rigless job upon contractor & client approval but in case of any failure in well integrity, contractor should provide drilling rig & required services in own responsibility & fix the problem.	Completion	4070	Rig Less	0.00	37.1	0.0	0.0	0.00	40.8	4070		0.00	52.04	
RIH with SL, set protection sleeve.	Completion	4070	Rig Less	0.00	37.1	0.0	0.0	0.00	40.8	4070			52.04	
RIH with CT to bottom & displace the well with diesel, POOH.	Completion	4070	Rig Less	0.00	37.1	0.0	0.0	0.00	40.8	4070			52.04	
RIH with SL, install Plug & Prong into NO-Go landing nipple, POOH.	Completion	4070	Rig Less	0.00	37.1	0.0	0.0	0.00	40.8	4070			52.04	
RIH with SL, retrieve protection sleeve.	Completion	4070	Rig Less	0.00	37.1	0.0	0.0	0.00	40.8	4070		0.00	52.04	
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV.	Completion	4070	Rig Less	0.00	37.1	0.0	0.0	0.00	40.8	4070		0.00	52.04	
Total Duration of Work-over:				*37.1*	*37.1*			*40.8*	*40.8*		*1249.0*	*52.0*	*52.0*	
NOTE:														
Total Duration for Cut & Slip DL in all hole section.	Cut & Slip	4070	4	0.17	37.3	0.2	4.4	0.18	41.0	4070	4.00	0.17	52.21	
Total Duration for Rig Service (Consider 0.5hr per day) in all hole section.	Rig Service	4070	21	0.88	38.2	1.0	23.1	0.96	42.0	4070		0.00	52.21	

Table 3-2: AZNS-85 Completion Time Breakdown

3.3 NPT Break-Down

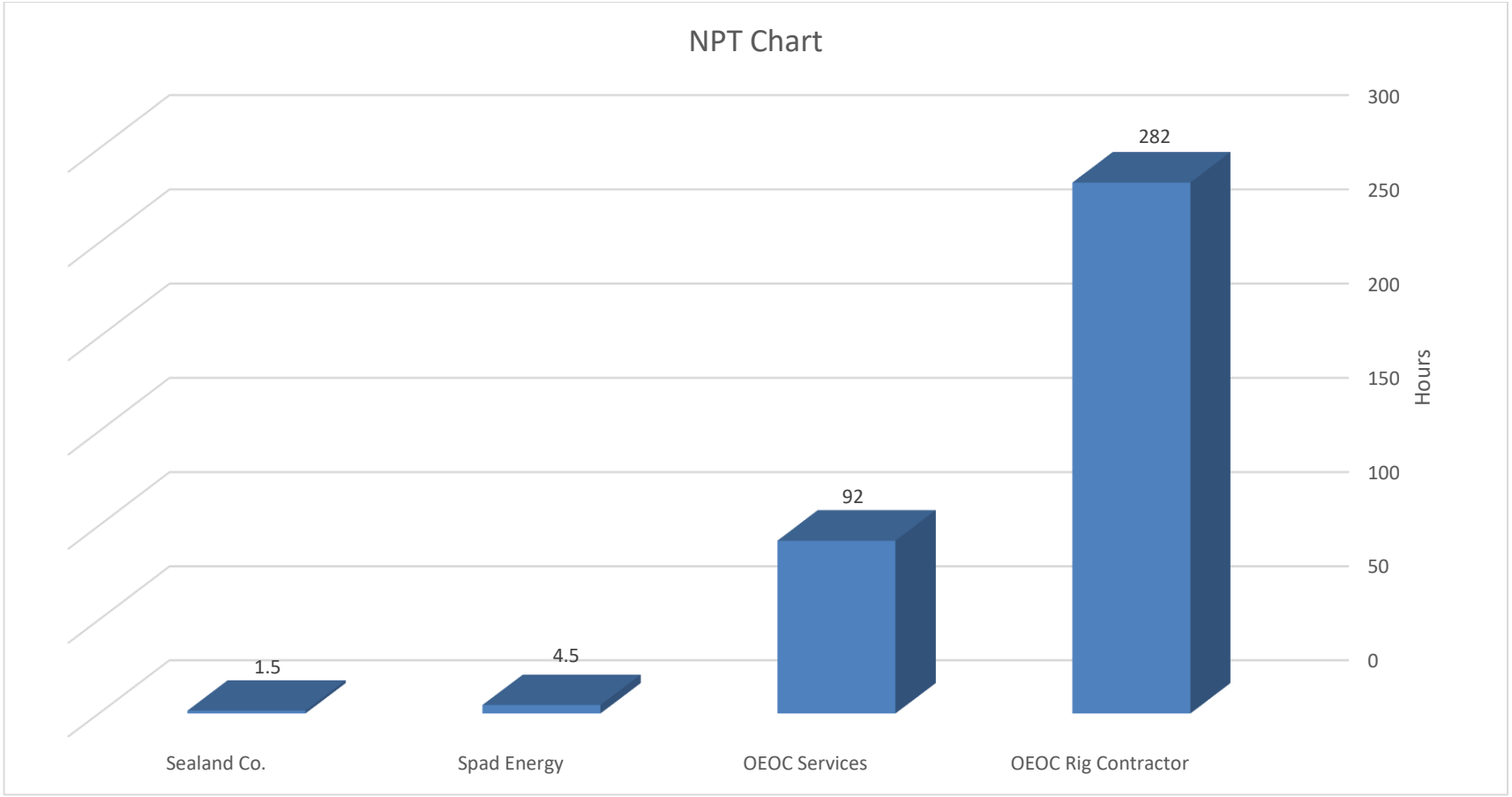


Figure 3-2: AZNS-85 NPT Break-Down Chart

Item	Company	Date	Problem event	Lost Time (hrs)
1	OEOC	1402 . 11. 07	P/U & M/U 5" DP & N/D Wellhead	14.0
2		1402 . 11. 08	N/D X-mass Tree	7.0
3		1402 . 11. 09	N/U BOP & test	19.0
4		1402 . 11. 17	Work on TDS	1.0
5	OEOC Service	1402 . 11. 22	Dry test packer failed. (Lost 1no drag block of pkr in hole)	18.0
6		1402 . 11. 23		24.0
7		1402 . 11. 24		19.0
5	OEOC	1402 . 11. 24	Work on drawworks	5.0
6	OEOC Service	1402 . 11. 25	Delay in operation	24.0
7		1402 . 11. 27	Dry test packer failed. (Lost 1no drag block of pkr in hole)	7.0
9	OEOC	1402 . 11. 30	Delay in operation	8.0
10		1402 . 12. 01	Work on mud pumps	1.0
11		1402 . 12. 03	Delay in operation & falling behind schedule	4.0
	Sealand Co.	1402 . 12 . 05	Failure VDR tools	1.5
	OEOC	1402 . 12 . 05	Falling behind schedule	10.0
		1402 . 12 . 06		24.0
		1402 . 12 . 07		17.5
		1402 . 12 . 09		24.0
		1402 . 12 . 10		7.0

			Drawworks failure	**7.5**
			Falling behind schedule	**4.0**
			Drawworks failure	**3.0**
			Falling behind schedule	**2.5**
		1402 . 12 . 11		**24.0**
		1402 . 12 . 12	**Wait on daylight**	**4.5**
		1402 . 12 . 13		**6.5**
		1402 . 12 . 15	**RIH w/ 5-7/8” J.mill for C/U inside 7” liner**	**1.5**
		1402 . 12 . 16		**24.0**
		1402 . 12 . 17		**12.5**
		1402 . 12 . 18	**Wait on weather**	**1.5**
			Wait on making completion fluid	**11.0**
		1402 . 12 . 19		**24.0**
		1402 . 12 . 20		**6.5**
		1402 . 12 . 21	**Wait on weather**	**1.5**
	Spad Energy	**1402 . 12 . 22**	**Work on pump truck test line leakage**	**4.5**
	OEOC	**1402 . 12 . 23**	**Work on SSD check valve**	**6.0**
Total NPT (hrs.)				**380.0**
Total NPT (days)				**16**

Table 3-3: AZNS-85 NPT Time BreakDown

4. Geological & Logging Reports

4.1 Logging and Testing Reports

Run	Date	Depth Interval (m)	Log Suite
1	24 – 02 – 2024	4045 to surface	SBT – VDL – GR – CCL
2	24 – 02 – 2024	2904 to surface	SBT – VDL – GR – CCL (repetition)
3	26 – 02 – 2024	4078 to 4045	SBT – VDL – GR – CCL (repetition)
4	27 – 02 – 2024	2907 to surface	CBIL – GR – CCL (9 – 5/8" CSG)
5	28 – 02 – 2024	Surface to 3030	Dummy run [CCL + 2* Gun]
6	01 – 03 – 2024	4076 to 3930	SBT – VDL – GR – CCL (repetition)
7	01 – 03 – 2024	4075 to 2860	CBIL – GR – CCL [liner 7"]
8	03 – 03 – 2024	4061 to surface	CCL – GR (perforation check)

Table 4-1: AZNS-85 Logging Information

5. Bottom Hole Assembly Record

Tool Description	Length (m)	Cum. Length (m)
BHA No. #1		
8 – 1/2" PDC	0.30	0.30
Bit Sub	1.10	1.40
6 – ½" DC	134.60	136.00
5" HWDP	27.84	163.84
Total Length (m)		**163.84**
Drilling & Mud Properties		
WOB (lb)		5 – 15
RPM		20 – 50
Torque (klb)		1 – 5
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		103.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #2		
5 – 7/8" B.bit	0.18	0.18
4 – ¾" Bit Sub	1.06	1.24
4 – ¾" DC	168.86	167.10
3 – ½" HWDP	55.29	222.39

3 – 1/2” DP	1123.57	1345.96
X.O.S	1.12	1347.08
Total Length (m)		**1347.08**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		103.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #3		
8 – 1/2" B.bit	0.23	0.23
9 – 5/8" Scraper	1.39	1.62
Bit Sub	1.06	2.68
6 – 1/2" DC	107.57	110.25
5" HWDP	27.84	138.09
Total Length (m)		**138.09**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-

SPP (psi)	-
Flow (gpm)	-
Mw (pcf)	103.00

Tool Description	Length (m)	Cum. Length (m)
BHA No. #4		
5” B.bit (first memory gauge)	1.26	1.26
5” Anchor Pipe	1.53	2.79
X.O.S	0.4	3.19
Middle Packer Element	1.33	4.52
Above Packer 9 – 5/8”	0.76	5.28
X.O.S 3-1/2” IF box * 4-1/2” IF pin	0.4	5.68
X.O.S 3-1/2” FH box * 3-1/2” IF pin	0.4	4.08
RH Release Safety Joint	0.73	6.81
Hydraulic Jar	1	7.81
5” M.BT (second memory gauge)	1.25	9.06
Hydro Spring	1.56	10.62
D.C.I.P	1.64	12.26
X.O.S 4-1/2” IF box * 3-1/2” IF pin	0.25	12.51
5” DP	9.56	22.07

Reverse Sub	0.4	22.47
6 – ½” DC	107.57	130.04
Total Length (m)		**130.04**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		103.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #5		
5 – 7/8" Junk Mill	0.93	0.93
Bit Sub	1.06	1.99
4 – 3/4" DC	165.86	167.85
3 – 1/2" HWDP	55.29	223.14
3 – 1/2" DP	1066.71	1289.85
X.O.S	1.12	1290.97
Total Length (m)		**1290.97**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-

Torque (klb)	-
SPP (psi)	-
Flow (gpm)	-
Mw (pcf)	103.00

Tool Description	Length (m)	Cum. Length (m)
BHA No. #6		
5 – 7/8" B.bit	0.18	0.18
7" Scraper	1.01	1.19
5 – 7/8” Bit Sub	1.06	2.25
4 – 3/4" DC	82.74	84.99
3 – 1/2" HWDP	55.29	140.28
3 – 1/2" DP	1028.71	1168.99
XOS	1.12	1170.11
6 – 1/2" D.pin	0.55	1170.66
9 – 5/8" Scraper	1.39	1172.05
6 – 1/2" Bit Sub	1.22	1173.27
Total Length (m)		**1173.27**

Drilling & Mud Properties	
WOB (lb)	-
RPM	-
Torque (klb)	-
SPP (psi)	-

Flow (gpm)		-
Mw (pcf)		103.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #7		
5 – 7/8" B.bit	0.18	0.18
4 – 3/4" Junk Sub	0.91	1.09
5" Junk Sub	0.78	1.87
4 – 3/4" D.pin	0.39	2.26
5" Junk Sub	0.92	3.18
4 – 3/4" DC	110.48	113.66
3 – 1/2" HWDP	55.29	168.95
3 – ½" DP	1067.27	1236.22
X.O.S	1.12	1237.34
Total Length (m)		**1237.34**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		103.00
Tool Description	**Length (m)**	**Cum. Length (m)**

BHA No. #8		
5 – 7/8" Junk Mill	0.93	0.93
5 – 1/2" Junk Sub	0.92	1.85
4 – 3/4" D.pin	0.28	2.13
5" Junk Sub	0.77	2.90
5 – 1/2" Junk Sub	0.92	3.82
4 – 3/4" DC	110.48	114.30
3 – 1/2" HWDP	55.29	169.59
3 – ½" DP	1067.27	1236.86
X.O.S	1.12	1237.98
Total Length (m)		**1237.98**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		103.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #9		
7 – 5/16" Polish mill	0.49	0.49
4 – ½" D.pin	0.55	1.04

6 – ½” Bit Sub	1.22	2.26
6 – ½” DC	80.58	82.84
5” HWDP	27.91	110.75
Total Length (m)		**110.75**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		103.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #10		
SN Packer	5.20	5.20
6 – ½” D.pin	0.55	5.75
6 – 1/2" Bit Sub	1.22	6.97
6 – 1/2" DC	134.86	141.83
5" HWDP	27.91	169.74
Total Length (m)		**169.74**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-

Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		103.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #11		
5 – 7/8” M.M. mill	0.93	0.93
5 – 7/8” Bit Sub	1.07	2.00
4 – ¾” DC	165.86	167.86
3 – ½” HWDP	55.29	223.15
3 – ½” DP	1067.27	1290.42
X.O.S	1.13	1291.55
Total Length (m)		**1291.55**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		103.00
Tool Description	**Length (m)**	**Cum. Length (m)**
BHA No. #12		

6 – 1/8” PDC Bit	0.21	0.21
5 – 7/8” Bit Sub (float in)	1.07	1.28
4 – ¾” DC	27.90	29.18
6 – ½” D.pin	0.28	29.46
7” Scraper	1.01	30.47
4 – ¾” Bit Sub	0.99	31.46
4 – ¾” DC	137.96	169.42
3 – ½” HWDP	55.29	224.71
3 – ½” DP	86.31	311.02
X.O.S	1.13	312.15
Total Length (m)		**312.15**
Drilling & Mud Properties		
WOB (lb)		-
RPM		-
Torque (klb)		-
SPP (psi)		-
Flow (gpm)		-
Mw (pcf)		103.00

6. Casing Information

Casing Spec.	20"	J – 55, 133^{ppf}, BTS	0 – 152^{m}
	13-3/8"	N – 80, 72^{ppf}, BTS	0 – 1098^{m}
		NKT, 72^{ppf}, BTS	1098 – 1277^{m}
	9-5/8"	110SS, 47^{ppf}, TPCQ	0 – 1844^{m}
		L – 80, 53.5^{ppf}, VAM	1844 – 3950^{m}
	7"	VM110SS, 35^{ppf}, N.VAM	3829 – 4090^{m}
	Tie-back #1	L – 80, 29^{ppf}, HSM1	3012 – 3829^{m}
	Tie-back #2	L – 80, 29^{ppf}, VAM	2907 – 3012^{m}
	SN Packer	L – 80, 29^{ppf}	2904 – 2907^{m}

Table 6-1: AZNS-85 Final Casing Data

7. Cement Job Report

Not Applicable.

8. Completion Report

8.1 Completion Tubing Information

Tubing Size (in)	Weight (ppf)	Grade	Ultimate Strengths	Connection	Collapse Pressure (psi)	Burst Pressure (psi)	Interval (m)
4 – 1/2	18.9	L – 80	233^{klb}	HSM3	13830	13380	0 – 2829.62
3 – 1/2	10.2	L – 80	440^{klb}	HSM3	12120	11560	2829.62 – 2963.38

Table 8-1: AZNS-85 Tubing Data

8.2 Completion Tools Specification Checklist

Item	Equipment Description	
1	**CONTROL LINE "WEIQIXIN":**	
	THINKNESS:	0.065"
	WORK PRESSURE:	15000 PSI
	O.D.:	0.25"
	TYPE:	SEAMLESS
	MATERIAL:	INC 825
	LENGTH:	120m/Roll
2	**UPPER FLOW COUPLING " WEIQIXIN" :**	
	OD:	**5.5"**
	ID:	3.640"
	THREADS:	4-1/2" 18.9 PPF, HSM3
	WORKING PRESSURE:	10000 PSI
	LENGTH:	6 FT
	MATERIAL:	13CR-110KSI
	LOWER FLOW COUPLING " WEIQIXIN" :	
	OD:	5.550"
	ID:	3.640"
	THREADS:	4-1/2" 18.9 PPF, HSM3
	WORKING PRESSURE:	10000 PSI
	LENGTH:	4 FT
	MATERIAL:	13CR-110KSI
3	**TUBING RETRIEVABL SAFETY VALVE " WEIQIXIN" :**	
	SETTING DEPTH:	63.90^{m}
	WORKING PRESSURE:	10000 PSI
	TEMPERATURE:	325 DEG F

	MAX O.D.:	7.5"
	MIN. I.D.:	3.437"
	PROFILE:	AF
	SELF-EQUALIZING FEATURE	HYDRAULIC CONTROL
	MATERIAL:	13CR-110KSI
	THREADS:	4-1/2" 18.9 PPF HSM3
	PROTECTION SLEEVE	
	PROFILE	AF
	MIN ID	2.9"
4	**TRAVELING JOINT " WEIQIXIN" :**	
	STROKE LENGTH	10 FT
	SHEAR PIN SET:	15000LB
	ELASTOMER:	AFLAS
	WORKING PRESSURE:	10000 psi
	TEMPERATURE:	300 DEG F
	TOOL OD:	5.315"
	TOOL ID:	2.922"
	THREADS:	3-1/2" 10.20 PPF HSM3 B*P
	MATERIAL:	13CR-110KSI
	TRANSFER TORQUE FEATURE	YES
5	**SLIDING SIDE DOOR " WEIQIXIN" :**	
	PROFILE:	X
	MAX.OD:	4.551"
	MIN ID:	2.750"
	PRESSURE RATING:	10000 PSI
	TEMPERATURE:	300 DEG F
	THREADS:	3-1/2" 10.20 PPF HSM3 B*P
	MATERIAL:	13CR-110KSI
	OPEN METHOD	DOWNWARD OPEN
6	**COMPLETION PACKER " WEIQIXIN" :**	
	TYPE:	PERMANENT
	CASING :	7" 32 PPF
	OD:	5.689"
	ID:	2.922"
	WORKING PRESSURE:	10000 PSI
	TEMPERATURE:	300 DEG F
	THREADS:	3-1/2" 10.20 PPF HSM3 B*P
	MATERIAL:	13CR-110KSI
	ELEMENT MATERIAL:	AFLAS
	4.0" ANCHOR SEAL	
	MAX OD (PACKING):	4"
	ID:	2.922"
	WORKING PRESSURE:	10000PSI
	TEMPERATURE:	300 DEG F
	THREADS:	3-1/2" 10.20 PPF HSM3 B
	TYPE:	RATCH LATCH

7	**LANDING NIPPLE " WEIQIXIN" :**	
	PROFILE TYPE:	XN
	WORKING PRESSURE:	10000 PSI
	MAX O.D.:	4.5"
	SEAL BORE:	2.75”
	MIN ID:	2.635"
	THREADS:	3-1/2" 10.20 PPF HSM3 B*P
	TEMPERATURE:	300 DEG F
	MATERIAL:	13CR-110KSI
8	**RE-ENTRY GUIDE SHOE " WEIQIXIN" :**	
	OD:	4.5"
	MIN ID:	2.922"
	TOP CONNECTION	3-1/2" 10.20 PPF HSM3 BOX
	MATERIAL:	13CR-110KSI
9	**3-1/2" CHEMICAL INJECTION" WEIQIXIN" :**	
	TYPE	TUBING RETRIVABLE
	MAX OD	5.00"
	MIN ID	2.922"
	MATERIAL:	13CR-110KSI
	TEMPERATURE	300 DEG F
	CONNECTION	3-1/2" 10.20 PPF HSM3 B*P
	CIV CONNECTION	NPT 3/8" BOX FOR CIV
	1.00"CIV VALVE	
	TYPE	DUAL CHECK VALVE WITH RUPTURE DISC
	WORKING PRESSURE	15000PSI
	MARERIAL	13CR-110KSI
	TEMPERATURE:	300 DEG F
	TOP CONNECTION	3/8" AUTOCLAVE
	BOTTOM CONNECTION	3/8" NPT
	RUPTURE DISC PRESSURE	6000PSI

Table 8-2: AZNS-85 Equipment Checklist

Item	Name	Description
1	**Tubing Hanger**	11" BOWL x 4-1/2" 18.9 PPF BOX TPCQ-EX CONNECTION BOTTOM & 4-1/2" 4-2G ACME BOX TOP WITH 4-1/2" EXTENDED NECK, WITH ONE CONTINUOUS 1/4" NPT C.LINE EXIT FOR SSSV, ONE 1/4" LINE EXIT FOR DHPG, INSTALLATION COMPLETE WITH RELATED FITTINGS AND PLUGS, WITH TWO CHANNEL FOR 3/8" CHEMICAL. BPV: 4" TYPE H THREAD PREPARATION, THREE FS SEAL TO PACK OFF THE UPPER NECK AND TWO S SEAL TO PACK OFF THE BODY OF HANGER, MATERIAL: 410SS. M.C: FF-0.5 T.C: U PSL:3G PR:2 COMPANY: MSP
2	**Fluted Adaptor**	4 1/2" 18.9 PPF, L-80 HSM3 PIN * TPCQ EX PIN, OD: 5.5'' ID: 3.64'' SLOTS: 6 * 1 INCH WIDTH

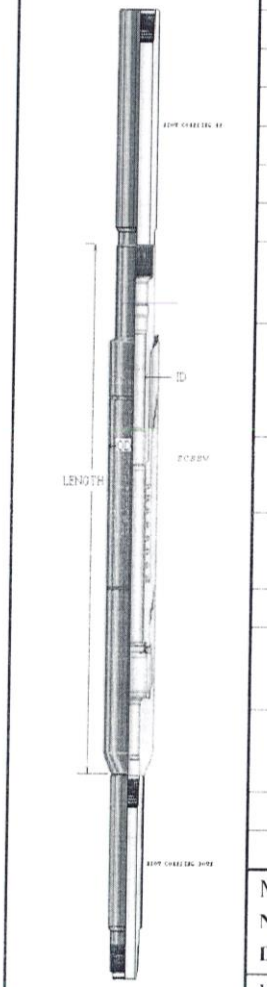
Table 8-3: AZNS-85 Equipment Checklist

8.3 Assembling Test Report

WEIQIXIN PETROLEUM MACHINERY(TIANJIN)CO.LTD

惟其信 WEIQIXIN

SCSSV Test Report



Type:	210W 343-AF04	
Serial Number:	2308 23038-04	
Connection Size:	4-1/2" 18.90 PPF HSM3 BOX × PIN	
Working Pressure	10,000Psi	
OD:	7.500"	
ID:	3.437"	
Length:	1.782 m	
Material:	S13CR-110 KSI	
Hydraulic Control Pressure	Full Open	Full Closed
	1800 Psi	800 Psi
8,000 Psi Hydraulic Control Pressure No Drop in 15 Minutes	ok	
8,000 Psi Pressure No Drop in 15 Minutes	ok	
8,000 Psi flapper Pressure No Drop in 15 Minutes	ok	
Top Connection	Length:	1.624 m
Top Flow Coupling:	Serial Number:	2307 Y230414-04
	Length:	1.726 m
Bottom Flow Coupling:	Serial Number:	2307 Y230415-04
	Length:	1.116 m
Bottom Connection:	Length:	1.162m
Well Name and Location:	AZNS-85	

MSA Project Manager:
Name/Title:
Date:

WQX:
Name/Title:
Date: 2024.03.08

Drilling Supervisor
South Azadegan Project
Rig No:
Date:
MSA
LPDS CONTRACTOR

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Figure 8-1: AZNS-85 TRSSSV Test Report

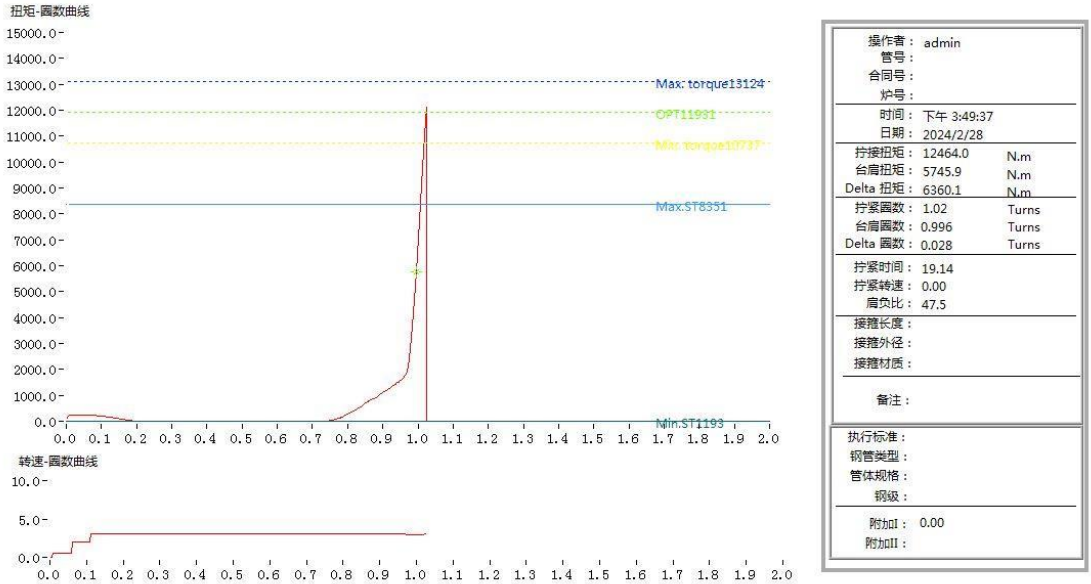


Figure 8-2: AZNS-85 Top Pup Jt – Top Flow Coupling Connection Graph

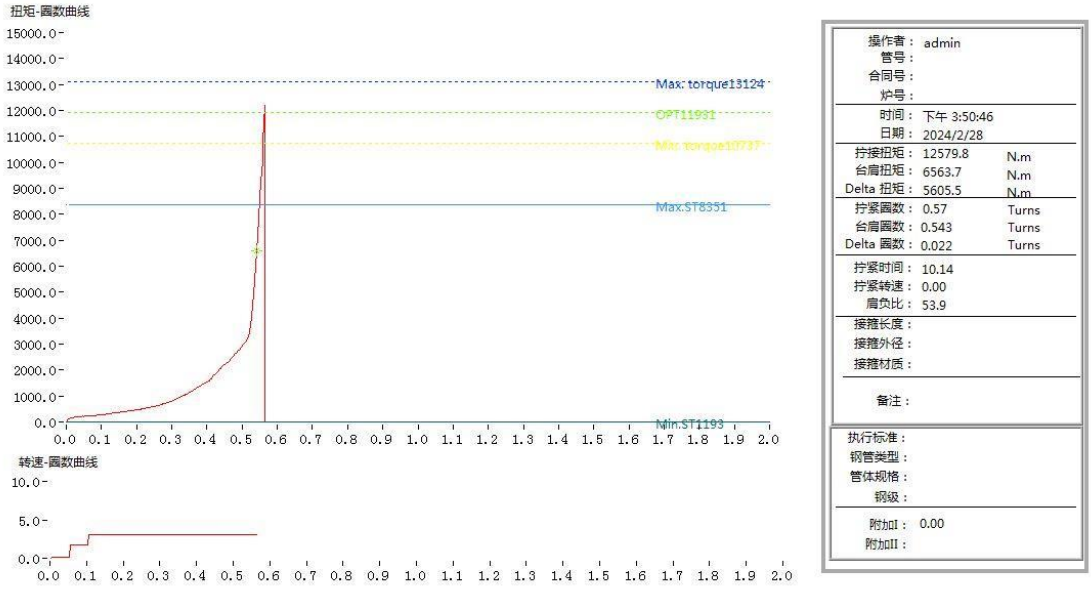


Figure 8-3: AZNS-85 Top Flow Coupling – TRSSSV Connection Graph

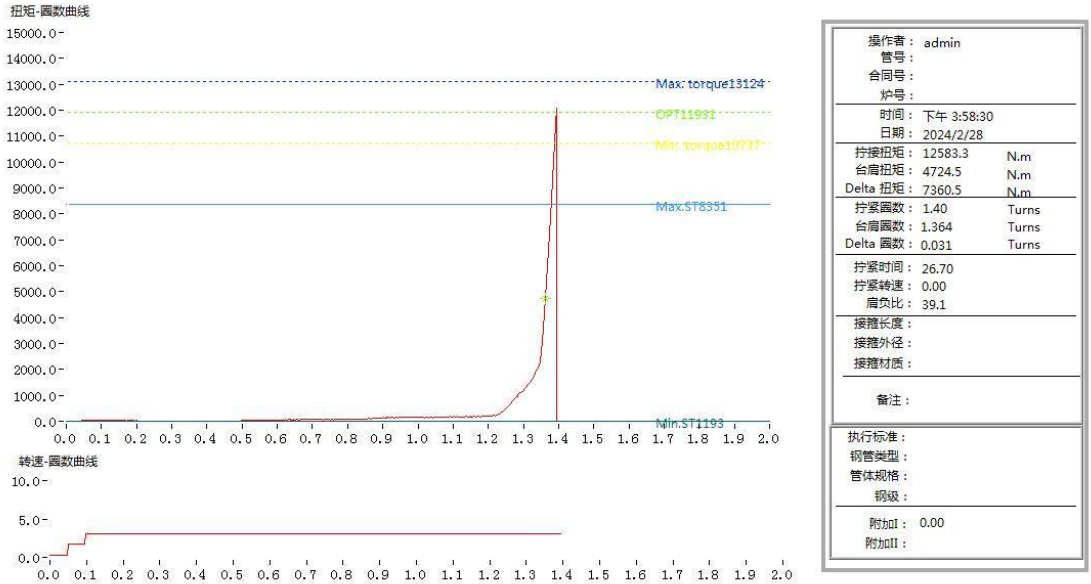


Figure 8-4: AZNS-85 TRSSSV – Bottom Flow Coupling Connection Graph

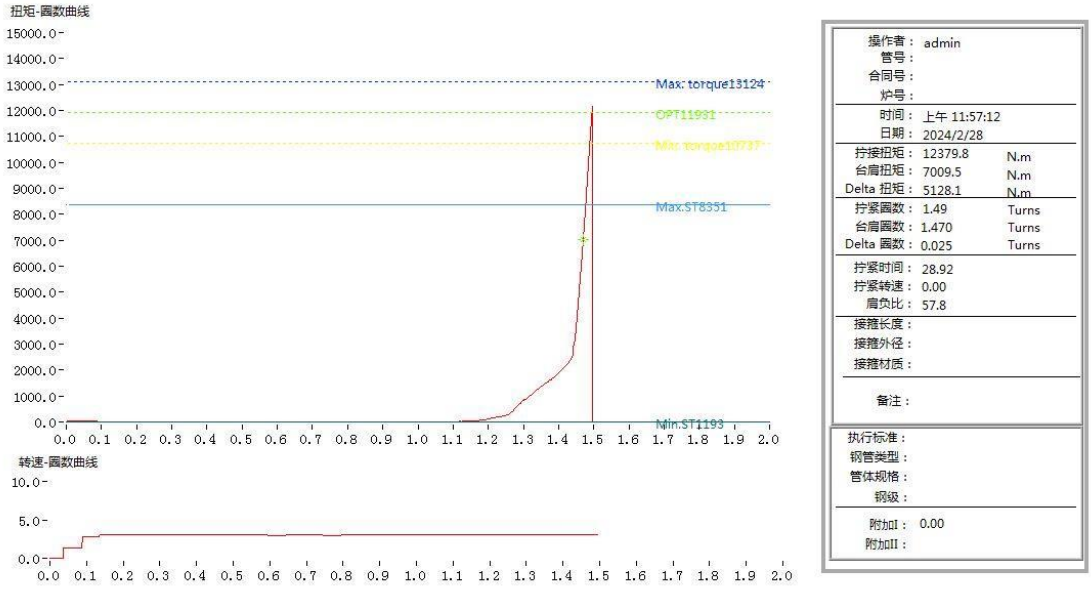


Figure 8-5: AZNS-85 Bottom Flow Coupling – Bottom Pup Jt Connection Graph

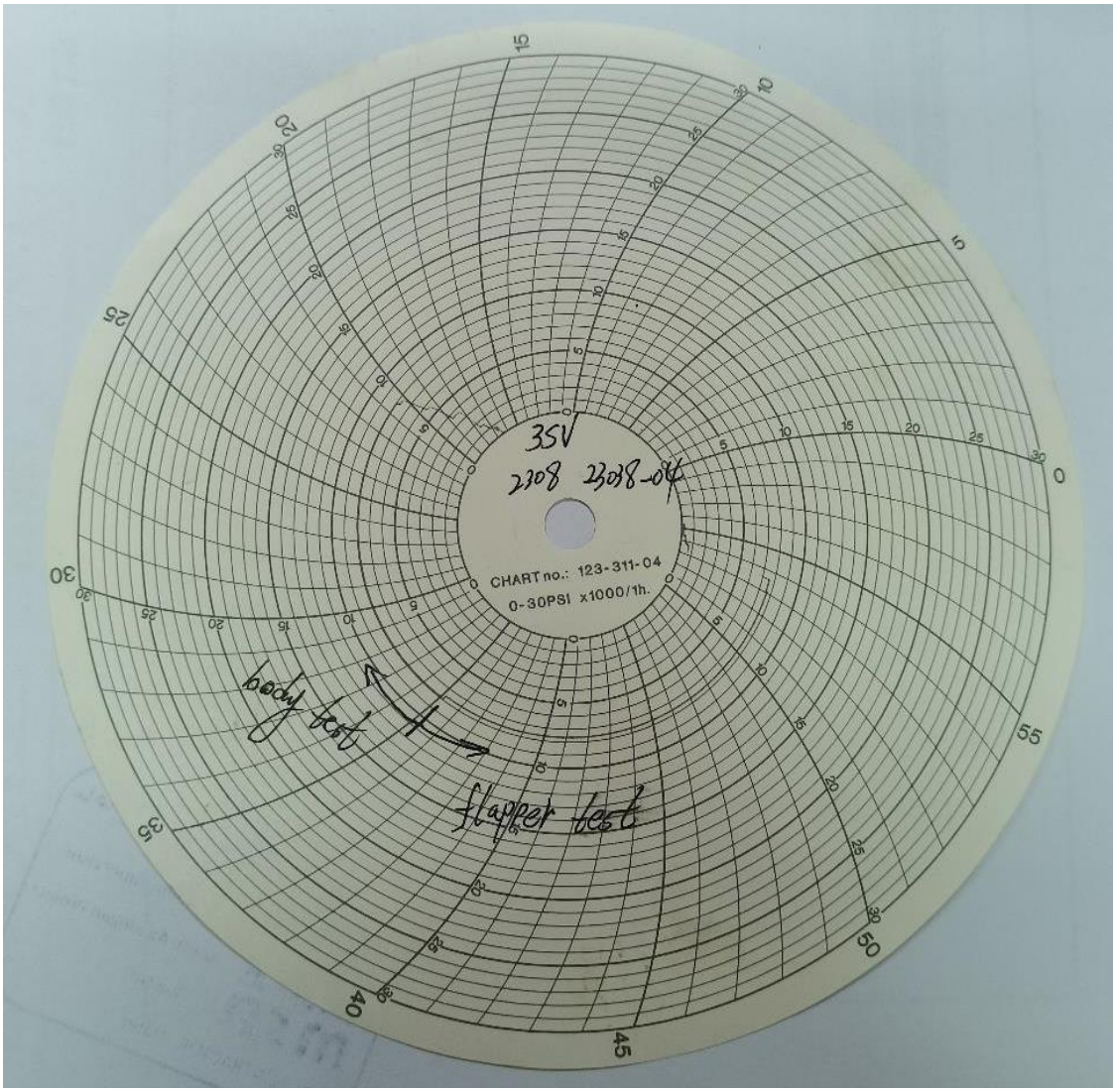


Figure 8-6: AZNS-85 TRSSSV Body Pressure Test Graph

4 1/2'' Short Flow Coupling

WQX 110W450-042,OD:5.500" ID:3.640" 4 FT,10.000 PSI S13CR-110KSI

4.1/2" 18.90PPF HSM-3 BxP (2307 Y230415-04)



4-1/2" X 3.437" TRSSV

WQX TRSSV 210W343-AF04,10.000 PSI 325 DEG F OD:7.500" , ID:3.437" S13CR-110KSI, INCONEL 718, 4.1/2" 18.90 PPF HSM3 BxP 2308 23038-04



4 1/2'' Long Flow Coupling

WQX 110W450-042,OD:5.500" ID:3.640"6 FT,10.000 PSI S13CR-110KSI

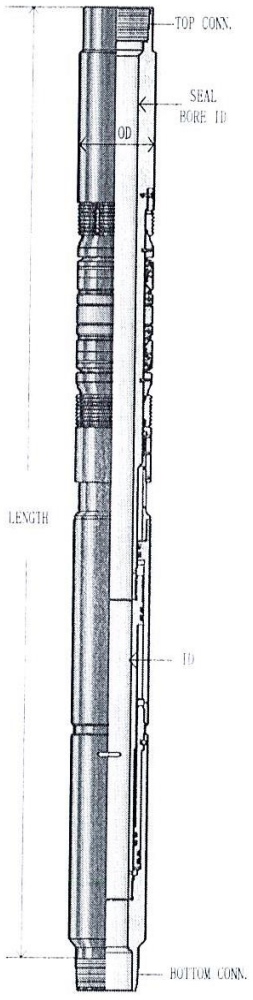
4.1/2" 18.90PPF HSM-3 BxP (2307 Y230414-03



WEIQIXIN PETROLEUM MACHINERY(TIANJIN)CO.LTD

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WSDP Hydraulic Set Permanent Packer Test Report



Type:	336W150-000	
Serial Number:	2308 23062-05	
Connection Size:	LEFT HAND X 3-1/2" 10.20 PPF HSM3 PIN	
Working Pressure	10,000 Psi	
OD:	5.910"	
ID:	2.922"	
Length:	4.440m	
Material:	S13CR-110KSI	
Temperature:	300 DEG F	
Element :	Aflas	
Setting Pressure:	3,500 Psi	
Seal Bore Size:	Upper Seal Bore ID:4.00" & Lower Seal Bore ID:2.922"	
Top Connection:	Length:	1.622m
Bottom Connection:	Length:	1.616 m
Well Name and Location:	AZNS-85	

MSA Project Manager:
Name/Title:
Date:

WQX:
Name/Title:
Date: 2024.03.08

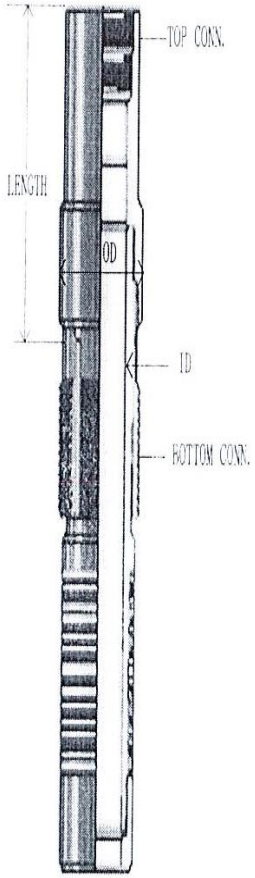
Drilling Supervisor
South Azadegan Project
Rig No:
LPDS CONTRACTOR Date:

Figure 8-7: AZNS-85 Permanent Packer Test Report

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

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Model WK Anchor Tubing Seal Report



Type:	360W150-000
Serial Number:	2308 23063-05
Connection Size:	3-1/2" 10.20 PPF HSM3 BOX X MULE SHOE GUIDE
OD:	5.689"
ID:	2.922"
Seal OD:	4.000"
Working pressure:	10,000Psi
Length:	0.228m
Material:	S13CR-110 KSI
Well Name and Location	AZNS-85

MSA Project Manager:
Name/Title:
Date:

WQX:
Name/Title:
Date: 2024.03.08

Drilling Supervisor
South Azadegan Project
Rig No:
Date:
LPDS CONTRACTOR

Figure 8-8: AZNS-85 Anchor Tubing Seal Test Report

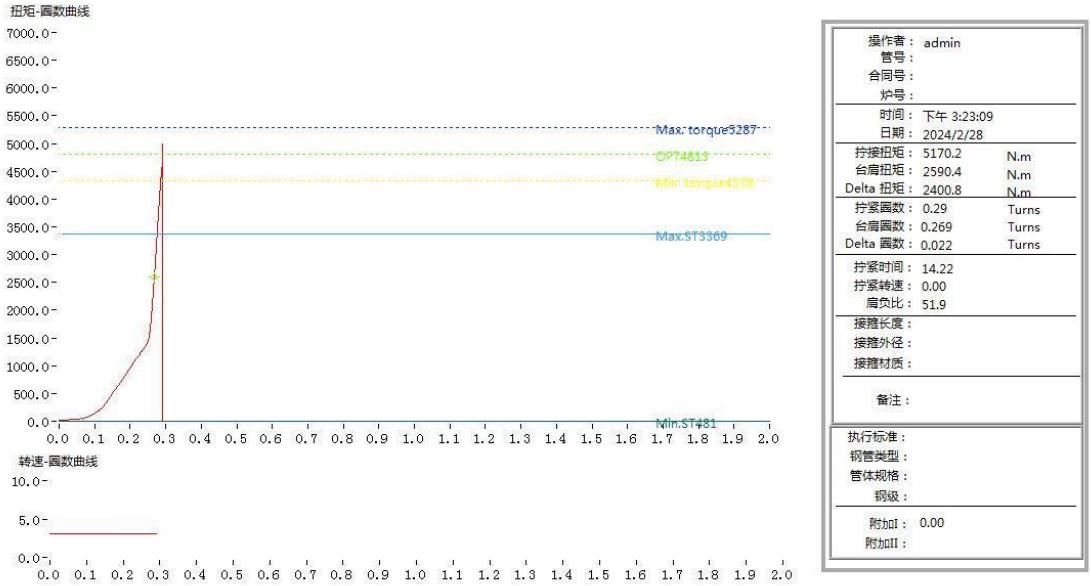


Figure 8-9: AZNS-85 Top Pup Jt – Anchor Seal Connection Graph

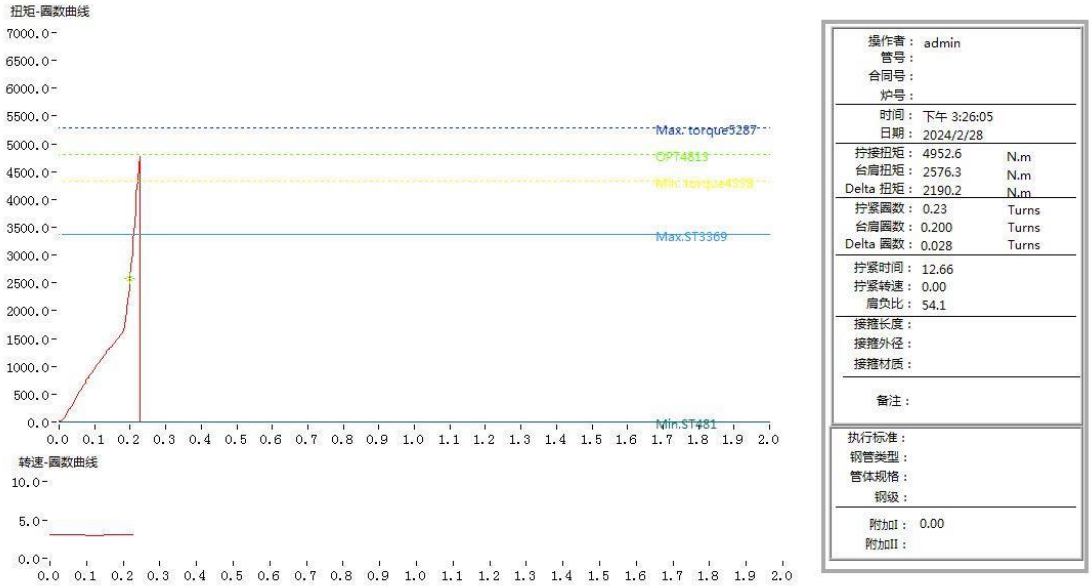


Figure 8-10: AZNS-85 Packer – Bottom Pup Jt Connection Graph

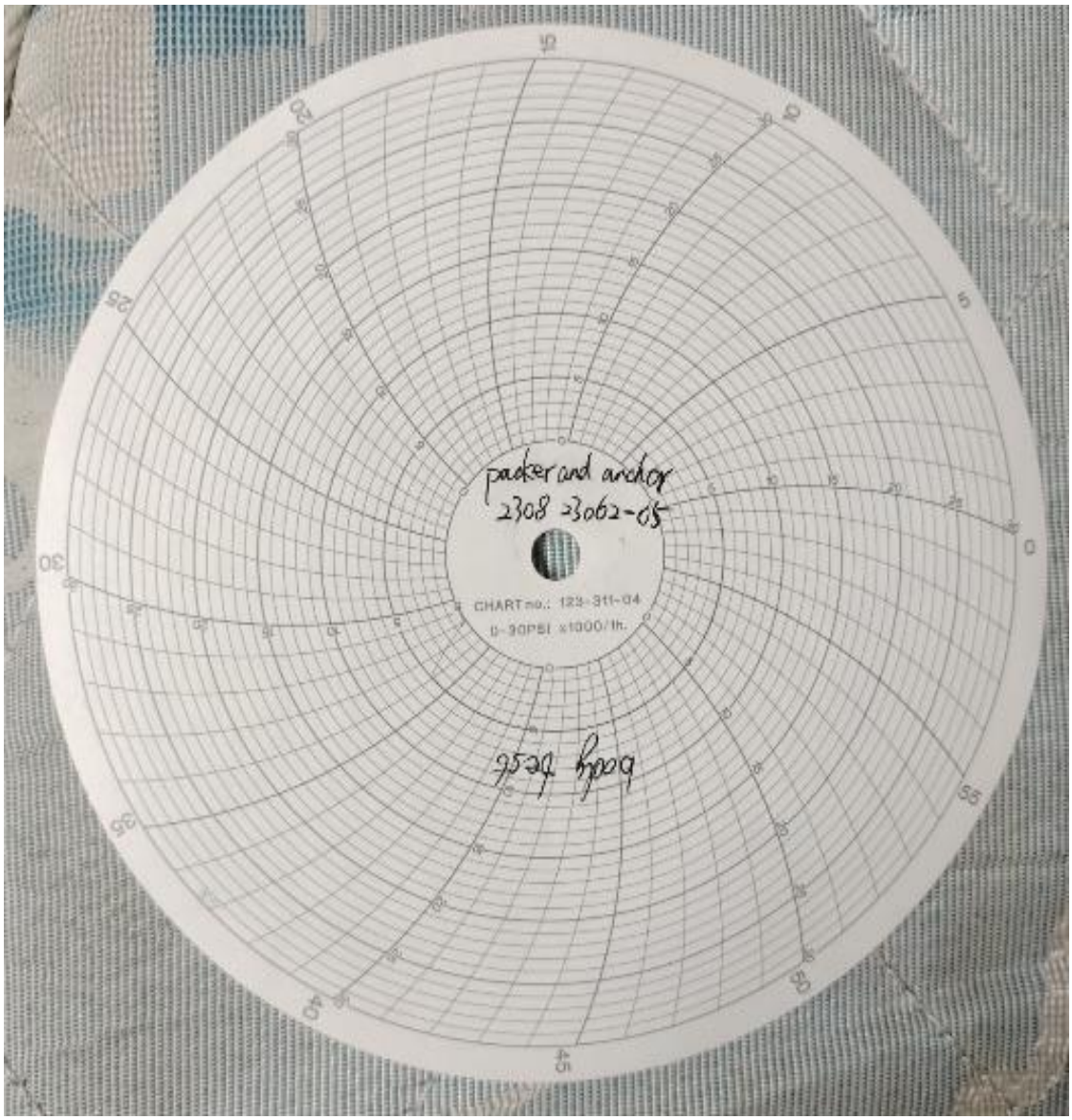


Figure 8-11: AZNS-85 Packer Body Pressure Test Graph

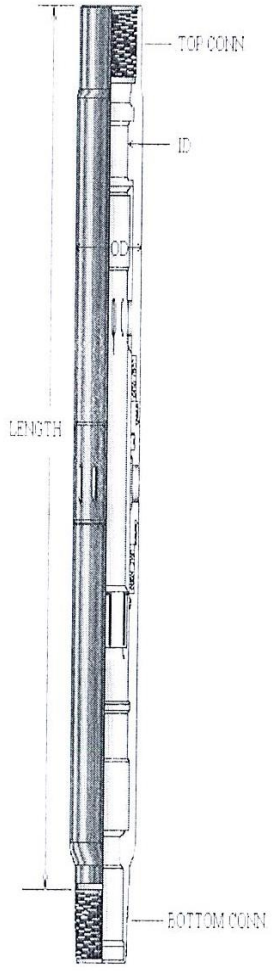
PACKER
WQX PACKER 336W150-000
10.000 PSI 300 DEG F OD:5.910" ID :4.000" ID:2.922"
3.1/2" 10.20 PPF HSM-3 PIN
2308 23062-05, S13CR-110KSI



WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

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Sliding Side Door Test Report



Type:	750W275-X39	
Serial Number:	2308 23039-04	
Connection Size:	3-1/2" 10.20 PPF HSM3 BOX × PIN	
Working Pressure	10,000 Psi	
OD:	4.551"	
ID:	2.750"	
Seal Bore ID :	2.750"	
Length:	1.197m	
Temperature	300 DEG F	
Material:	S13CR-110KSI	
Profile:	X	
Shifting Direction:	DOWNWARD OPEN	
Top Connection:	Length:	1.823 m
Bottom Connection:	Length:	9.588m
8,000 Psi Pressure No Drop in 15 Minutes	Ok	
8,000 Psi blanking plug Pressure No Drop in 15 Minutes	Ok	
Well Name and Location:	AZNS-85	

MSA Project Manager:
Name/Title:
Date:

WQX:
Name/Title:
Date: 2024. 03. 08

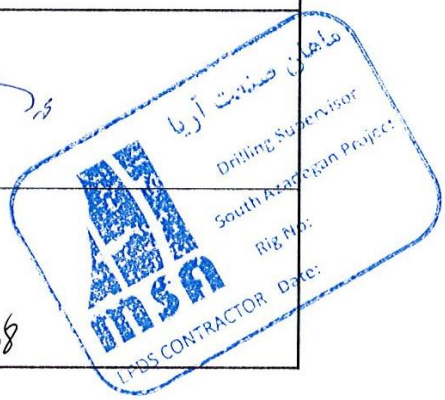


Figure 8-12: AZNS-85 SSD Test Report

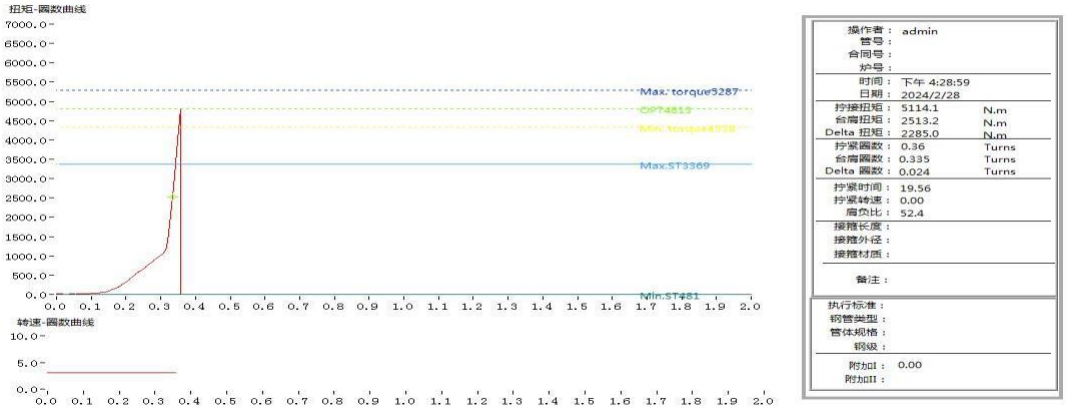


Figure 8-13: AZNS-85 Top Pup Jt – SSD Connection Graph

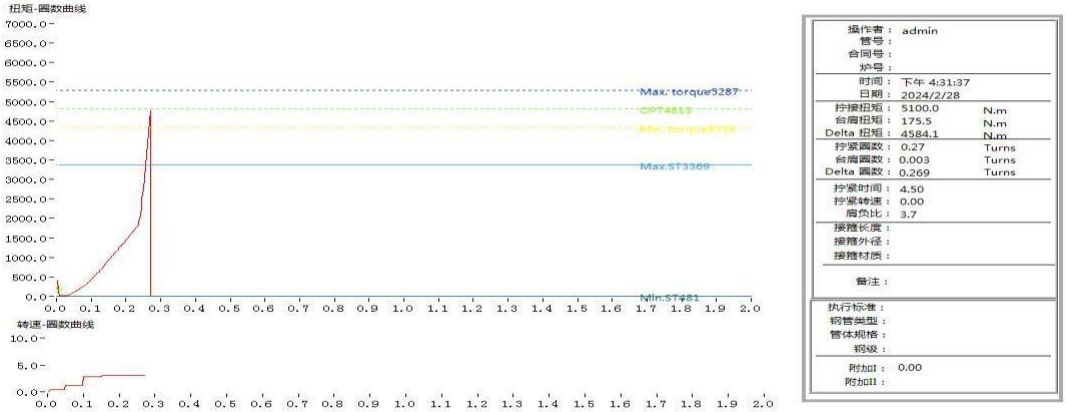


Figure 8-14: AZNS-85 SSD – Bottom Pup Jt Connection Graph

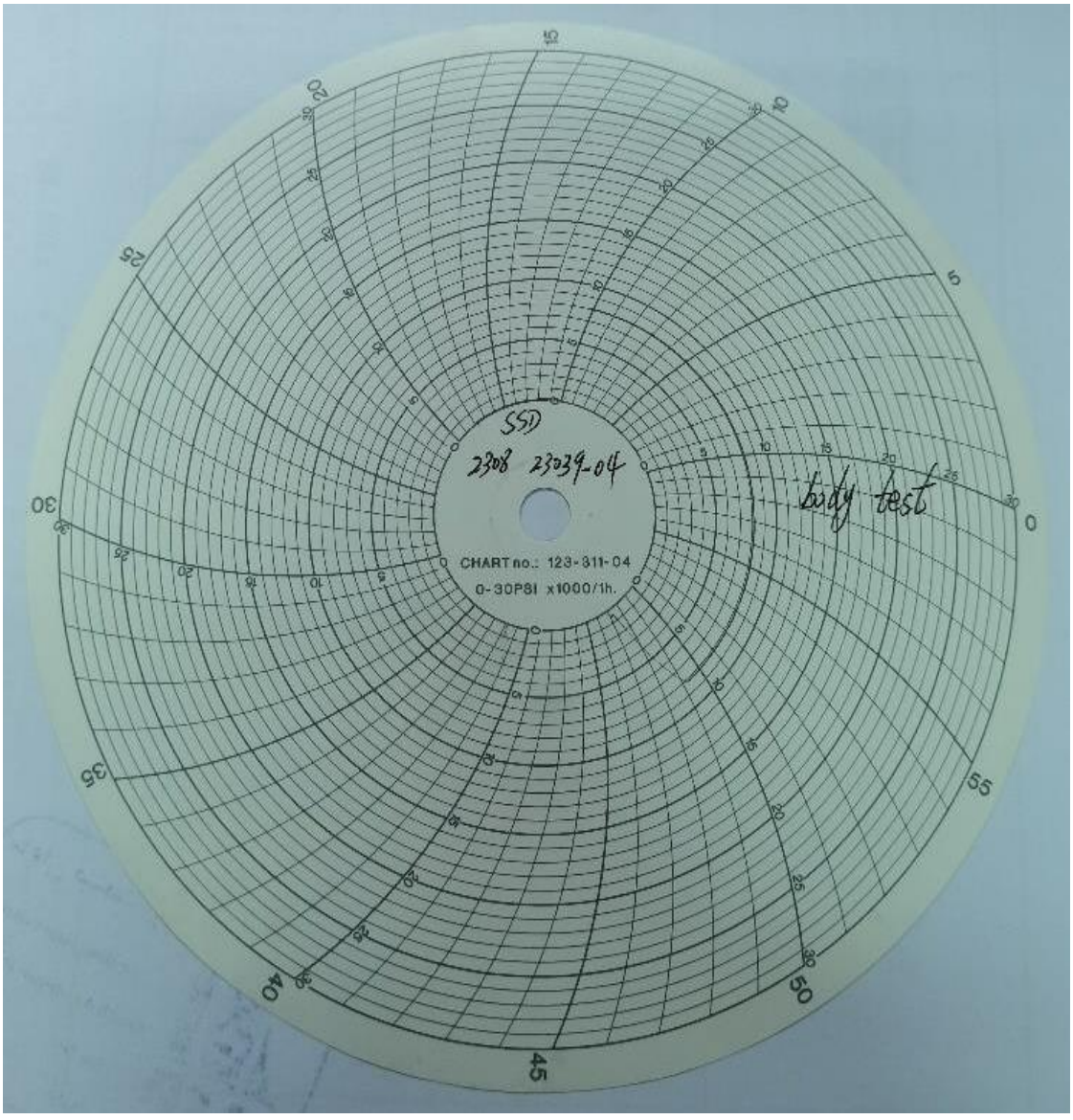


Figure 8-15: AZNS-85 SSD Body Pressure Test Graph

4) 3-1/2" SLIDING SIDE DOOR

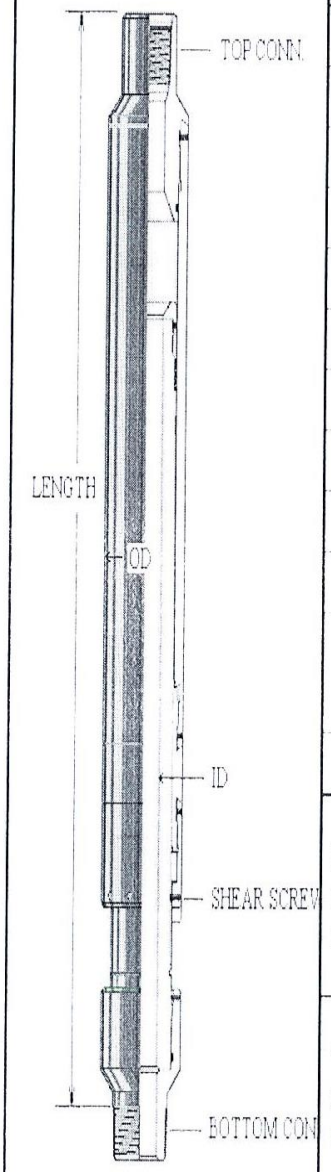
WQX 750W 275-X39, 10.000 PSI 300 DEG F
OD:4.551" ID:2.750" , 3-1/2" 10.20 PPF HSM-3 BxP
2308 23039-04 , S13CR-110 KSI



WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

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Expansion Joint Test Report



Type:	370W350-030	
Serial Number:	2308 23040-04	
Connection Size:	3-1/2" 10.20 PPF HSM3 BOX × PIN	
Working Pressure	10,000 Psi	
OD:	5.315"	
ID:	2.922"	
Length:	4.258m(CLOSE)/7.320 m(OPEN)	
Material	S13CR-110KSI	
Share Screw:	2	
Top Connection:	Length:	1.897 m
Bottom Connection:	Length:	1.741 m
8.000 Psi Pressure No Drop in 15 Minutes	ok	
Well Name and Location:	AZNS-85	

MSA Project Manager:
Name/Title:
Date:

WQX:
Name/Title:
Date: 2024.03.08

Drilling Supervisor
South Azadegan Project
Rig No:
Date:
EPDS CONTRACTOR

Figure 8-16: AZNS-85 EX Jt Test Report

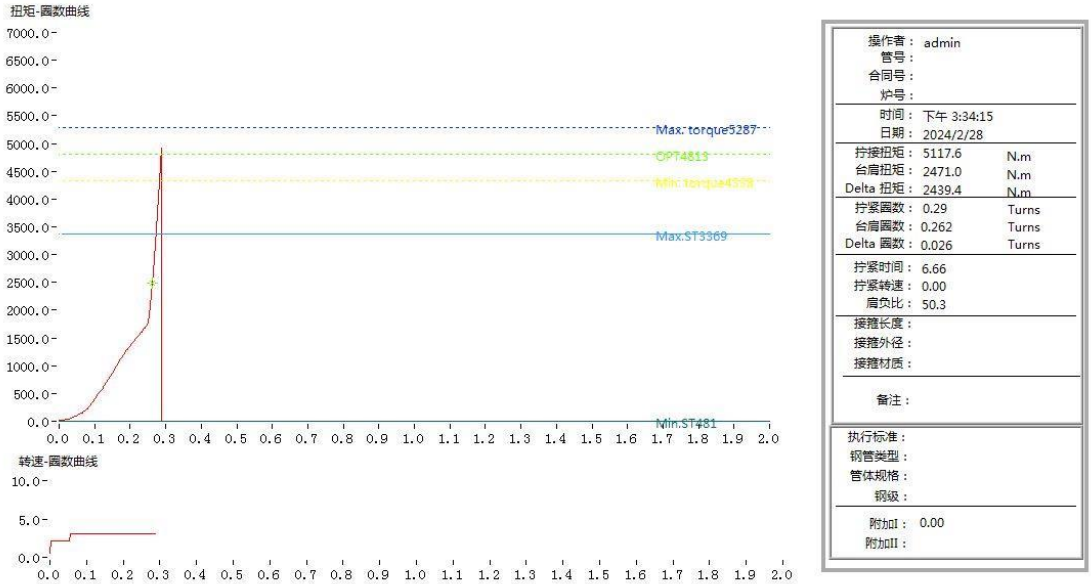


Figure 8-17: AZNS-85 Top Pup Jt – EX Jt Connection Graph

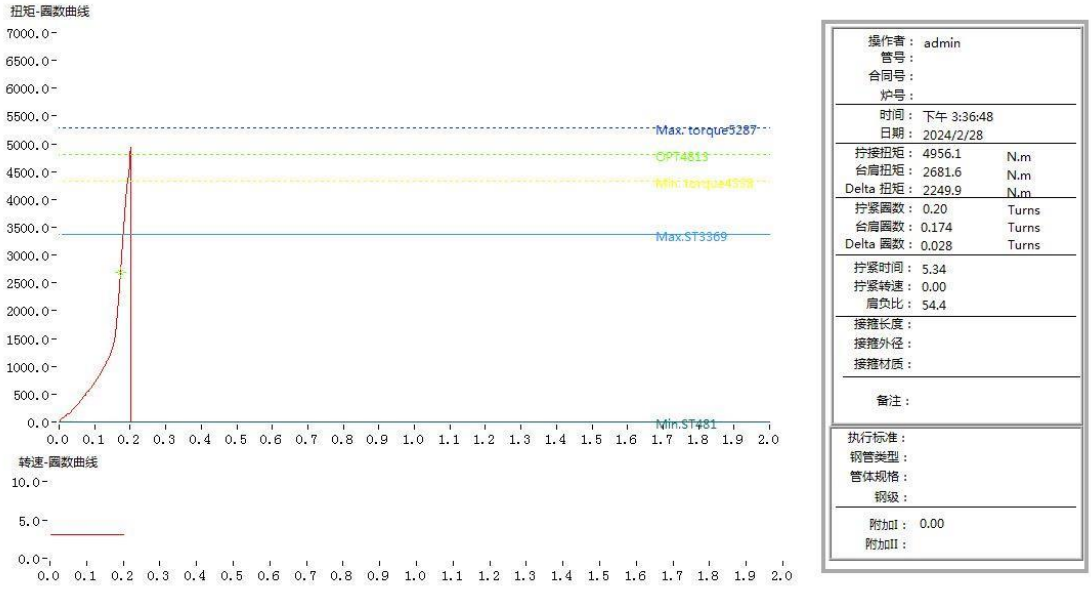


Figure 8-18: AZNS-85 EX Jt – Bottom Pup Jt Connection Graph

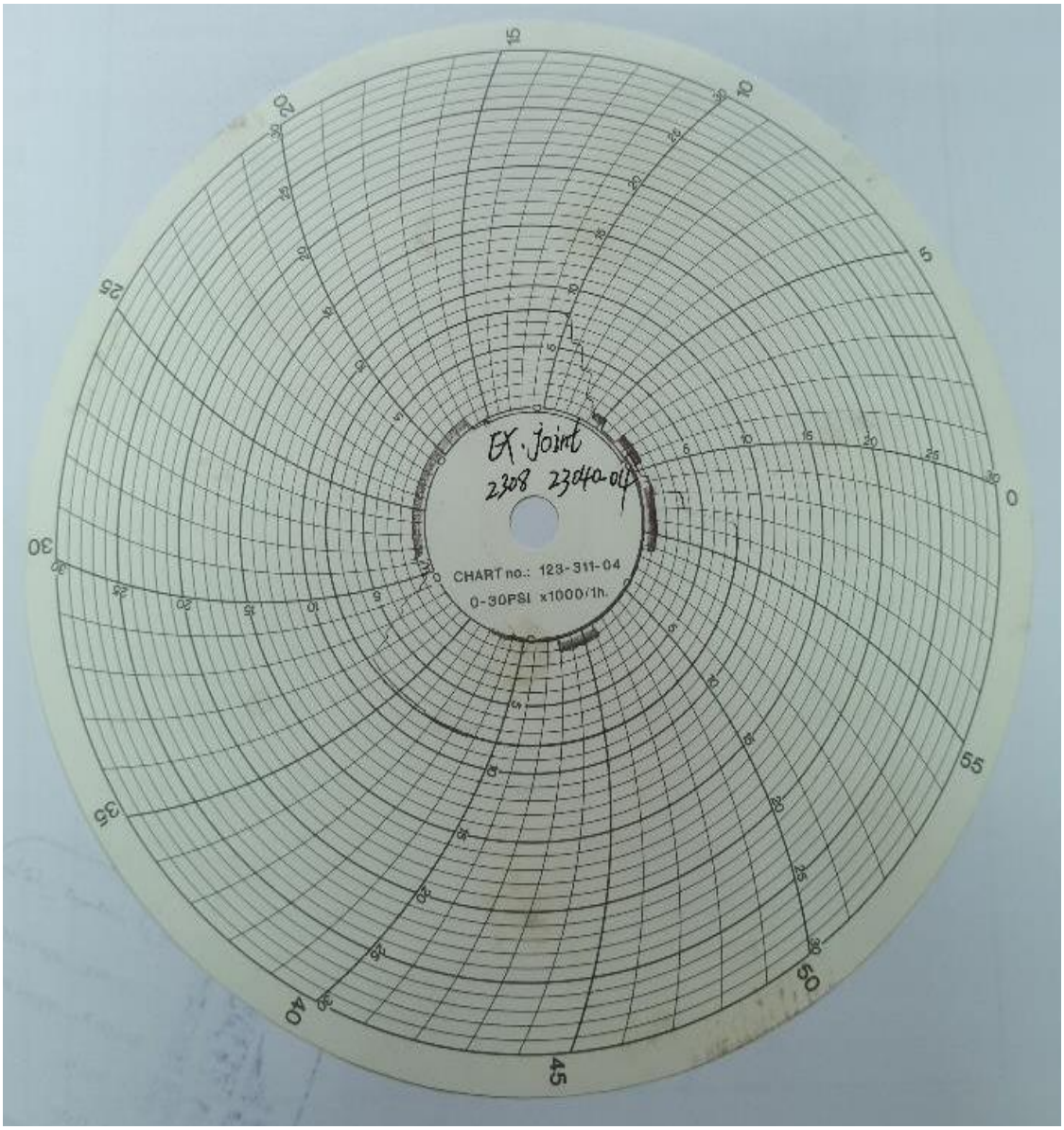


Figure 8-19: AZNS-85 EX Jt Body Pressure Test Graph

9) 3.1/2" EXP. JOINT

WQX 370W350-030,10.000 PSI 300 DEG F , OD:5.315" ID : 2.922" 3-1/2" 10,20 PPF HSM-3 BxP

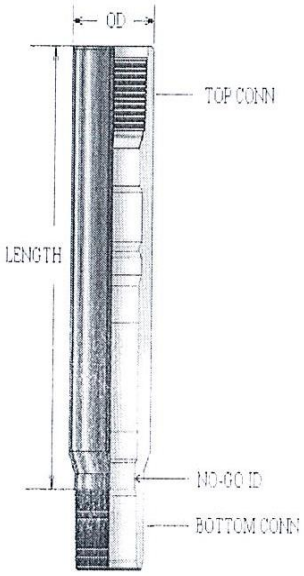
2308 23040-04 , S13CR-110KSI



WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

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Landing Nipple Test Report



Type:	706W275-N24	
Serial Number:	2307 Y230416-04	
Connection Size:	3-1/2"10.22 PPF HSM3 BOX × PIN	
OD:	4.50"	
ID:	2.75"	
NO-GO ID:	2.635"	
Length:	0.373m	
Material:	S13CR-110 KSI	
Top Connection:	Length:	9.512m
8,000 Psi blanking plug Pressure No Drop in 15 Minutes	Ok	
Well Name and Location	AZNS-85	

MSA Project Manager:
Name/Title:
Date:

WQX:
Name/Title:
Date: 2024.05.08

ماهان صنعت آریا
Drilling Supervisor
South Azadegan Project
Rig No:
Date:
EPDS CONTRACTOR

Figure 8-20: AZNS-85 Landing Nipple Test Report

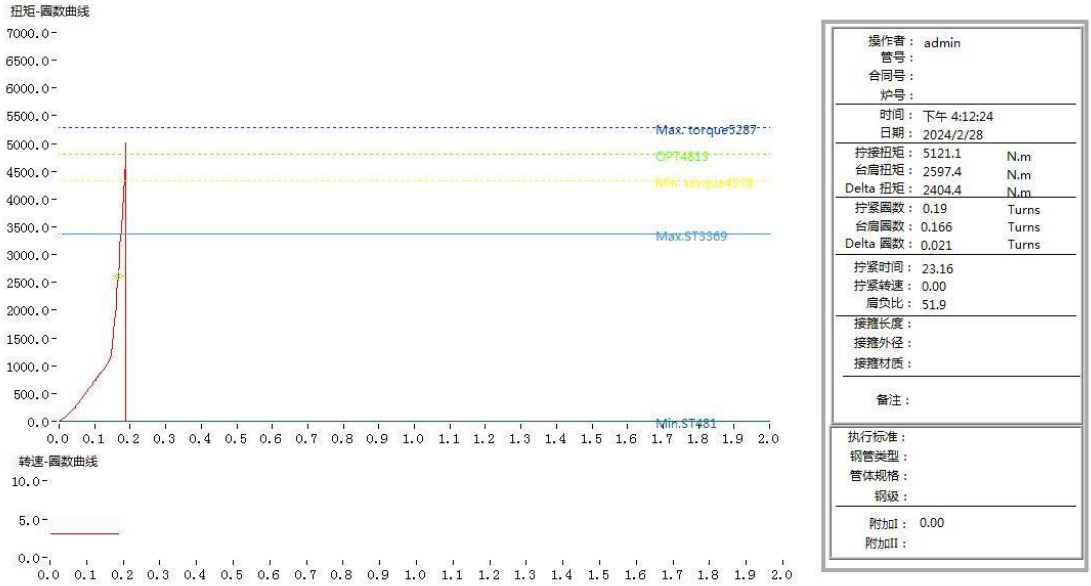


Figure 8-21: AZNS-85 Top Jt – Landing Nipple Connection Graph

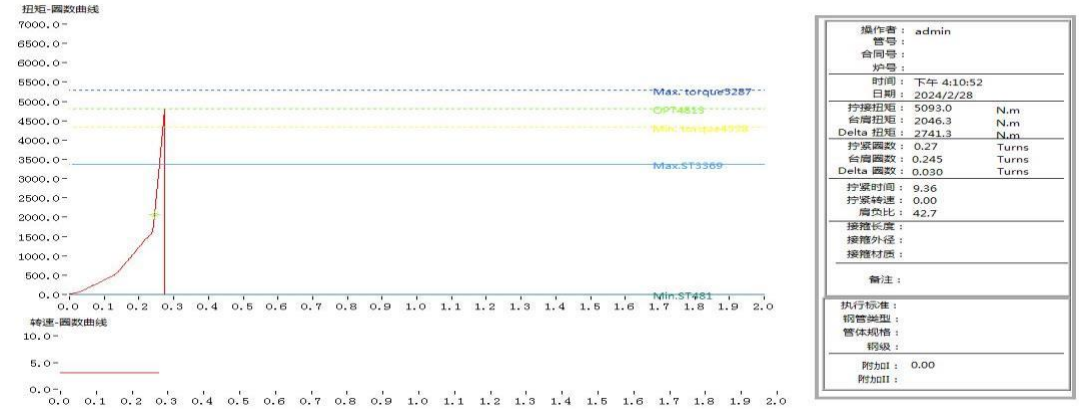


Figure 8-22: AZNS-85 Landing Nipple – Mule Shoe Connection Graph



Figure 8-23: AZNS-85 Landing Nipple & Mule Shoe Body Pressure Test Graph

1) 3-1/2" RE-ENTRY GUID SHOE

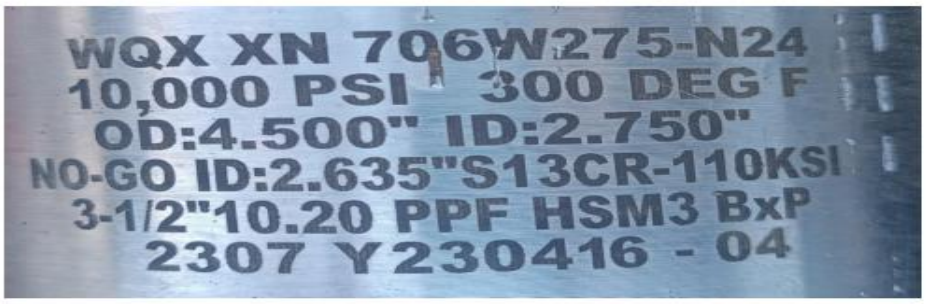
WQX 824 W350-021 OD:4.250" ID:2.922"

3.1/2" 10.20 PPF HSM-3 BOX 2307 Y230448-4



2) 3 ½" NO-GO LANDING NIPPLE

WQX XN 706W275-N24 10.000PSI 300 DEG F
OD:4.500" ID: 2.750"
NO-GO ID:2.635" S12CR-110KSI 3.1/2 10.20 PPF HSM-3 BxP
2307 Y230416-04

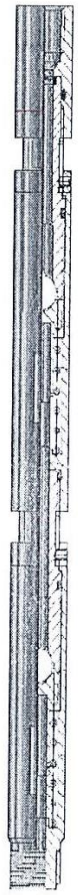




WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

惟其信 WEIQIXIN

Chemical Injection Valve Test Report



Type:	802W100-601
Serial Number:	2306 23068-05
Connection Size:	3/8" JAM NUT BOX x 3/8" NPT PIN
OD:	1"
ID:	0.299"
Rupture DISK:	6000Psi
Spring open pressure:	500Psi
Length:	0.310m
Material:	S13CR-110 KSI
8,000 Psi Pressure No Drop in 15 Minutes	Ok
Well Name and Location	AZNS-85

MSA Project Manager:
Name/Title:
Date:

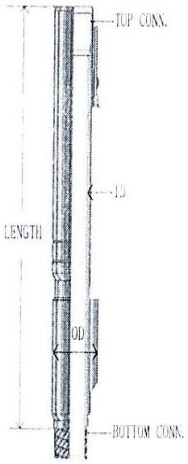
WQX:
Name/Title:
Date: 2024.03.08

ماهان صنعت آریا
Drilling Supervisor
South Azadegan Project
Rig No:
Date:
msa
EPDS CONTRACTOR

Figure 8-24: AZNS-85 First CIV Test Report

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

Chemical Injection Mandrel Test Report



Type:	800W350-028	
Serial Number:	2307 23067-05	
Connection Size:	3-1/2" 10.2PPF HSM3 BOX × PIN	
OD:	5.000"	
ID:	2.922"	
Working pressure:	15,000Psi	
Temperature:	300 DEG F	
Length:	0.727m	
Top Connection:	Length:	1.821 m
Bottom Connection:	Length:	9.588 m
Material:	S13CR-110KSI	
8,000 Psi Pressure No Drop in 15 Minutes	Ok	
Well Name and Location	AZNS-85	

MSA Project Manager:
Name/Title:
Date:

WQX:
Name/Title:
Date:
2024. 03. 08

Drilling Supervisor
South Azadegan Project
Rig No:
Date:
HPDS CONTRACTOR

Figure 8-25: AZNS-85 First CIM Test Report

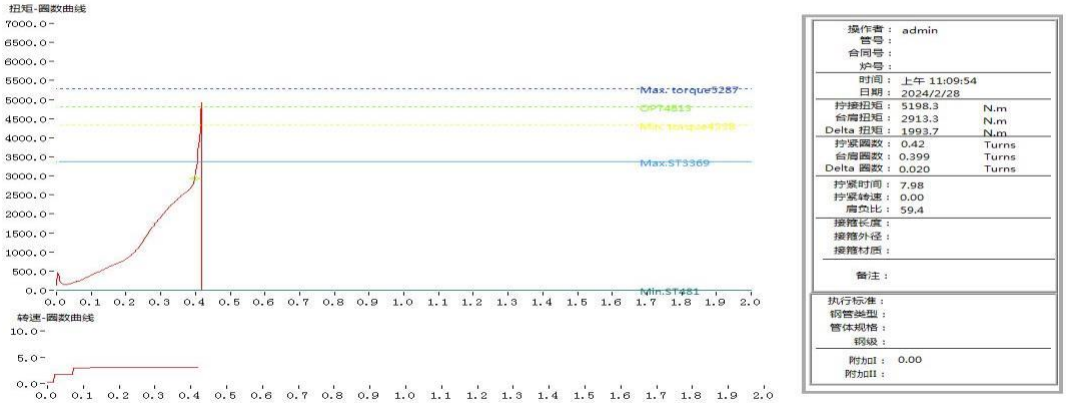


Figure 8-26: AZNS-85 Top Pup Jt – CIM #1 Connection Graph

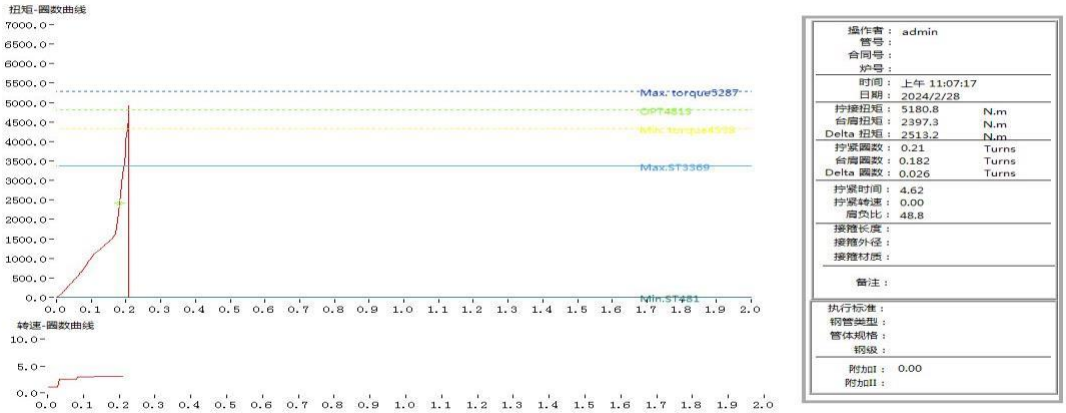


Figure 8-27: AZNS-85 CIM #1 – Bottom Jt Connection Graph

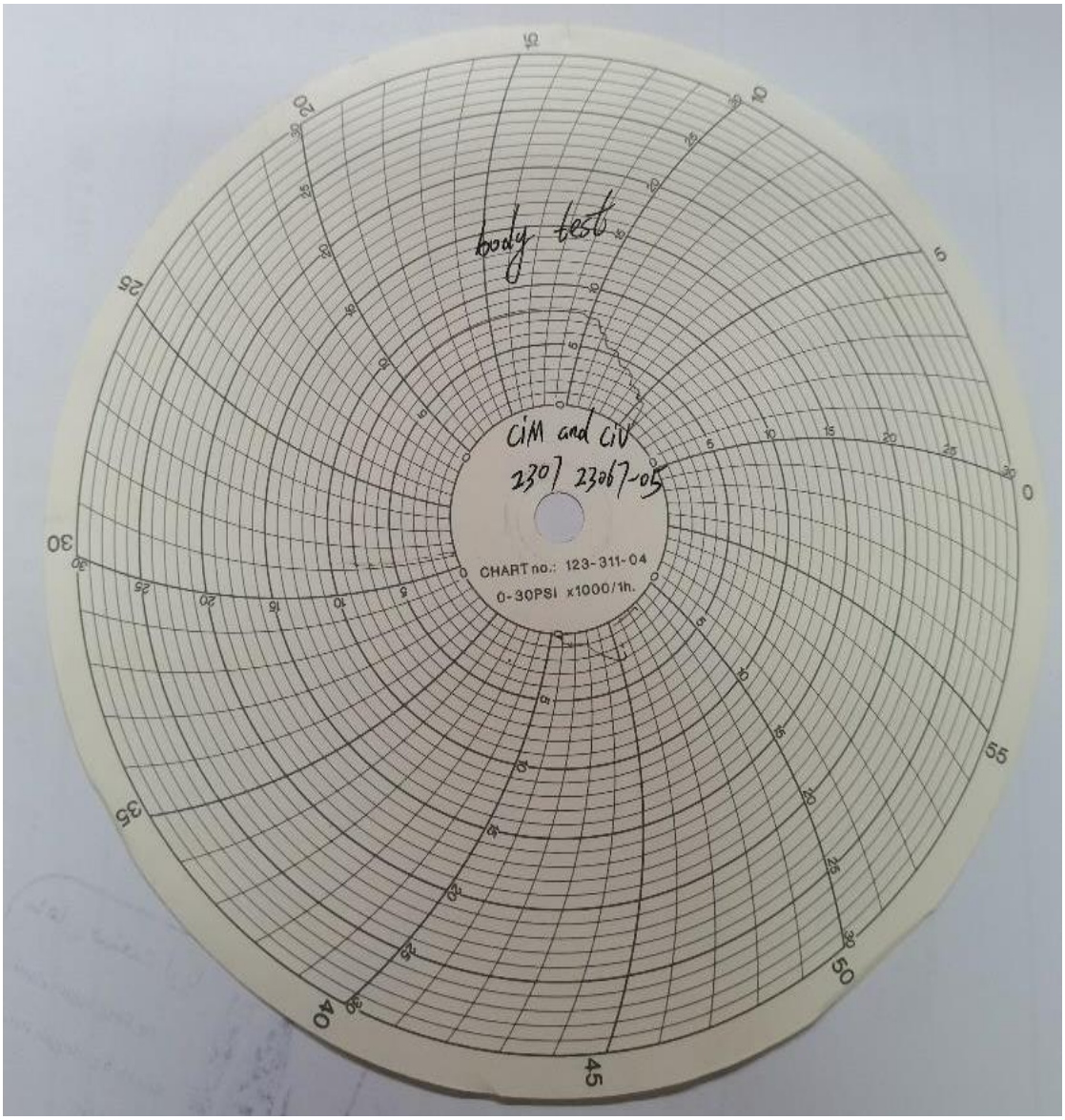


Figure 8-28: AZNS-85 First CIM Body Pressure Test Graph

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

惟其信 WEIQIXIN

Chemical Injection Valve Test Report



Type:	802W100-601
Serial Number:	2306 23068-06
Connection Size:	3/8" JAM NUT BOX x 3/8" NPT PIN
OD:	1"
ID:	0.299"
Rupture DISK:	6000Psi
Spring open pressure:	500Psi
Length:	0.310m
Material:	S13CR-110 KSI
8.000 Psi Pressure No Drop in 15 Minutes	Ok
Well Name and Location	AZNS-85

MSA Project Manager:
Name/Title:
Date:

WQX:
Name/Title:
Date: 2024.01.08

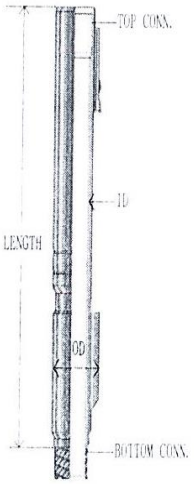
ماهان صنعت آریا
Drilling Supervisor
South Azadegan Project
Rig No:
MSA EPDS CONTRACTOR Date:

Figure 8-29: AZNS-85 Second CIV Test Report

WEIQIXIN PETROLEUM MACHINERY(TIANJIN) CO.LTD

惟其信 WEIQIXIN

Chemical Injection Mandrel Test Report



Type:	800W350-028	
Serial Number:	2307 23067-06	
Connection Size:	3-1/2" 10.2PPF HSM3 BOX × PIN	
OD:	5.000"	
ID:	2.922"	
Working pressure:	15,000Psi	
Temperature:	300 DEG F	
Length:	0.727m	
Top Connection:	Length:	1.821 m
Bottom Connection:	Length:	9.452 m
Material:	S13CR-110KSI	
8,000 Psi Pressure No Drop in 15 Minutes	Ok	
Well Name and Location	AZNS-85	

MSA Project Manager:
Name/Title:
Date:

WQX:
Name/Title:
Date: 2024.03.08

ماهان صنعت آریا
Drilling Supervisor
South Azadegan Project
Rig No:
Date:
MSA
LPDS CONTRACTOR

Figure 8-30: AZNS-85 Second CIM Test Report

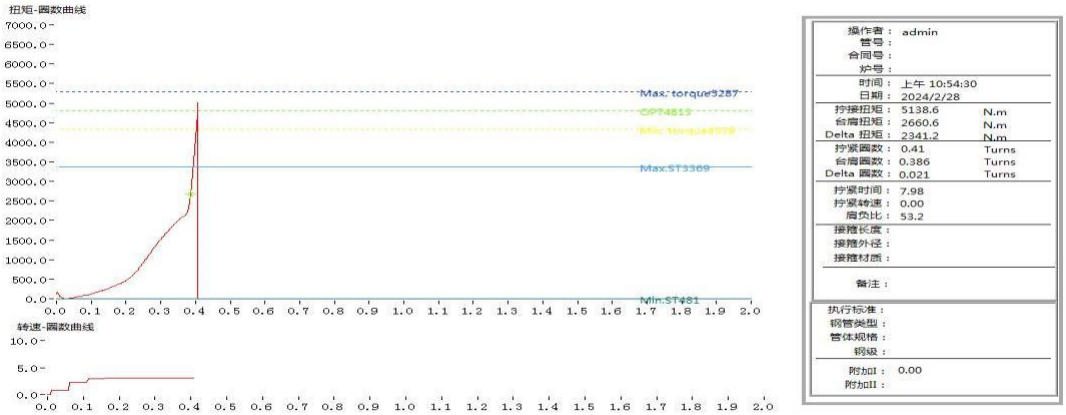


Figure 8-31: AZNS-85 Top Pup Jt – CIM #2 Connection Graph

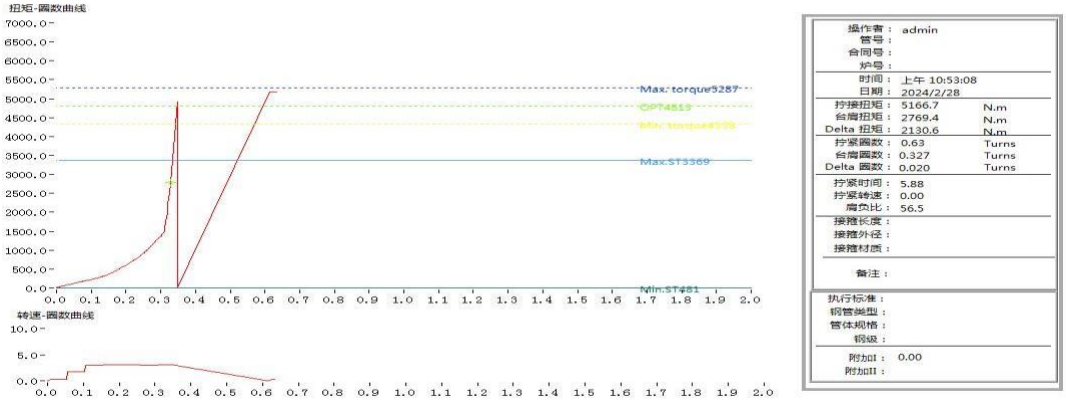


Figure 8-32: AZNS-85 CIM #2 – Bottom Jt Connection Graph

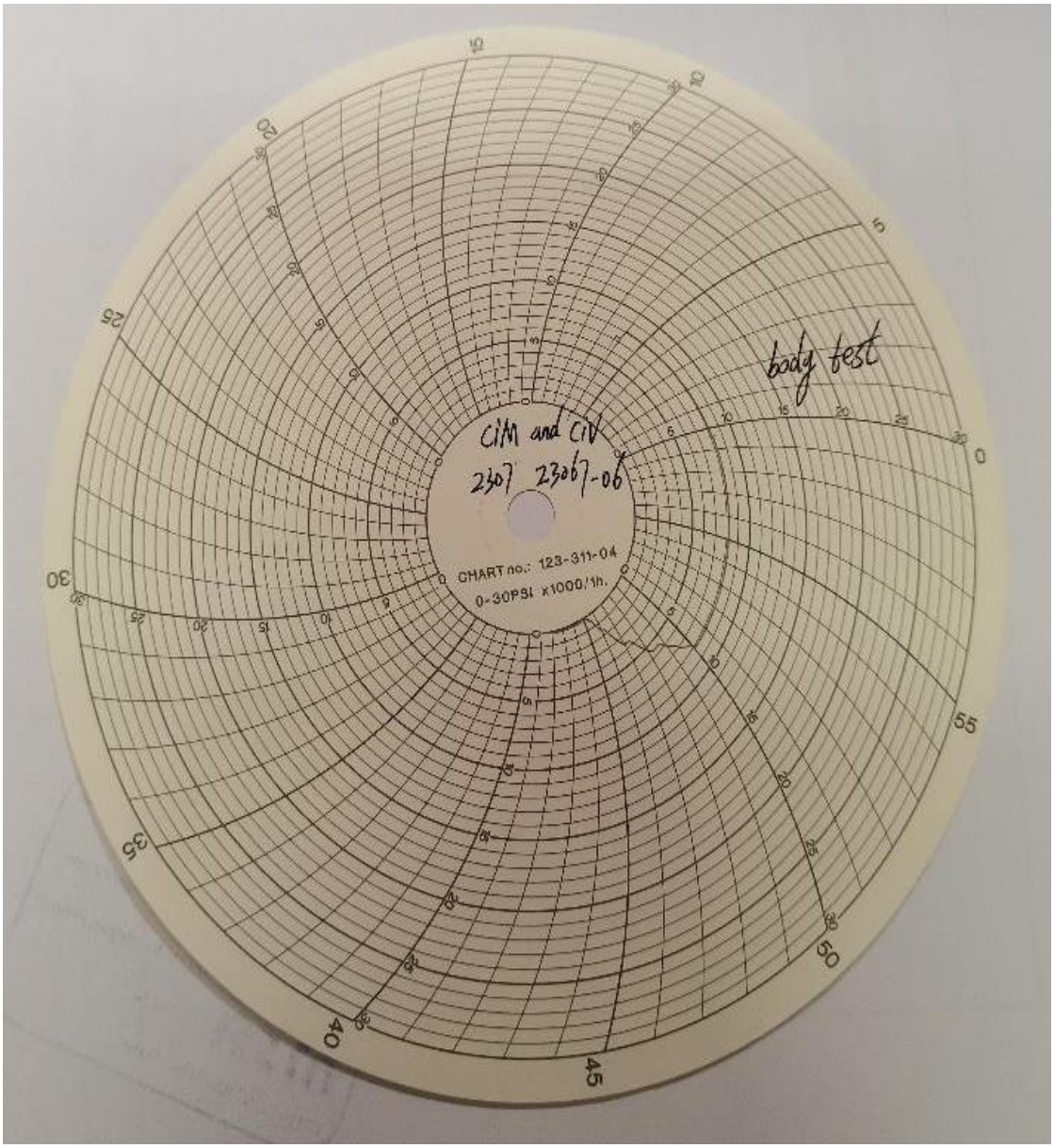
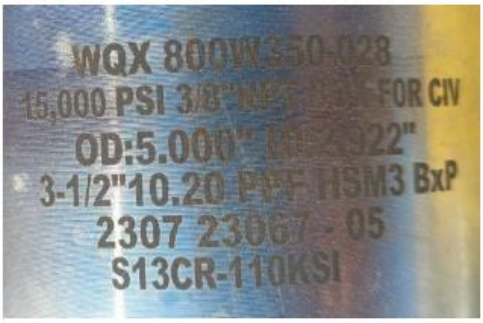


Figure 8-33: AZNS-85 Second CIM Body Pressure Test Graph

5) CHEMICAL INJECTION MANDREL & CHEMICAL INJECTION VALVE (QTY 2)
WQX 800W350-028, 15000 PSI 3/8"NPT BOX FOR CIV, OD:5000" ID:2.922"
3.1/2" 10.20 PPF HSM-3 BxP, (2307 23067-05 , 2307 23067-06)

WQX 802W100-601, 15000 PSI 300 DEG F 3/8" NPT PIN (2306 23068-05, 2306 23068-06)









8.4 Completion Operation Checklist

Item	Operation Name	Remark	Check Mark✓
1	Scraper Trip	Packer depth should be scraped	√
2	Pre-job preparation	Pre-connection All Down Hole Equipment With Upper & Lower Pup Joints On Confirmed Thread-makeup at Workshop, And Pressure Test Prior To Being Transported To Well Site.	√
		Retrieve Wear Bushing, M/U Jet Sub And Flush Bop Stack & Well Head With Fresh Water.	√
		All Crossovers, Pup Joints For Space Out Measured Put In Right Place.	√
		Lift Control Line And Protectors On Rig Floor.	√
		Hold A Pre-Job Safety Meeting For Running Upper Completion String.	√
3	Assembling and RIH	M/U 3-1/2" Re-entry Guide Shoe And 3-1/2" BTM No Go Landing Nipple	√
		Check The Conditions Landing Nipple To Make Sure There Is No Damage, Observe There Is No Any Trash Trapped Inside.	√
		Assemble Landing Nipple.	√
		RIH 1 Joint Tubing As Per Tally	√
		P/U And M/U Hydraulic-Set PERMANENT Production Packer.	√
		Much More Care Should Be Taken When Packer Is Passing Well Head And Liner Hanger While RIH,	√
		Assemble Ratch Latch If Packer	-
		Slack Off Packer Assembly, Double Check The Quantity Of Shear Pins,	√

		Remove The Tape On The Element, Check If Any Damages On It.	
		RIH 1 Joint Tubing As Per Tally	√
		M/U Pup Joint And SSD (close Position) To Tubing String With check valve Inside.	√
		Pressure Test Tubing String 7000psi/15min Every 50 JTs.	√
		RIH Completion String and M/U traveling joint as per running tally.	√
		M/U 2 NO. CIV as per running tally& Press Test to 2000 Psi.	-
		Cont. Running Tubing as per Tally while Install Cross protector for the 2 NO. 3/8”control line.	-
		M/U TRSSSV	√
		Install control line to TRSSSV and pressure test control line 8000psi x 15mins. And then keep 5000psi pressure while running to ensure TRSSSV open. Install TRSSSV protector for the 1/4” control line. Continue to run 4-1/2” tubing as per running tally.	√
4	M/U and Space out Tubing Hanger	Pick up & M/U tubing hanger with double pin crossover. M/U landing joint. Check string weight upward and downward.	√
		Roughly cut control line about 2 meters above tubing hanger, wrap 2-3 turns and then feed through Tubing Hanger.	√
		Check string weight upward and downward and Space out Tubing Hanger 1.5 m (Half stroke) above tubing head spool	√
Item	**Operation Name**	**Remark**	**Check Mark✓**
5	Set production	R/Up Slick Line. Remove Check valve inside SSD	√
		RIH and Set check vlve in Landing Nipple. POOH to surface.	√

	packer	Pressurize inside tubing to 500psi to ensure traveling joint is full open. Increase Pressure gradually to SET PACKER. Increase pressure to 7000 Psi to Test String.	√
7	Land Tubing hanger	Move down completion string and apply required weight to shear traveling joint load pins and Land Tubing hanger in T.H.S. Slack off completely & Record the weight.	√
		Drain the BOP stack through the gate valves of tubing head before landing hanger in THS. Leave the valves open.	√
		Tie Down Screws on THS should be fully open before Operation	√
8	N/D BOP & N/U X-mas tree	Tighten tie down screws.	√
		Test inside string w/ 5000 psi, 15 min.	√
		R/Up Slick Line. RIH with slick line and Ret. Check valve from Landing Nipple. POOH to surface.	√
		Test Annulus w/ 2000 Psi 15min.	√
		Close BOP ram and test above tubing hanger (annulus between BOP and landing joint) to 1000psi.	√
		B/O & L/D landing joint.	√
		Set 4” 2Way in TBG Hanger & Test it, Back off and L/D Running Tool.	√
		N/D BOP.	√
		Feed control line, through head bonnet.	√
		Check hanger for any damage and pressure test control line to 5000psi	√
		Pressure test chemical injection line & rapture the disc after installing	-

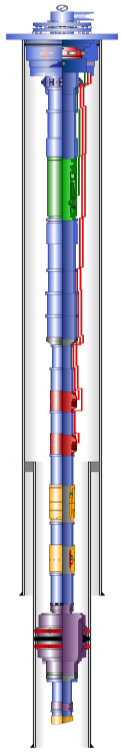
		Bonnet.	
		N/U x-mass tree as program.(Bottom Master)	√
		Perform required tests for X-mass tree. (Bottom Master)	√
		Connect from X-mass tree Kill valve to the pump unit & test X-mass tree body to 1000/7500^{psi}.	√
		Remove 4" two way, install 4" one way	√
		Close TRSSSV	√
		Release the rig. Clean the well site.	√

Table 8-4: AZNS-85 Operation Checklist

8.5 Completion Sketch & Tally

AZNS-85 Completion Sketch

Contractor	MSA / OEOC	Liner & Tubing Data					Client	PEDEC / PPL
Field	AZADEGAN	Description	7"Liner	7"Liner Tie-Back	3 1/2" Tubing	4 1/2" Tubing	7" Tie back #1 L. Lap	3012
Well Number:	AZNS-85	OD[in]	7.00	7.00	3.50	4.50	7" Tie back #2 L. Lap	2907
Rig :	201-OEOC	Nominal weight [lb/ft]	35.00	29.00	10.20	18.90	7" Liner Lap:	3829
RTE-GLE	3.95	Grade	110SS	L-80	L-80	L-80	7" Liner Shoe:	4090
RTE-THF	13.45	Thread	N.VAM	VAM	HSM3	HSM3	Total Depth:	4090
Rig Spudded Date :		Inside diameter [in]	6.184	6.184	2.915	3.640	X-mass Tree:	MSP-T Type
End of Operation		API Drift Diameter	6.059	6.059	2.797	3.515	Size & Rating:	4-1/16" 10K

Schematic		Description	Max OD	Min ID	Length(m)	BTM(m)	TOP(m)
	1	Elevation (Floor - TBG Head Flange)				10.544	9.994
	2	TBG HANGER MSP, 10K, 11" * 4 1/2" ACME Box up * 4 1/2" TPCQ-EX Box Down , M/C: FF-0.5 T/C: UPSL: 3G PR: 2, 4" Type "H" Two Way	11	3.68	0.150	10.694	10.544
	3	4-1/2" FLUTED SWAGE ADAPTER D/P, 18.9# TPCQ-EX PIN ON TOP X HSM3 PIN ON DOWN* SUPER 13CR-110KSI, WEIQIXIN Co	6.1	3.64	1.018	11.712	10.694
	4	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	1.625	13.337	11.712
	5	5 JOINTs TUBING 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	47.201	60.538	13.337
	6	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	1.625	62.163	60.538
	7	4-1/2" FLOW COUPLING,18.9#,SUPER 13CR-110KSI,HSM3 , 6 FT, WEIQIXIN Co	5.5	3.64	1.725	63.888	62.163
	8	TRSSV-10 Tubing Retrievable Saftey Valve, 4 1/2", 18.9# W/ 3.437" Seal bore,HSM3 , SUPER 13CR-110KSI, 10K WP With AF Profile, Flapper Type, WEIQIXIN Co	7.5	3.437	1.780	65.668	63.888
	9	4-1/2" FLOW COUPLING, 18.9#,SUPER 13CR-110KSI,HSM3 , 4' FT, WEIQIXIN Co	5.5	3.64	1.115	66.783	65.668
	10	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	1.623	68.406	66.783
	11	286 JOINTs TUBING 4-1/2"HSM3 18.9#, L-80	5.118	3.64	2761.201	2829.607	68.406
	12	X-over 4-1/2"HSM3, 18.9# BOX * 3-1/2"HSM3 , 10.2# PIN,	5.118	2.922	1.120	2830.727	2829.607
	13	6 JOINTs TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	56.931	2887.658	2830.727
	14	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.618	2889.276	2887.658
	15	3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 , 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC, WEIQIXIN Co	5	2.922	0.725	2890.001	2889.276
	16	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.515	2899.516	2890.001
	17	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.903	2901.419	2899.516
	18	3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 , 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC, WEIQIXIN Co	5	2.922	0.725	2902.144	2901.419
	19	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.515	2911.659	2902.144
	20	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.898	2913.557	2911.659
	21	3-1/2" EXP. JOINT,HSM3 , SUPER 13CR-110KSI, 10'STK (Half Open) Splind Rotational Lock System Any Position, Shear Pin Set 20,000 LBS, WEIQIXIN Co	5.135	2.922	5.760	2919.317	2913.557
	22	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.900	2921.217	2919.317
	23	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.525	2930.742	2921.217
	24	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.703	2932.445	2930.742
	25	3-1/2" NON-ELASTOMERIC SLIDING SIDE DOOR, SUPER 13CR-110KSI, HSM3 With Type X Selective Profile (PROFILE: 2.750" X, DOWNWARD OPEN), WEIQIXIN Co	4.551	2.75	1.273	2933.718	2932.445
	26	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.778	2935.496	2933.718
	27	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.430	2944.926	2935.496
	28	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.824	2946.750	2944.926
	29	3-1/2" ANCHOR SEAL ASSEMBLY,HSM3, 10.2#,SUPER 13CR-110KSI, 4" PACKING OD, WEIQIXIN Co	4.000	2.922	0.230	2946.980	2946.750
	30	7" HYDRAULIC SET PERMANENT PRODUCTION PACKER, MODEL WSDP TO BE RUN IN 7.00" 29-32 PPF CASING, SUPER 13CR-110KSI, UPPER SEAL BORE: 4", WEIQIXIN Co	5.910	2.922	4.445	2951.425	2946.980
	31	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.250	2.922	1.822	2953.247	2951.425
	32	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.513	2962.760	2953.247
	33	13-1/2" BOTTOM NO-GO LANDING,HSM3 , 10.2#, SUPER 13CR-110KSI W/ Seal Bore 2.75" TYPE XN, WEIQIXIN Co	4.5	2.562	0.370	2963.130	2962.760
	34	3-1/2" WIRLINE RE-ENTRY GUIDE, 10.2 #,HSM3 Con,SUPER 13CR-110KSI, WEIQIXIN Co	4.25	2.922	0.240	2963.370	2963.130
		END OF COMPLETION STRING					2963.370

Tubing Retrievable Saftey Valve	65.668	63.888
Traveling Joint	2919.317	2913.557
CIV 1	2890.001	2889.276
CIV 2	2902.144	2901.419
SSD	2933.718	2932.445
7" HYDRAULIC SET PERMANENT PRODUCTION PACKER	2951.425	2946.980
NO-GO Landing Nipple	2963.130	2962.760
Entry Guide	2963.370	2963.130

Figure 8-34: AZNS-85 Completion Sketch

MSA

Field :	Azadegan		Well NO:	AZNS85		Rig NO:	OEOC201	Drig CO:	OEOC
PEDEC, PPL			Completion String						

Grade	ID "	Drift "	Weight lb/ft	Burst psi	Collaspse psi	Yield klbs	Coupling "	TQ ft x lb	Nominal Dia. OD
L-80	2.922	2.797	10.2	11,560.0	12120	233	3.962	3200-3550-3900	3-1/2"
L-80	3.640	3.515	18.9	13,380.0	13820	440	5.137	7920-8800-9680	4-1/2"

BTM NO GO Depth:	2962.76	Formation Name :	Fahliyan	Hook Weight :	58 k lb
BTM of String	2963.37	Formation Top :	3987	VOLUME (TOTAL)	123.13
Packer Depth :	2946.98			Mud Weight (pcf) :	104

Number on jt	Grade	Run In No.	Jt Length (m)	Cum Length (m)	Dist to RT (m)	M/D weight (Klbs)	VOL(BBL)	Remarks	
								joint no	IN / OUT
					2963.370	67.00			
3-1/2" WIRLINE RE-ENTRY GUIDE, 10.2 #,HSM3 Con,SUPER 13CR-110KSI, WEIQIXIN Co	3-1/2" Mule Shoe		0.240	0.24	2963.130	67.01	0.01		
3-1/2" BOTTOM NO-GO LANDING,HSM3 , 10.2#, SUPER 13CR-110KSI W/ Seal Bore 2.75" TYPE XN, WEIQIXIN Co	3-1/2" BTM NO GO		0.370	0.61	2962.760	67.02	0.02		
TUBING 3-1/2", 10.2#, L-80,HSM3	TUBING 3-1/2", 10.2#, L-80,HSM3	1	9.513	10.12	2953.247	67.27	0.28	A1	
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.822	11.95	2951.425	67.31	0.33		
7" HYDRAULIC SET PERMANENT PRODUCTION PACKER, MODEL WSDP TO BE RUN IN 7.00" 29-32 PPF CASING, SUPER 13CR-110KSI, UPPER SEAL BORE: 4", WEIQIXIN Co	PKR		4.445	16.39	2946.980	67.43	0.45	1	
3-1/2" ANCHOR SEAL ASSEMBLY,HSM3, 10.2#,SUPER 13CR-110KSI, 4" PACKING OD, WEIQIXIN Co	Ratch Lach		0.230	16.62	2946.750	67.44	0.45		
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.824	18.44	2944.926	67.49	0.50		
TUBING 3-1/2", 10.2#, L-80,HSM3	TUBING 3-1/2", 10.2#, L-80,HSM3	2	9.430	27.87	2935.496	67.73	0.76	A2	
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.778	29.65	2933.718	67.78	0.81		
3-1/2" NON-ELASTOMERIC SLIDING SIDE DOOR, SUPER 13CR-110KSI, HSM3 With Type X Selective Profile (PROFILE: 2.750" X, DOWNWARD OPEN), WEIQIXIN Co	SSD		1.273	30.93	2932.445	67.81	0.84		
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.703	32.63	2930.742	67.86	0.89		
TUBING 3-1/2", 10.2#, L-80,HSM3	TUBING 3-1/2", 10.2#, L-80,HSM3	3	9.525	42.15	2921.217	68.11	1.15	A3	
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.900	44.05	2919.317	68.16	1.20		
3-1/2" EXP. JOINT,HSM3 , SUPER 13CR-110KSI, 10'STK (Fully Open) Splind Rotational Lock System Any Position, Shear Pin Set 20,000 LBS, WEIQIXIN Co	Traveling Jt		5.760	49.81	2913.557	68.31	1.36		
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.898	51.71	2911.659	68.36	1.41		
TUBING 3-1/2", 10.2#, L-80,HSM3	TUBING 3-1/2", 10.2#, L-80,HSM3	4	9.515	61.23	2902.144	68.61	1.67	A4	
3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 , 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC, WEIQIXIN Co	CIV #1		0.725	61.95	2901.419	68.63	1.69		
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.903	63.85	2899.516	68.68	1.74		
TUBING 3-1/2", 10.2#, L-80,HSM3	TUBING 3-1/2", 10.2#, L-80,HSM3	5	9.515	73.37	2890.001	68.93	2.00	A5	Protector
3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 , 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC, WEIQIXIN Co	CIV #2		0.725	74.09	2889.276	68.95	2.02		
3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	Pup Joint 3-1/2", 10.2#,HSM3		1.618	75.71	2887.658	68.99	2.06		
TUBING 3-1/2", 10.2#, L-80,HSM3	TUBING 3-1/2", 10.2#, L-80,HSM3	6	9.513	85.23	2878.145	69.25	2.32	1	Protector
TUBING 3-1/2", 10.2#, L-80,HSM3	TUBING 3-1/2", 10.2#, L-80,HSM3	7	9.430	94.66	2868.715	69.49	2.58	2	Protector
TUBING 3-1/2", 10.2#, L-80,HSM3	TUBING 3-1/2", 10.2#, L-80,HSM3	8	9.525	104.18	2859.190	69.75	2.84	3	Protector
TUBING 3-1/2", 10.2#, L-80,HSM3	TUBING 3-1/2", 10.2#, L-80,HSM3	9	9.515	113.70	2849.675	70.00	3.09	4	Protector
TUBING 3-1/2", 10.2#, L-80,HSM3	TUBING 3-1/2", 10.2#, L-80,HSM3	10	9.518	123.21	2840.157	70.25	3.35	5	Protector
TUBING 3-1/2", 10.2#, L-80,HSM3	TUBING 3-1/2", 10.2#, L-80,HSM3	11	9.430	132.64	2830.727	70.49	3.61	6	Protector
X-over 4-1/2"HSM3, 18.9# BOX * 3-1/2"HSM3 , 10.2# PIN,	XOS 4-1/2", HSM3 B X 3-1/2" HSM3, P	-	1.120	133.76	2829.607	70.55	3.64		

TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	12	9.577	143.34	2820.030	71.02	4.04	1	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	13	9.525	152.87	2810.505	71.48	4.45	2	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	14	9.544	162.41	2800.961	71.95	4.85	3	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	15	9.528	171.94	2791.433	72.41	5.25	4	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	16	9.308	181.25	2782.125	72.87	5.65	5	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	17	9.528	190.77	2772.597	73.33	6.05	6	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	18	9.428	200.20	2763.169	73.79	6.45	7	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	19	9.183	209.38	2753.986	74.24	6.83	8	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	20	9.525	218.91	2744.461	74.71	7.24	9	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	21	9.526	228.44	2734.935	75.17	7.64	10	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	22	9.525	237.96	2725.410	75.64	8.04	11	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	23	9.523	247.48	2715.887	76.10	8.44	12	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	24	9.525	257.01	2706.362	76.57	8.84	13	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	25	9.418	266.43	2696.944	77.03	9.24	14	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	26	9.522	275.95	2687.422	77.49	9.64	15	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	27	9.522	285.47	2677.900	77.96	10.05	16	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	28	9.523	294.99	2668.377	78.42	10.45	17	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	29	9.528	304.52	2658.849	78.89	10.85	18	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	30	9.528	314.05	2649.321	79.35	11.25	19	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	31	9.428	323.48	2639.893	79.81	11.65	20	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	32	9.520	333.00	2630.373	80.28	12.05	21	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	33	9.534	342.53	2620.839	80.74	12.46	22	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	34	9.526	352.06	2611.313	81.21	12.86	23	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	35	9.538	361.60	2601.775	81.67	13.26	24	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	36	9.525	371.12	2592.250	82.14	13.66	25	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	37	9.518	380.64	2582.732	82.60	14.07	26	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	38	9.524	390.16	2573.208	83.07	14.47	27	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	39	9.548	399.71	2563.660	83.53	14.87	28	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	40	9.518	409.23	2554.142	84.00	15.27	29	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	41	9.516	418.74	2544.626	84.46	15.67	30	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	42	9.522	428.27	2535.104	84.93	16.08	31	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	43	9.520	437.79	2525.584	85.39	16.48	32	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	44	9.186	446.97	2516.398	85.84	16.87	33	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	45	9.528	456.50	2506.870	86.31	17.27	34	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	46	9.525	466.03	2497.345	86.77	17.67	35	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	47	9.528	475.55	2487.817	87.24	18.07	36	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	48	9.535	485.09	2478.282	87.70	18.48	37	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	49	9.545	494.63	2468.737	88.17	18.88	38	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	50	9.534	504.17	2459.203	88.63	19.28	39	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	51	9.570	513.74	2449.633	89.10	19.69	40	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	52	9.328	523.07	2440.305	89.56	20.08	41	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	53	9.438	532.50	2430.867	90.02	20.48	42	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	54	9.519	542.02	2421.348	90.48	20.88	43	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	55	9.408	551.43	2411.940	90.94	21.28	44	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	56	9.528	560.96	2402.412	91.41	21.68	45	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	57	9.528	570.49	2392.884	91.87	22.08	46	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	58	9.528	580.01	2383.356	92.34	22.49	47	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	59	9.583	589.60	2373.773	92.80	22.89	48	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	60	9.528	599.13	2364.245	93.27	23.29	49	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	61	9.525	608.65	2354.720	93.73	23.69	50	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	62	9.535	618.19	2345.185	94.20	24.10	51	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	63	9.526	627.71	2335.659	94.67	24.50	52	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	64	9.566	637.28	2326.093	95.13	24.90	53	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	65	9.094	646.37	2316.999	95.58	25.29	54	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	66	9.524	655.90	2307.475	96.04	25.69	55	Protector

TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	67	9.528	665.42	2297.947	96.51	26.09	56	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	68	9.523	674.95	2288.424	96.97	26.49	57	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	69	9.523	684.47	2278.901	97.44	26.90	58	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	70	9.528	694.00	2269.373	97.90	27.30	59	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	71	9.528	703.53	2259.845	98.37	27.70	60	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	72	9.209	712.73	2250.636	98.82	28.09	61	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	73	9.573	722.31	2241.063	99.28	28.49	62	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	74	9.548	731.86	2231.515	99.75	28.90	63	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	75	9.563	741.42	2221.952	100.22	29.30	64	TEST/Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	76	9.573	750.99	2212.379	100.68	29.71	65	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	77	9.536	760.53	2202.843	101.15	30.11	66	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	78	9.530	770.06	2193.313	101.62	30.51	67	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	79	9.310	779.37	2184.003	102.07	30.90	68	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	80	9.416	788.78	2174.587	102.53	31.30	69	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	81	9.566	798.35	2165.021	103.00	31.71	70	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	82	9.421	807.77	2155.600	103.46	32.10	71	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	83	9.519	817.29	2146.081	103.92	32.51	72	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	84	9.444	826.73	2136.637	104.38	32.90	73	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	85	9.524	836.26	2127.113	104.85	33.31	74	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	86	9.540	845.80	2117.573	105.31	33.71	75	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	87	9.540	855.34	2108.033	105.78	34.11	76	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	88	9.528	864.87	2098.505	106.24	34.51	77	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	89	9.546	874.41	2088.959	106.71	34.92	78	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	90	9.523	883.93	2079.436	107.17	35.32	79	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	91	9.527	893.46	2069.909	107.64	35.72	80	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	92	9.564	903.03	2060.345	108.11	36.13	81	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	93	9.540	912.57	2050.805	108.57	36.53	82	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	94	9.538	922.10	2041.267	109.04	36.93	83	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	95	9.570	931.67	2031.697	109.51	37.34	84	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	96	9.573	941.25	2022.124	109.97	37.74	85	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	97	9.525	950.77	2012.599	110.44	38.14	86	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	98	9.551	960.32	2003.048	110.90	38.55	87	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	99	9.531	969.85	1993.517	111.37	38.95	88	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	100	9.531	979.38	1983.986	111.83	39.35	89	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	101	9.461	988.85	1974.525	112.30	39.75	103	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	102	9.458	998.30	1965.067	112.76	40.15	104	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	103	9.323	1007.63	1955.744	113.21	40.54	105	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	104	9.306	1016.93	1946.438	113.67	40.94	106	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	105	9.370	1026.30	1937.068	114.13	41.33	107	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	106	9.524	1035.83	1927.544	114.59	41.73	108	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	107	9.560	1045.39	1917.984	115.06	42.14	109	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	108	9.530	1054.92	1908.454	115.52	42.54	110	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	109	9.528	1064.44	1898.926	115.99	42.94	111	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	110	9.526	1073.97	1889.400	116.45	43.35	112	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	111	9.526	1083.50	1879.874	116.92	43.75	113	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	112	9.526	1093.02	1870.348	117.38	44.15	114	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	113	9.528	1102.55	1860.820	117.85	44.55	115	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	114	9.333	1111.88	1851.487	118.30	44.95	116	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	115	9.533	1121.42	1841.954	118.77	45.35	117	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	116	9.524	1130.94	1832.430	119.23	45.75	118	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	117	9.394	1140.33	1823.036	119.69	46.15	119	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	118	9.525	1149.86	1813.511	120.16	46.55	120	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	119	9.548	1159.41	1803.963	120.62	46.95	121	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	120	9.533	1168.94	1794.430	121.09	47.36	122	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	121	9.531	1178.47	1784.899	121.55	47.76	123	Protector

TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	122	9.529	1188.00	1775.370	122.02	48.16	124	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	123	9.546	1197.55	1765.824	122.49	48.56	125	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	124	9.528	1207.07	1756.296	122.95	48.97	126	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	125	9.524	1216.60	1746.772	123.42	49.37	127	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	126	9.437	1226.04	1737.335	123.88	49.77	128	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	127	9.540	1235.58	1727.795	124.34	50.17	129	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	128	9.551	1245.13	1718.244	124.81	50.57	130	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	129	9.408	1254.53	1708.836	125.27	50.97	131	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	130	9.530	1264.06	1699.306	125.73	51.37	132	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	131	9.530	1273.59	1689.776	126.20	51.78	133	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	132	9.198	1282.79	1680.578	126.65	52.16	134	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	133	9.531	1292.32	1671.047	127.11	52.57	135	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	134	9.530	1301.85	1661.517	127.58	52.97	136	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	135	9.532	1311.39	1651.985	128.04	53.37	139	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	136	9.190	1320.58	1642.795	128.49	53.76	140	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	137	9.365	1329.94	1633.430	128.95	54.15	141	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	138	9.435	1339.38	1623.995	129.41	54.55	142	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	139	9.528	1348.90	1614.467	129.88	54.96	143	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	140	9.530	1358.43	1604.937	130.34	55.36	144	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	141	9.546	1367.98	1595.391	130.81	55.76	145	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	142	9.528	1377.51	1585.863	131.27	56.16	146	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	143	9.531	1387.04	1576.332	131.74	56.57	147	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	144	9.531	1396.57	1566.801	132.20	56.97	149	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	145	9.529	1406.10	1557.272	132.67	57.37	150	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	146	9.536	1415.63	1547.736	133.13	57.77	151	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	147	9.529	1425.16	1538.207	133.60	58.18	152	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	148	9.532	1434.70	1528.675	134.06	58.58	153	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	149	9.534	1444.23	1519.141	134.53	58.98	155	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	150	9.525	1453.75	1509.616	134.99	59.38	156	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	151	9.528	1463.28	1500.088	135.46	59.79	157	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	152	9.531	1472.81	1490.557	135.93	60.19	158	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	153	9.542	1482.36	1481.015	136.39	60.59	160	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	154	9.573	1491.93	1471.442	136.86	61.00	162	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	155	9.530	1501.46	1461.912	137.32	61.40	164	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	156	9.524	1510.98	1452.388	137.79	61.80	165	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	157	9.562	1520.54	1442.826	138.26	62.20	166	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	158	9.428	1529.97	1433.398	138.72	62.60	167	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	159	9.565	1539.54	1423.833	139.18	63.01	168	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	160	9.583	1549.12	1414.250	139.65	63.41	169	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	161	9.458	1558.58	1404.792	140.11	63.81	170	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	162	9.544	1568.12	1395.248	140.58	64.21	171	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	163	9.546	1577.67	1385.702	141.04	64.62	172	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	164	9.526	1587.19	1376.176	141.51	65.02	173	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	165	9.565	1596.76	1366.611	141.98	65.42	174	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	166	9.573	1606.33	1357.038	142.44	65.83	175	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	167	9.408	1615.74	1347.630	142.90	66.22	176	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	168	9.304	1625.04	1338.326	143.36	66.62	177	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	169	9.525	1634.57	1328.801	143.82	67.02	178	Protector

TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	170	9.527	1644.10	1319.274	144.29	67.42	179	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	171	9.236	1653.33	1310.038	144.74	67.81	180	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	172	9.412	1662.74	1300.626	145.20	68.21	181	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	173	9.538	1672.28	1291.088	145.66	68.61	182	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	174	9.111	1681.39	1281.977	146.11	69.00	183	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	175	9.523	1690.92	1272.454	146.57	69.40	184	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	176	9.533	1700.45	1262.921	147.04	69.80	185	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	177	9.214	1709.66	1253.707	147.49	70.19	186	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	178	9.540	1719.20	1244.167	147.95	70.59	187	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	179	9.523	1728.73	1234.644	148.42	71.00	188	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	180	9.526	1738.25	1225.118	148.88	71.40	189	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	181	9.536	1747.79	1215.582	149.35	71.80	190	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	182	9.524	1757.31	1206.058	149.82	72.20	191	TEST/Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	183	9.538	1766.85	1196.520	150.28	72.61	193	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	184	9.535	1776.39	1186.985	150.75	73.01	194	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	185	9.304	1785.69	1177.681	151.20	73.40	195	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	186	9.521	1795.21	1168.160	151.67	73.80	196	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	187	9.552	1804.76	1158.608	152.13	74.21	197	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	188	9.524	1814.29	1149.084	152.60	74.61	198	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	189	9.524	1823.81	1139.560	153.06	75.01	199	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	190	9.524	1833.33	1130.036	153.53	75.41	200	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	191	9.524	1842.86	1120.512	153.99	75.82	201	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	192	9.573	1852.43	1110.939	154.46	76.22	202	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	193	9.532	1861.96	1101.407	154.92	76.62	203	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	194	9.538	1871.50	1091.869	155.39	77.02	204	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	195	9.524	1881.03	1082.345	155.86	77.43	205	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	196	9.084	1890.11	1073.261	156.30	77.81	206	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	197	9.530	1899.64	1063.731	156.76	78.21	207	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	198	9.404	1909.04	1054.327	157.22	78.61	208	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	199	9.524	1918.57	1044.803	157.69	79.01	209	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	200	9.574	1928.14	1035.229	158.16	79.42	210	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	201	9.572	1937.71	1025.657	158.62	79.82	211	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	202	9.528	1947.24	1016.129	159.09	80.22	212	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	203	9.572	1956.81	1006.557	159.56	80.63	213	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	204	9.520	1966.33	997.037	160.02	81.03	214	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	205	9.402	1975.74	987.635	160.48	81.43	215	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	206	9.527	1985.26	978.108	160.94	81.83	217	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	207	9.528	1994.79	968.580	161.41	82.23	218	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	208	9.343	2004.13	959.237	161.87	82.63	220	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	209	9.523	2013.66	949.714	162.33	83.03	221	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	210	9.525	2023.18	940.189	162.80	83.43	222	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	211	9.528	2032.71	930.661	163.26	83.83	223	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	212	9.413	2042.12	921.248	163.72	84.23	224	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	213	9.548	2051.67	911.700	164.19	84.63	225	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	214	9.413	2061.08	902.287	164.65	85.03	226	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	215	9.328	2070.41	892.959	165.10	85.42	227	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	216	9.535	2079.95	883.424	165.57	85.83	229	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	217	9.433	2089.38	873.991	166.03	86.23	230	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	218	9.523	2098.90	864.468	166.49	86.63	231	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	219	9.540	2108.44	854.928	166.96	87.03	232	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	220	9.413	2117.86	845.515	167.42	87.43	233	Protector

TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	221	9.523	2127.38	835.992	167.88	87.83	234	TEST/Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	222	9.523	2136.90	826.469	168.35	88.23	235	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	223	9.525	2146.43	816.944	168.81	88.63	236	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	224	9.358	2155.78	807.586	169.27	89.03	238	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	225	9.563	2165.35	798.023	169.74	89.43	239	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	226	9.543	2174.89	788.480	170.20	89.84	240	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	227	9.533	2184.42	778.947	170.67	90.24	241	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	228	9.520	2193.94	769.427	171.13	90.64	242	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	229	9.070	2203.01	760.357	171.58	91.02	243	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	230	9.378	2212.39	750.979	172.03	91.42	245	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	231	9.393	2221.78	741.586	172.49	91.82	246	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	232	9.428	2231.21	732.158	172.95	92.22	248	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	233	9.308	2240.52	722.850	173.41	92.61	249	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	234	9.520	2250.04	713.330	173.87	93.01	250	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	235	9.501	2259.54	703.829	174.34	93.41	251	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	236	9.528	2269.07	694.301	174.80	93.81	252	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	237	9.520	2278.59	684.781	175.27	94.22	253	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	238	9.518	2288.11	675.263	175.73	94.62	254	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	239	9.528	2297.64	665.735	176.20	95.02	256	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	240	9.528	2307.16	656.207	176.66	95.42	259	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	241	9.293	2316.46	646.914	177.11	95.82	260	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	242	9.528	2325.98	637.386	177.58	96.22	262	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	243	9.423	2335.41	627.963	178.04	96.62	263	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	244	9.523	2344.93	618.440	178.50	97.02	265	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	245	9.521	2354.45	608.919	178.97	97.42	266	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	246	9.525	2363.98	599.394	179.43	97.82	269	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	247	9.550	2373.53	589.844	179.90	98.23	270	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	248	9.528	2383.05	580.316	180.37	98.63	271	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	249	9.528	2392.58	570.788	180.83	99.03	272	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	250	9.531	2402.11	561.257	181.30	99.43	275	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	251	9.523	2411.64	551.734	181.76	99.83	277	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	252	9.573	2421.21	542.161	182.23	100.24	279	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	253	9.318	2430.53	532.843	182.68	100.63	283	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	254	9.244	2439.77	523.599	183.13	101.02	285	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	255	9.583	2449.35	514.016	183.60	101.43	287	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	256	9.578	2458.93	504.438	184.07	101.83	288	Protector

TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	257	9.523	2468.46	494.915	184.54	102.23	289	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	258	9.525	2477.98	485.390	185.00	102.64	290	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	259	9.523	2487.50	475.867	185.47	103.04	291	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	260	9.528	2497.03	466.339	185.93	103.44	292	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	261	9.553	2506.58	456.786	186.40	103.84	293	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	262	9.528	2516.11	447.258	186.86	104.25	294	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	263	9.531	2525.64	437.727	187.33	104.65	295	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	264	9.528	2535.17	428.199	187.79	105.05	296	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	265	9.531	2544.70	418.668	188.26	105.45	298	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	266	9.520	2554.22	409.148	188.72	105.86	300	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	267	9.515	2563.74	399.633	189.19	106.26	303	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	268	9.550	2573.29	390.083	189.65	106.66	304	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	269	9.523	2582.81	380.560	190.12	107.06	305	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	270	9.523	2592.33	371.037	190.58	107.47	306	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	271	9.528	2601.86	361.509	191.05	107.87	307	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	272	9.523	2611.38	351.986	191.51	108.27	308	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	273	9.523	2620.91	342.463	191.98	108.67	309	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	274	9.523	2630.43	332.940	192.44	109.07	310	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	275	9.394	2639.82	323.546	192.90	109.47	311	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	276	9.153	2648.98	314.393	193.35	109.86	312	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	277	9.553	2658.53	304.840	193.82	110.26	314	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	278	9.540	2668.07	295.300	194.28	110.66	315	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	279	9.549	2677.62	285.751	194.75	111.07	316	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	280	9.526	2687.15	276.225	195.21	111.47	317	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	281	9.528	2696.67	266.697	195.68	111.87	318	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	282	9.528	2706.20	257.169	196.14	112.27	319	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	283	9.528	2715.73	247.641	196.61	112.68	320	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	284	9.553	2725.28	238.088	197.07	113.08	321	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	285	9.178	2734.46	228.910	197.52	113.47	322	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	286	9.528	2743.99	219.382	197.99	113.87	323	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	287	9.320	2753.31	210.062	198.44	114.26	324	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	288	9.403	2762.71	200.659	198.90	114.66	326	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	289	9.520	2772.23	191.139	199.37	115.06	327	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	290	9.525	2781.76	181.614	199.83	115.46	328	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	291	9.350	2791.11	172.264	200.29	115.86	329	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	292	9.528	2800.63	162.736	200.75	116.26	330	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	293	9.528	2810.16	153.208	201.22	116.66	331	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	294	9.438	2819.60	143.770	201.68	117.06	333	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	295	9.528	2829.13	134.242	202.14	117.47	334	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	296	9.553	2838.68	124.689	202.61	117.87	335	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	297	9.530	2848.21	115.159	203.08	118.27	336	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	298	9.418	2857.63	105.741	203.54	118.67	338	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	299	9.148	2866.78	96.593	203.98	119.06	339	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	300	9.531	2876.31	87.062	204.45	119.46	340	Protector
TUBING 4-1/2", 18.9#, L-80,HSM4	TUBING 4-1/2", 18.9#, L-80,HSM4	301	9.238	2885.55	77.824	204.90	119.85	341	Protector
TUBING 4-1/2", 18.9#, L-80,HSM5	TUBING 4-1/2", 18.9#, L-80,HSM5	302	9.418	2894.96	68.406	205.36	120.25	97	Protector

PUP JOINT 4 1/2",HSM3	PUP JOINT 4 1/2",HSM3		1.623	2896.59	66.783	205.44	120.31		
Short Flow Coupling	Short Flow Coupling		1.115	2897.70	65.668	205.49	120.36		
TRSSV	TRSSV		1.780	2899.48	63.888	205.58	120.44		
Long Flow Coupling	Long Flow Coupling		1.725	2901.21	62.163	205.66	120.51		
PUP JOINT 4 1/2",HSM3	PUP JOINT 4 1/2",HSM3		1.625	2902.83	60.538	205.74	120.58		
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	303	9.471	2912.30	51.067	206.20	120.98	98	Test/Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	304	9.423	2921.73	41.644	206.66	121.38	99	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	305	9.449	2931.18	32.195	207.13	121.77	100	Protector
TUBING 4-1/2", 18.9#, L-80,HSM3	TUBING 4-1/2", 18.9#, L-80,HSM3	306	9.429	2940.60	22.766	207.59	122.17	101	Protector
TUBING 4-1/2", 18.9#, L-80,HSM4	TUBING 4-1/2", 18.9#, L-80,HSM4	307	9.429	2950.03	13.337	208.05	122.57	102	Protector
PUP JOINT 4 1/2",HSM3	PUP JOINT 4 1/2",HSM3		1.625	2951.66	11.712	208.13	122.64		
Fluted Swage	Fluted Swage		1.018	2952.68	10.694	208.18	122.68		
TBG HGR	TBG HGR		0.150	2952.83	10.544	208.18	122.69		
Tubing Hanger Running Tool	Tubing Hanger Running Tool		0.550	2953.38	9.994	208.21	122.71		
L/J	L/J		10.000	2963.38	-0.006	208.70	123.13		Final Press Test

Table 8-5: AZNS-85 Completion Tally

8.6 Completion String Pressure Test

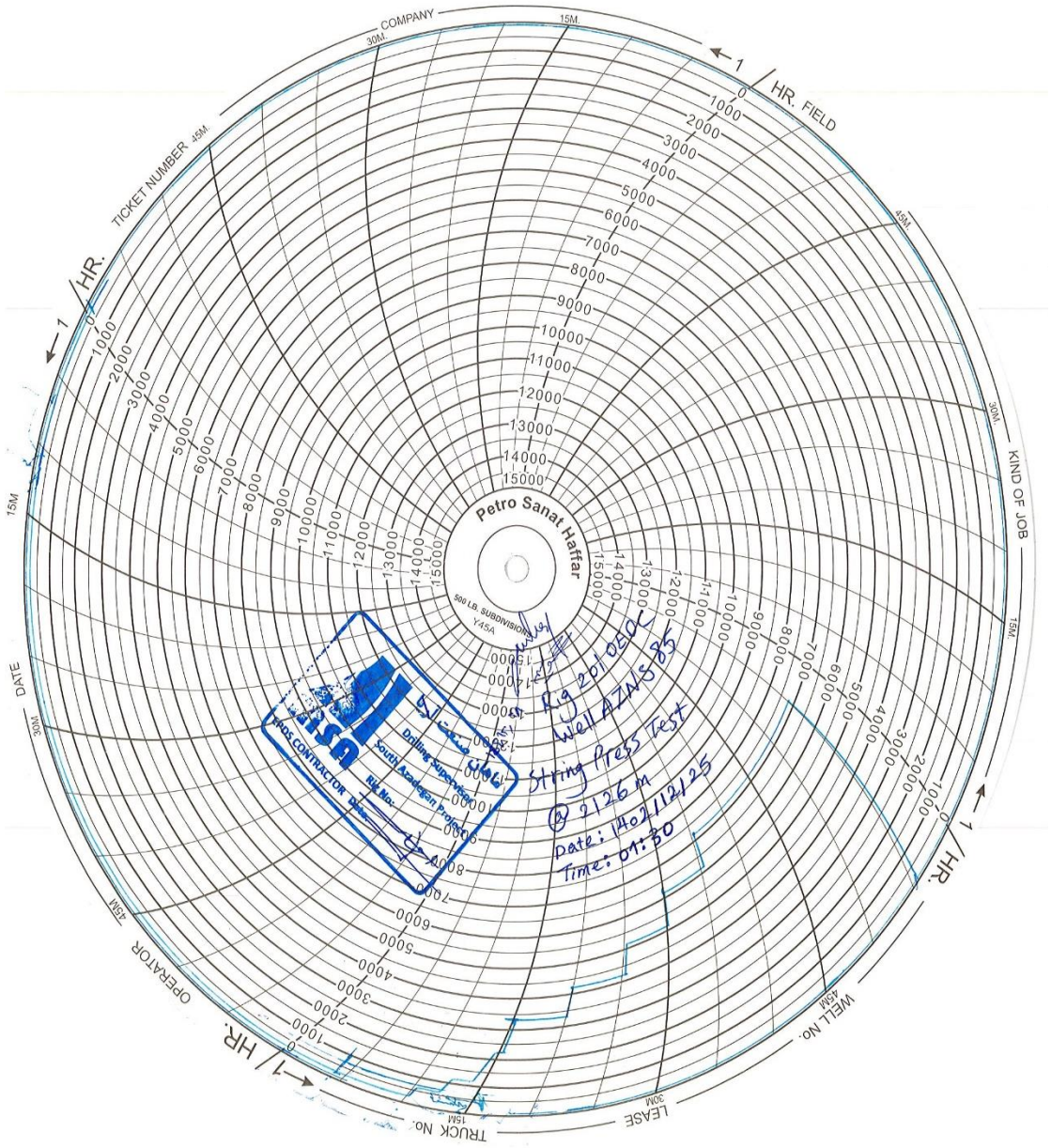


Figure 8-35: AZNS-85 Completion String Pressure Test Graph

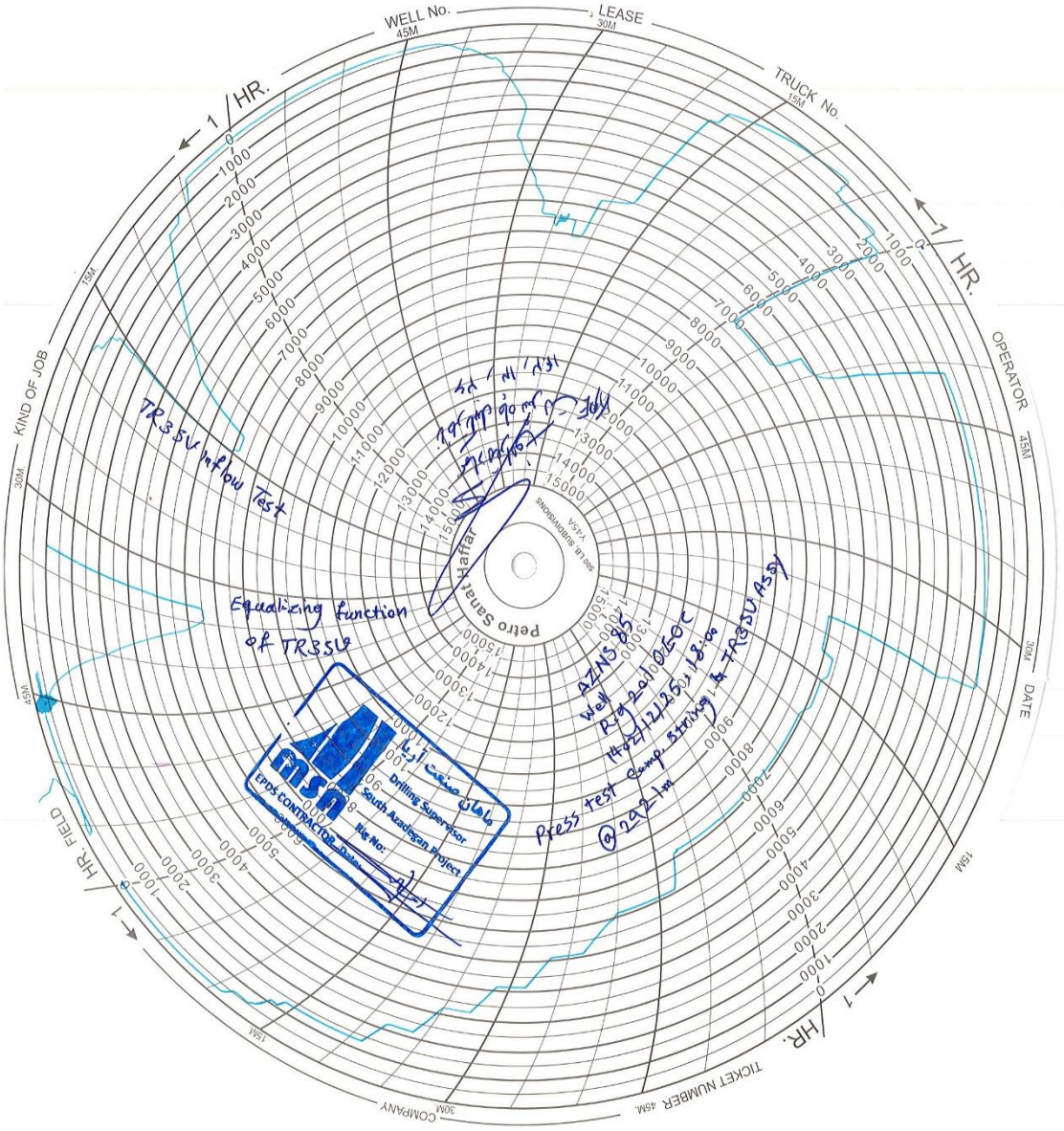


Figure 8-36: AZNS-85 Completion String & TRSSSV Pressure Test Graph

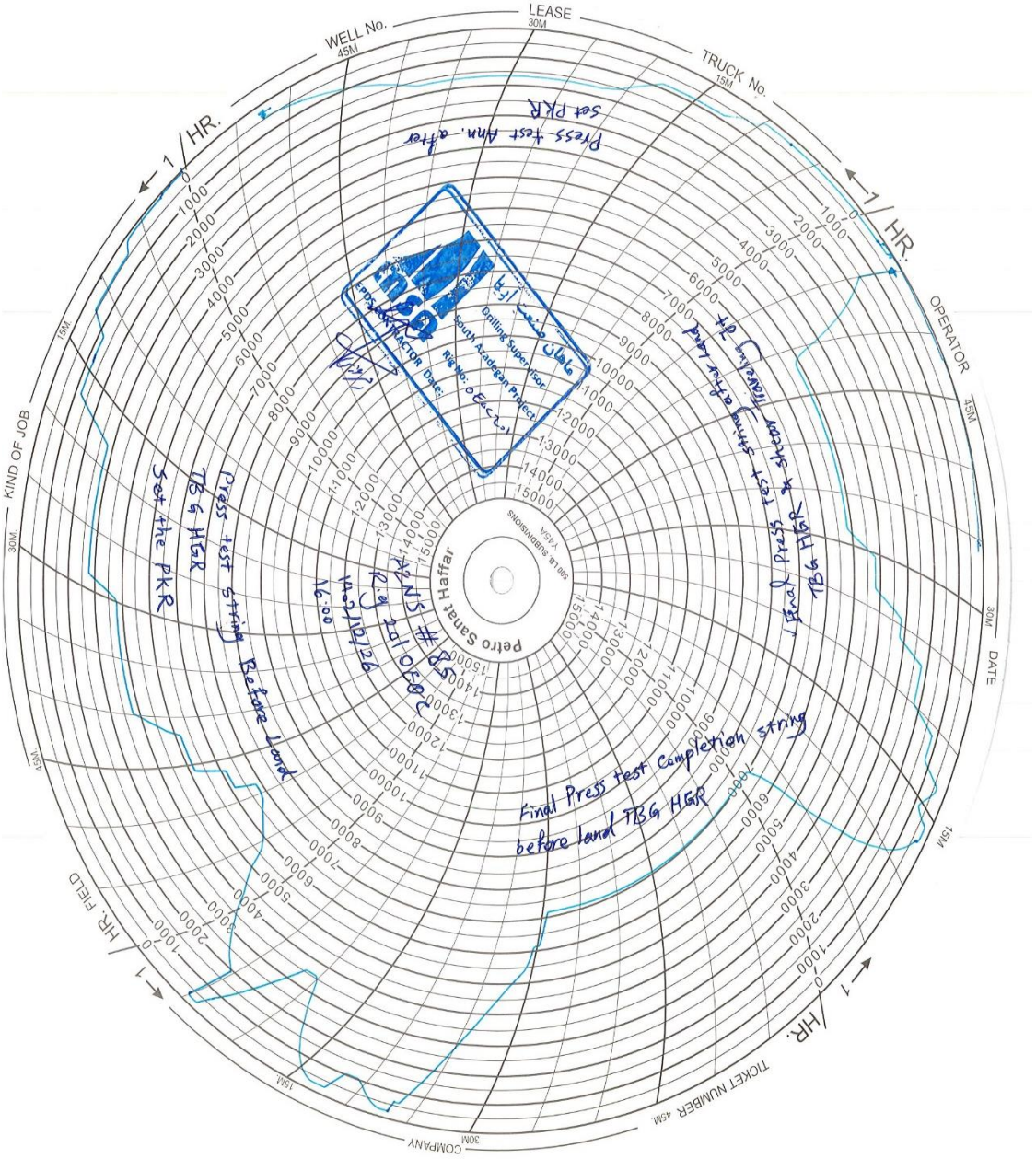


Figure 8-37: AZNS-85 Setting Packer, Annulus & Final Completion String Pressure Test

9. Wellhead Report

9.1 Final Well Status

Well Head

Item	Specification (size, pressure rating, M/C)	Test Pressure	Test Duration
C.H.H	20-3/4'' 3K, DD-0.5 PSL1, PR1	Already Installed	
C.H.S	20-3/4'' 3K * 13-5/8'' 5K , DD-0.5, PSL1, PR1	Already Installed	
MSP Drilex T.H.S.	13-5/8'' 5K * 11''-10K, FF-0.5, PSL3, PR2	Already Installed	
MSP Drilex X-MASS TREE(BTM)	4-1/16''-10K, FF-0.5, PSL3G, PR2	8000 psi	15 MIN

Wellbore Completion Fluids

Well bore	Fluid Type	Weight (pcf)
Cased Hole	$CaBr_2$	103 – 104

Well Head Pressure Record

Annulus	Pressure (psi)	Date / Time
Wellhead	0	1402.12.28
Tubing- 9 5/8"	0	1402.12.28
9 5/8" – 13 3/8"	0	1402.12.28
13 3/8" – 20"	0	1402.12.28

Barriers

Barrier Type	Depth	Pressure Test	Date Tested
Close TRSSSV	63.90 m	500 psi (inflow test after finish flow the well)	1402.12.26

Temporary Installed Equipment:

After Completing the well, Plug & Prong have been installed in No-Go landing nipple

Table 9-1: AZNS-85 Final Well Status

9.2 Wellhead & X-mass tree Specification

Well Type	Specification	Check and Remark
MSP Drilex Well Head 10,000 Psi for Fahliyan Wells	**TUBING HEAD ADAPTOR** BOTTOM STUDDED FLANGE 11", 10000 PSI, BX158 TOP STUDDED FLANGE 4-1/16" 10000 PSI BX155 FULL BORE PREPARED WITH TWO S-SEAL AND ONE METAL TO METAL SEAL TO PACK TOP NECK OF TUBING HANGER C/W RELATED TEST AND INJECTION PORTS PREPARTION. C/W TWO CONTINUOUS FEED THROUGH TYPE 1/4" NPT CONTROL LINE EXITS PREPARTION (ONE FOR 1/4" TRSSSV CL AND ONE FOR 1/4" DOWNHOLE PRESSURE GAUGE) AND TWO 3/8" NPT CONTINUOUS FEED THROUGH TYPE LINE PORTS FOR CHEMCIAL INJECTION LINE EXIT INCLUDING RELATED SWAGE LOCK FITTINGS C/WITH ONE 4-1/16", 10K WP FULL BORE MANUAL GATE VALVE FORGED BODY AND BONNET, ALLOY STAINLESS STEEL 410SS GATE & SEAT AISI 410 COATED WITH TUNGSTEN CARBIDE USING HVOF METHOD STEM: INCONEL 718 WITH BACK SEAT CAPABILITY, NON RISING STEM METAL TO METAL SEAL BETWEEN SEAT AND GATE C/W HAND WHEEL FORGED BODY ALLOY STEEL AISI 410 MAXIMUM HEIGHT: 800MM M/C:FF-0.5 T/C:U PSL:3G PR:2 CROSS T-TYPE SOLID BLOCK CROSS T-TYPE SOLID BLOCK, BOTTOM FLANGED 4-1/16" API 10000 PSI WP TOP FLANGED STUDDED 4-1/16" API 10000 PSI WP FOR TREE CAP WITH TWO STUDDED SIDE OUTLETS 4-1/16" 10000PSI WP ELEVATED AT 900 DEG AND TWO 4-1/16", 10000 PSI MANUAL GATE VALVE AT THE TOP AND BOTTOM AND ONE 4-1/16", 10000 PSI WP SAFETY VALVE EQUIPPED WITH HYDRAULIC WIRE CUT	

	ACTUATOR AS UPPER MASTER VALVE MATERIAL OF BODY,BONNET: AISI 410 SEAT, GATE: AISI 410SS HARD FACED WITH TUNGSTEN CARBIDE UASING HVOP METHOD MATERIAL OF STEM: INCONEL 718 M/C:FF-0.5 T/C:U PSL:3G PR:2G. MANUAL GATE VALVE API 4-1/16" 10000 PSI WP. FLANGED ENDS AND FULL BORE METAL TO METAL SEAL BETWEEN BODY/SEAT/GATE WITH BACK SEAT CAPABILITY, NON RISING STEM C/W HANDWHEELS MATERIAL OF BODY,BONNET,SEAT,GATE: AISI 410SS MATERIAL OF STEM: INCONEL 718 GATE&SEAT HARD FACED WITH TUNGSTEN CARBIDE BY HOVF METHOD M/C:FF-0.5 T/C:U PSL:3G PR:2 X-MAS TREE CAP 4-1/16" API 10000 PSI WP, FLANGED BOTTOM (4-1/16", BORE) 8-3/8"-4ACME-2G THREAD TOP, BLANKING PLUG TAPPED WITH 1/2" NPT THREAD, FORGED BODY, MATERIAL: STAINLESS STEEL 410 M/C:FF-0.5 T/C:U PSL:3G PR:2	

Table 9-2: AZNS-85 Equipment Specification

9.3 Wellhead Schematic

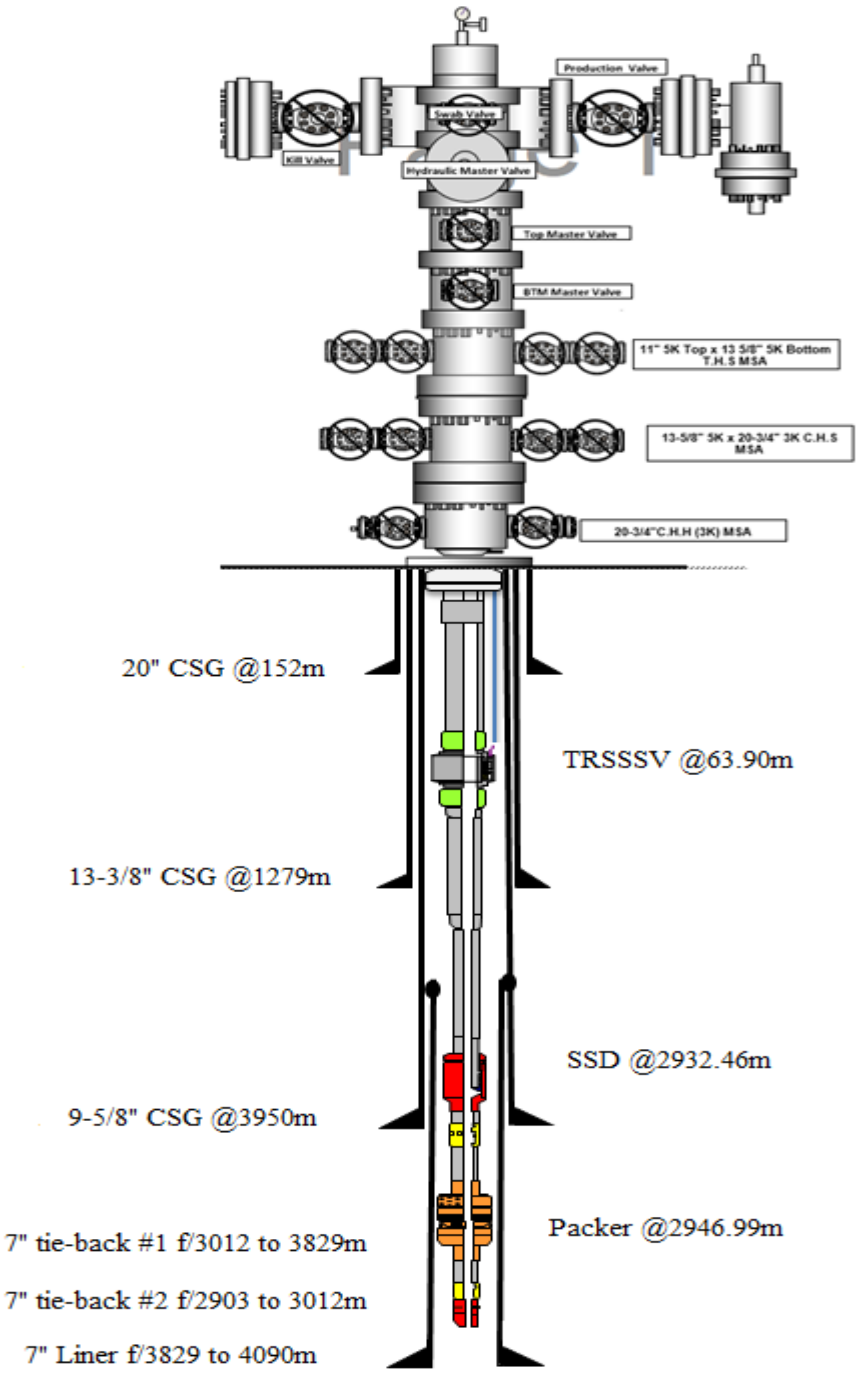


Figure 9-1: AZNS-85 Wellhead Sketch

9.4 X-mass Tree Specification (Section D)

Item	Equipment	M/C	T/C	PSL	Qty
1	BTM Master Valve 11” * 4 1/16” 10K	FF-0.5	U	3G	1
2	TOP Master Valve 4 1/16” 10K	FF-0.5	U	3G	1
3	Surface Safety Valve 4 1/16” 10K	FF-0.5	U	3G	1
4	Deeping Valve 4 1/16” 10K	FF-0.5	U	3G	1
5	Wing gate valve-1 (Kill Valve) 4 1/16" 10K	FF-0.5	U	3G	1
6	Wing gate valve-2 (production valve) 10K	FF-0.5	U	3G	1
7	41/16" Choke valve 4 1/16" 10K	FF-0.5	U	3G	1
8	Christ Mas Tree	FF-0.5	U	3G	1

Table 9-3: AZNS-85 Section "D" Specification

9.5 X-mass Tree Pressure Test Report (Section D – befor install)

Item	Test	Check Mark and Result
1	Pressure test for the kill Valve w/ 1000 & 8000 Psi / 15 min	OK
2	Pressure test for the SSV w/ 1000 & 8000 Psi / 15 min	OK
3	Pressure test for the top master valve w/ 1000 & 8000 Psi / 15 min	OK
4	Pressure test for Swab valve w/ 1000 & 8000 Psi / 15 min	OK
5	Pressure test for the Production valve w/ 1000 & 8000 Psi / 15 min	OK

Table 9-4: AZNS-85 Section "D" Test Report Befor Installation

9.6 X-mass Tree Pressure Test Report (Section D – after install)

Item	Test	Check mark and Result
1	Pressure test X-MAS TREE solid block and bottom section w/1000 & 8000^{psi} for 15min.	OK
2	Pressure test TBG HGR metal seal w/800 & 8000^{psi} for 5 & 15min.	OK
3	Pressure test neck of tubing hanger metal seal W/1000 &8000 psi for 15min	OK

4	Pressure test F-seal & ring gasket between THS & THA (BMV) to 8000^{psi} for 15min.	OK
5	Pressure test CL o 8000^{psi}.	OK
6	Pressure test CIV lines to 2000^{psi}.	OK

Table 9-5: AZNS-85 Section "D" Test Report After Installation

9.7 Wellhead Related Test Reports

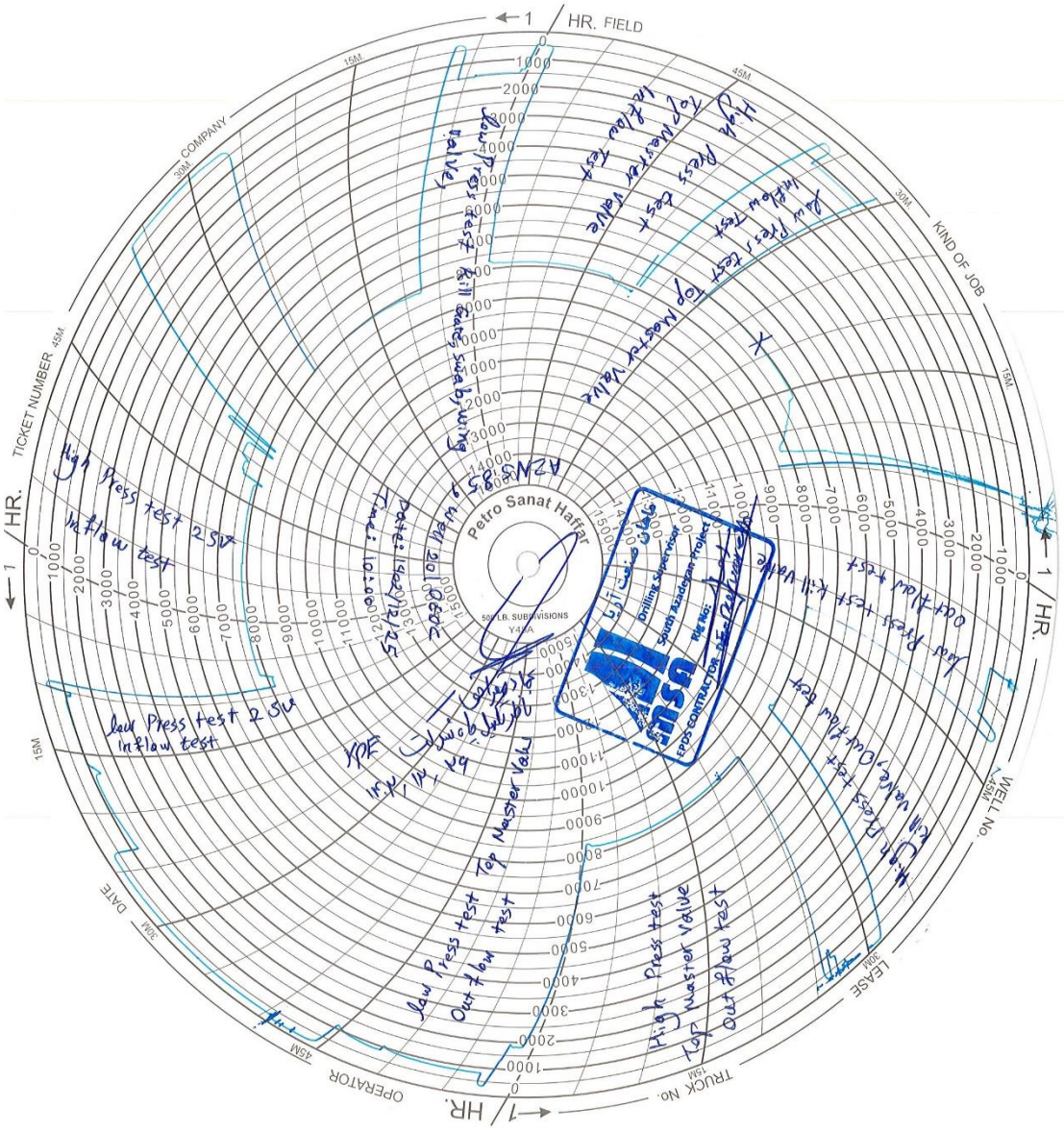


Figure 9-2: AZNS-85 X-Mass Tree Body Pressure Test Graph Before Installation

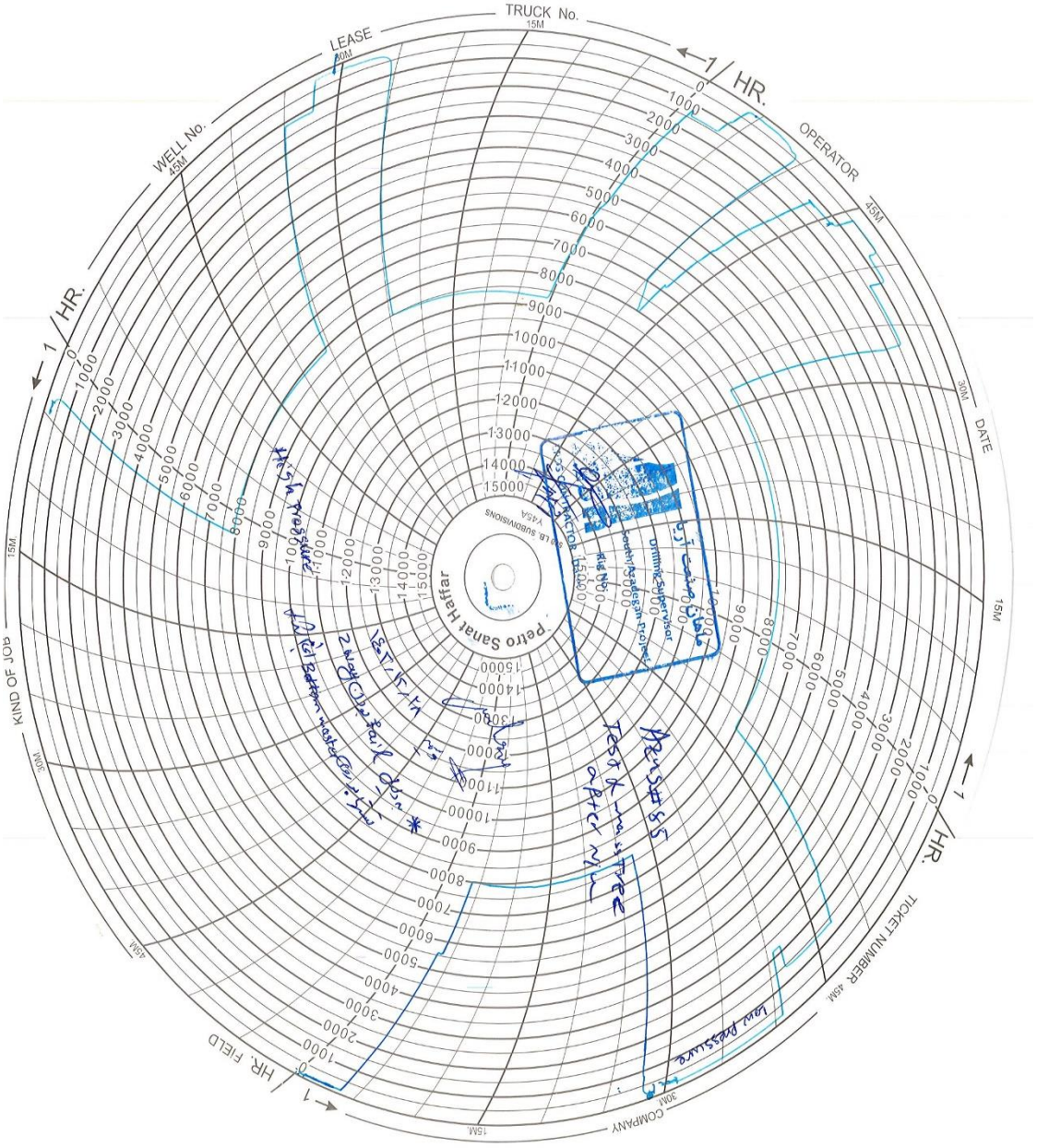


Figure 9-3: AZNS-85 X-Mass Tree Body Pressure Test Graph After Installation

تاریخ: 1402/12/25

پیوست: دارد

بسمه تعالی

موضوع: صورتجلسه تست سطحی تاج چاه 85 ازادگان جنوبی، قبل از نصب در دکل 201 عملیات اکتشاف نفت

بدینوسیله در مورخه 1402/12/25 تست های سطحی تاج چاه X-MASS TREE SOLID BLOCK T TYPE 4-1/16" 10K, MSP DRILEX مرتبط به چاه شماره 85 آزادگان جنوبی با حضور نماینده محترم کارفرما (پتروپارس) در محل دستگاه 201 عملیات اکتشاف نفت، قبل از نصب بر روی تاج چاه به شرح ذیل انجام پذیرفت:

- BODY PRESSURE TEST OF X-MASS TREE SOLID BLOCK W/ 1000PIS, 5MIN & 8000 PSI, 15MIN "OK" SN: 0269167C N180004
- INFLOW PRESSURE TEST OF 4-1/16" PRODUCTION (WING) VALVE W/ 1000PIS, 5MIN & 8000 PSI, 15MIN "OK"
- INFLOW PRESSURE TEST OF 4-1/16" DIPPING (SWAB) VALVE W/ 1000PIS, 5MIN & 8000 PSI, 15MIN "OK".
- INFLOW & OUT FLOW PRESSURE TEST OF 4-1/16" KILL VALVE W/ 1000PIS, 5MIN & 8000 PSI, 15MIN "OK" SN: 002929B N180086
- INFLOW PRESSURE TEST OF 4-1/16" SURFACE SAFETY VALVE W/ 1000PIS, 5MIN & 8000 PSI, 15MIN "OK"
- INFLOW & OUT FLOW PRESSURE TEST OF 4-1/16" TOP MASTER VALVE W/ 1000PIS, 5MIN & 8000 PSI, 15MIN "OK"
- PRESSURE TEST OF 2SV ACTUATOR W/ 2500 PSI, 15 MIN "OK"

– گراف مربوط به تست فشار پیوست صورتجلسه گردید.

– لازم به ذکر است طبق قرارداد، روی تاج چاه کاهنده ثابت نصب گردید.

نماینده محترم تکمیل چاه کارفرما

شرکت نفت کیش

ناظر عملیات حفاری

شرکت ماهان صنعت آریا

ماهان صنعت آریا
Drilling Supervisor
South Azadegan Project
Rig No:
EPDS CONTRACTOR Date:

E-Pervaresh

Figure 9-4: AZNS-85 Wellhead MOM Befor Installation

تاریخ: ۱۴۰۲/۱۲/۲۷
پیوست: دارد

بسمه تعالی

موضوع: صورتجلسه تست های پس از نصب تاج و ولهد چاه ۸۵ ازادگان جنوبی بعد از نصب در دکل ۲۰۱ عملیات اکتشاف نفت

بدینوسیله در مورخه ۱۴۰۲/۱۲/۲۷ تست های تاج چاه X-MASS TREE SOLID BLOCK T TYPE 4-1/16" 10K, MSP و ولهد چاه شماره ۸۵ ازادگان جنوبی با حضور نماینده محترم تکمیل چاه کارفرما و نماینده عملیات حفاری ماهان DRILEX صنعت آریا در محل دستگاه حفاری ۲۰۱ عملیات اکتشاف نفت بعد از نصب بر روی چاه به شرح ذیل انجام پذیرفت:

- BODY PRESSURE TEST OF X-MASS TREE SOLID BLOCK W/ 10K 11" x 4-1/16" 10K TUBING HEAD ADAPTOR (BTM MASTER VALVE) 10K MSP/DRILEX W/ 8000 PSI FOR 15MIN "OK" SN: Q269167C N180004

4-1/16" TOP MASTER VALVE SN: 008929L N180001 / 4-1/16" TOP MASTER VALVE SN: 008929L N180004 / 4-1/16" PRODUCTION (WING) VALVE SN: 0029233 S210019 / 4-1/16" DIPPING (SWAB) VALVE SN: 008929K N180004 / 4-1/16" KILL VALVE SN: 002929B S210033 / 4-1/16" SURFACE SAFETY VALVE SN: 0269167C N180004

- PRESSURE TEST TBG HGR METAL SEAL TO 1000PSI FOR 5MIN, 8000PSI FOR 15MIN "OK"
- PRESS TEST F-SEAL & RING GASKET BETWEEN T.H.S & TUBING HEAD ADAPTOR (BTM MASTER VALVE) TO 8000 PSI FOR 15 MIN "OK"
- PRESSURE TEST NECK OF TBG HGR F-SEAL TO 8000PSI FOR 15MIN "OK"
- PRESS TEST 1/4" CONTROL LINE TO 8000 PSI FOR 15MIN "OK"
- PRESS TEST 3/8"CONTROL INJECTION LINES TO 2000 PSI "OK"

– فشار تمامی دالیزها بررسی گردید که فشار آنها صفر بود.
– بعد از اتمام نصب تاج چاه درون آویزه لوله مغزی TWO WAY CHECK VALVE جهت انگیزش چاه خارج گردید.
– خروجی کنترل لاین شیر ایمنی درون چاهی درون چاهی در جهت شیر اصلی پایینی می باشد و همچنین CIV شماره یک در خلاف جهت ویست پیت و CIV شماره دو در جهت ویست پیت می باشد و همچنین بر روی دو CIV و کنترل لاین پلاگ کپ نصب گردید.
– لازم به ذکر است روی تاج چاه کاهنده ثابت نصب گردید و سایزهای مختلف CHOKE BEAN کاهنده ثابت تاج چاه طبق قرداد می بایست تحویل کارفرما گردد.
– گراف مربوط به تست فشار پیوست صورتجله گردید.

ماهان صنعت آریا
Drilling Supervisor
South Azadegan Project
Rig No: 0E01-201
EPDC CONTRACTOR Date:

ناظر عملیات حفاری
شرکت ماهان صنعت آریا

نماینده محترم تکمیل چاه کارفرما
شرکت نفت کیش

۱۴۰۲/۱۲/۲۸

Figure 9-5: AZNS-85 Wellhead MOM After Installation

10. Drilling Fluid Information

10.1 Interval Remark

Operation Remark

P/U & M/U 5" DPs & RIH with 8 – 1/2" D/O BHA to 2347^{m}. Continued ream down to 2763^{m}, commenced D/O cement to 2907^{m}, POOH to surface.

RIH with 5 – 7/8" BHA to 2904^{m}, ream down inside 7" liner & D/O plug, then RIH with ream down to 3665^{m}, tag cement plug, POOH to surface.

RIH with RTTS & performed job.

Performed scraper trip.

Performed dry test "NO Succeed".

RIH with fishimg magnet to 3665^{m}, then POOH to surface & catch it (Drag Block).

Performed dry test.

Performed fishing job with magnet, "No Succeed".

RIH with 5 – 7/8" junk mill to 3664.5^{m} (0.5^{m} above TOF), took weight, started milling fish & D/O cement to 3879^{m}, hard wash & ream down ti 4078^{m}, POOH to surface.

Performed tandem scraper job & displaced in hole with 30^{bbl} $CaCl_2$ brine in 7" liner, then POOH to surface.

Perfomed logging job.

RIH with 5 – 7/8" BHA + junk sub to 4087^{m}, work on junk & POOH to surface, while pumped 30^{bbl} non-solid fluid @ liner lap.

Perfomed logging job.

RIH with 5 – 7/8" BHA to 4075^{m}, spotted 200^{bbl} fesh mud then POOH to surface.

Performed W/L job.

Performed perforation job.

RIH with SN Packer to 2907^{m} & set it @ this point, then POOH to surface.

RIH with 5 – 7/8" milling BHA to 4065^{m}, below perforation interval, then POOH to surface.

Performed scraper job & displaced mud in hole with 104^{pcf} completion fluid @3100^{m}, then POOH to surface.

Ran completion string as per program.

Drilling Fluid Remark

The Mw increased up to by adding Limestone Powder up & hematite to 103^{pcf} at end of this interval.

Treated frequently with essential chemical such as lime as Alkalinity controller to adjust pH, Starch LV for control FL, Soda Ash for control hardness, Xanthan gum as Viscosifier and Citric acid & sodium bicarbonate for CMT drilling.

10.2 Drilling Fluid type

KCl Pol Mud		Drilling Start	Range		Drilling End
Mw (min)	pcf	89	89	103	103
Funnel Viscosity	sec/qt	40	31	48	42
PV	cps	24	15	24	16
YP	lb/100sq.ft	27	11	27	14
API Fluid Loss	cc	NC	NC		NC
pH	-	9.7	9.2	10.9	10.3
Solid	%	30	23	30	28
Cl (NaCl)	mg/L	105,000	73,000	106,000	106,000
Total Hardness	mg/L	3,720	3,560	6,000	5,600
Ca Hardness	mg/L	3,600	3,400	4,800	3,600

Table 10-1: AZNS-85 Fluid Properties Recap Table

10.3 Material Consumption Analysis

Product	Units Used	Total Wt		Effective Concentration
Caustic Soda	8	55	440 lb	0.30 ppb
Citric Acid	20	55	1,100 lb	0.75 ppb
Hematite	100	2,200	220,000 lb	150.85 ppb
Lime	2	55	110 lb	0.08 ppb
Limestone	82	2,200	180,400 lb	123.70 ppb
Salt (NaCl)	45	2,200	99,000 lb	67.88 ppb
Starch LV	47	55	2,585 lb	1.77 ppb
Soda Ash	4	55	220 lb	0.15 ppb
Sodium Bicarbonate	46	55	2,530 lb	1.73 ppb
Xanthan Gum	63	55	3,465 lb	2.38 ppb
NOTE: Used xanthan gum for treated old mud, prepared fresh mud & mixed Hi-Vis Pill.				

Initial Volume	0.0 bbl
Total Volume Built	1,458.4 bbl
Volume Received	1,730.0 bbl
Volume Returned	0.0 bbl
Total Volume Consumed	1,436 bbl
Volume Transfer to Next Section	1,752.4 bbl
Meter Drilled	0 m

Table 10-2: AZNS-85 Material Consumption Analysis Table

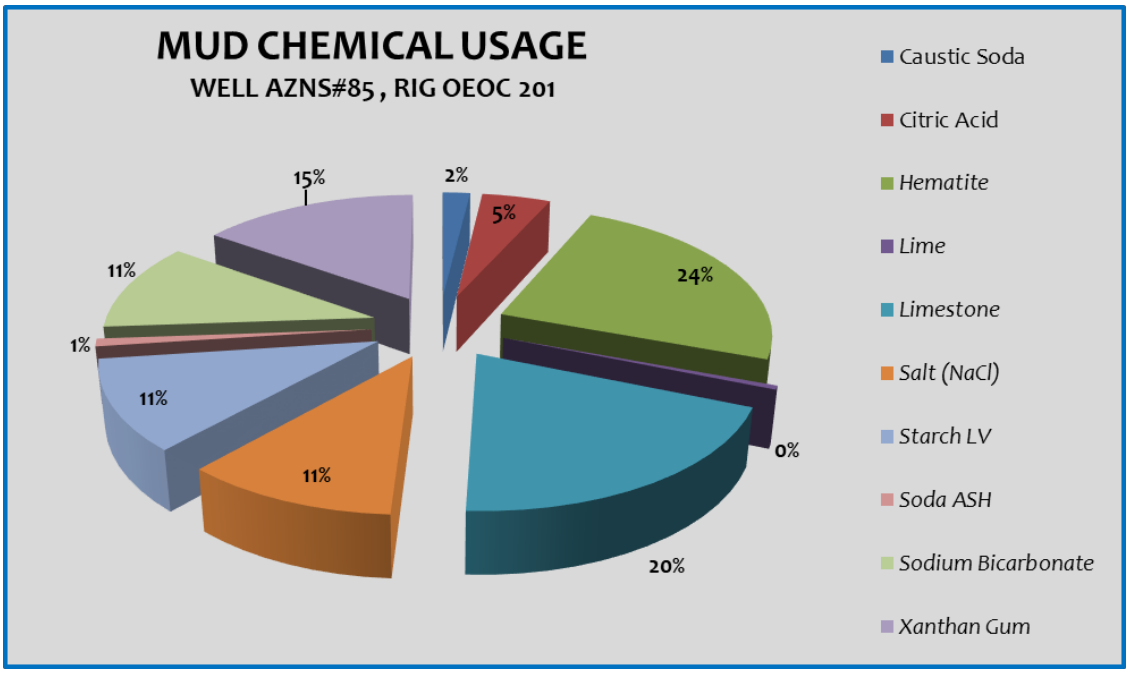


Figure 10-1: AZNS-85 Mud Chemical Usage

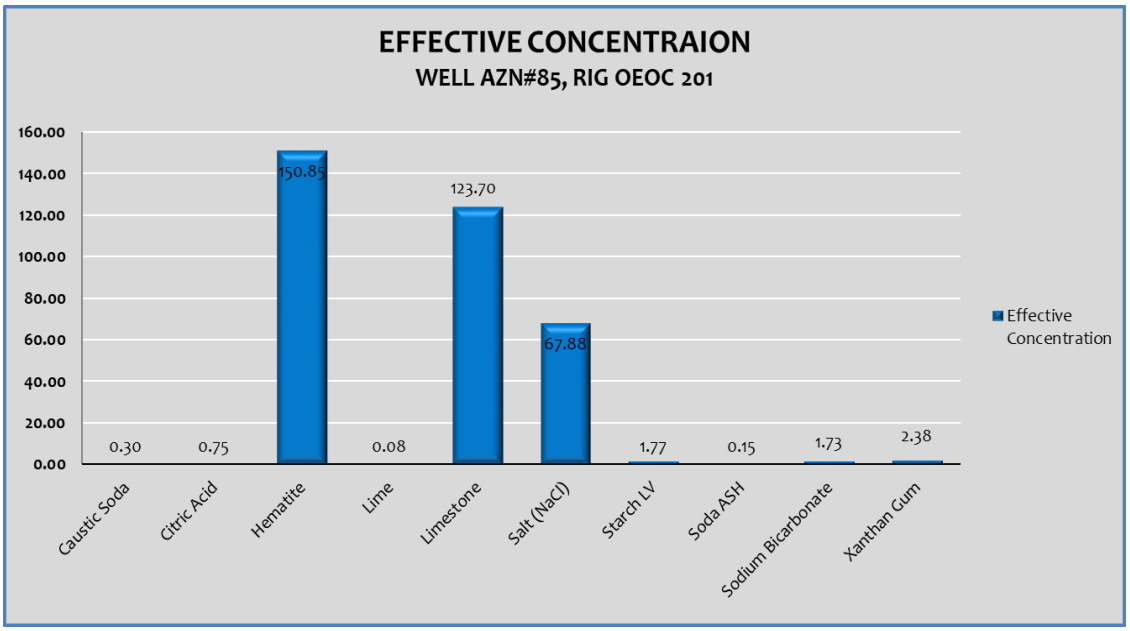


Figure 10-2: AZNS-85 Effective Concentration

10.4 Volume Analysis

Received Volume (bbl)	1,730.0
Volume Return (bbl)	0.0
Volume Built (bbl)	1,458.4
Surface Losses/Dumped (bbl)	1,436.0
Formation Losses (bbl)	0.0
Total Losses (bbl)	1,436.0
Volume Transfer Out (bbl)	1,752.4

Table 10-3: AZNS-85 Volume Analysis Table

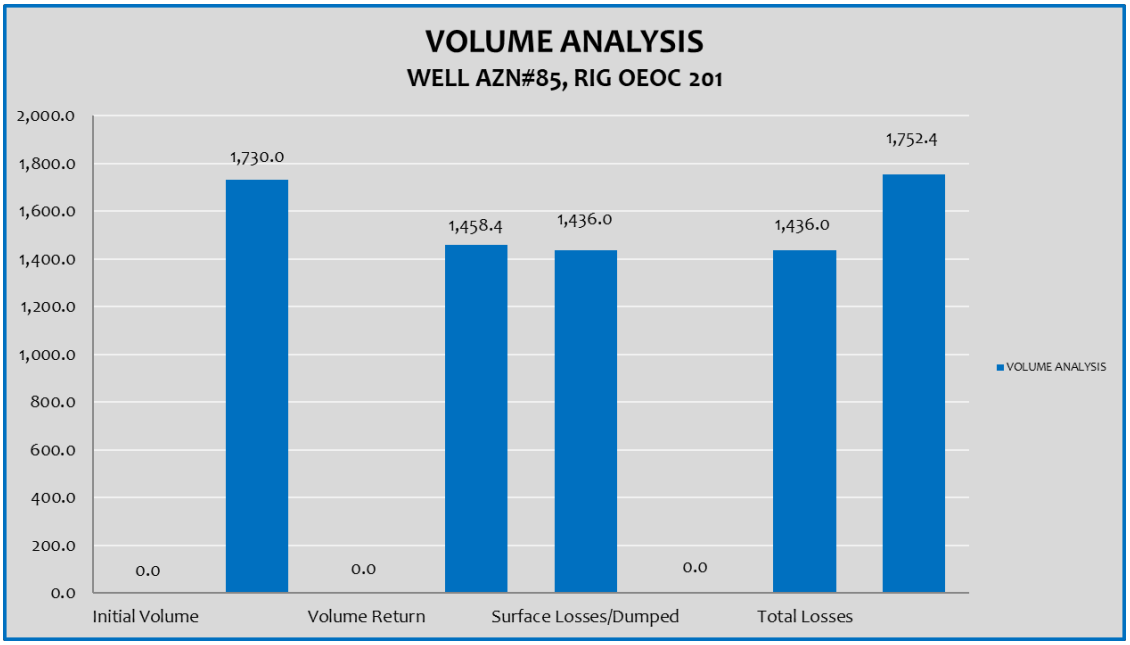


Figure 10-3: AZNS-85 Volume Analysis Chart

10.5 Daily volume Breakdown

Date/DMR.N		Volume of Built	Surface Losses	Formation Losses
25-Jan-24	01	0.0	0	0
26-Jan-24	02	575.3	0	0
27-Jan-24	03	0.0	0	0
28-Jan-24	04	0.0	0	0
29-Jan-24	05	0.0	0	0
30-Jan-24	06	90.0	0	0
31-Jan-24	07	100.9	12	0
1-Feb-24	08	1.5	111	0
2-Feb-24	09	1.2	99	0
3-Feb-24	10	104.7	49	0
4-Feb-24	11	2.0	32	0
5-Feb-24	12	0.4	40	0
6-Feb-24	13	0.5	49	0
7-Feb-24	14	79.3	105	0
8-Feb-24	15	0.0	0	0
9-Feb-24	16	1.5	38	0
10-Feb-24	17	0.0	0	0
11-Feb-24	18	0.0	0	0
12-Feb-24	19	8.0	46	0
13-Feb-24	20	0.0	25	0
14-Feb-24	21	0.2	0	0
15-Feb-24	22	0.0	0	0
16-Feb-24	23	2.2	28	0
17-Feb-24	24	0.6	10	0
18-Feb-24	25	0.5	0	0
19-Feb-24	26	0.5	51	0
20-Feb-24	27	0.2	62	0
21-Feb-24	28	142.8	100	0
22-Feb-24	29	1.6	262	0
23-Feb-24	30	0.0	26	0
24-Feb-24	31	101.3	0	0
25-Feb-24	32	1.3	11	0
26-Feb-24	33	0.0	15	0
27-Feb-24	34	0.0	0	0
28-Feb-24	35	241.9	0	0
29-Feb-24	36	0.0	33	0
1-Mar-24	37	0.0	10	0
2-Mar-24	38	0.0	0	0
3-Mar-24	39	0.0	123	0
4-Mar-24	40	0.0	78	0
5-Mar-24	41	0.0	11	0
6-Mar-24	42	0.0	0	0

7-Mar-24	43	0.0	10	0
8-Mar-24	44	0.0	0	0
9-Mar-24	45	0.0	0	0
Total (bbl)		Volume of Built	Surface Losses	Formation Losses
		1,458.4	1,436	0

Table 10-4: AZNS-85 Daily Volume Break Down Table

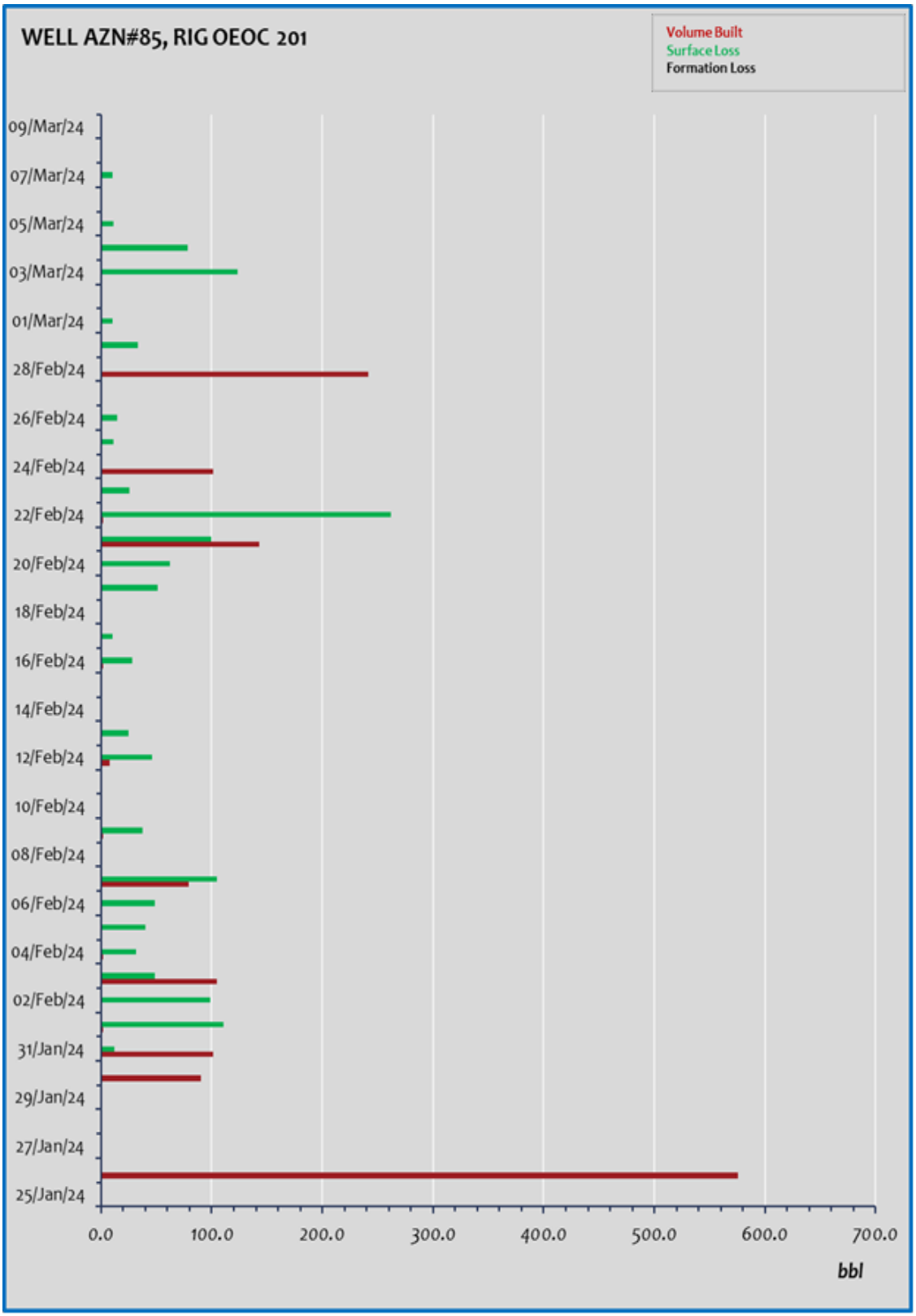


Figure 10-4: AZNS-85 Daily Volume Break Down Chart

10.6 Loss Distribution

Date/DMR.N		Volume of Built	Dump/Cont amination	Shakers	Evaporation	Centrifuge	Trip	Left In Hole	Pit Cleaning	Formation	Other
25-Jan-24	01	0.0	0	0	0	0	0	0	0	0	0
26-Jan-24	02	575.3	0	0	0	0	0	0	0	0	0
27-Jan-24	03	0.0	0	0	0	0	0	0	0	0	0
28-Jan-24	04	0.0	0	0	0	0	0	0	0	0	0
29-Jan-24	05	0.0	0	0	0	0	0	0	0	0	0
30-Jan-24	06	90.0	0	0	0	0	0	0	0	0	0
31-Jan-24	07	100.9	0	12	0	0	0	0	0	0	0
1-Feb-24	08	1.5	0	24	0	0	0	65	0	0	22
2-Feb-24	09	1.2	0	28	0	0	0	46	0	0	25
3-Feb-24	10	104.7	0	27	0	0	0	0	0	0	22
4-Feb-24	11	2.0	0	10	0	0	15	0	0	0	7
5-Feb-24	12	0.4	0	25	0	0	0	0	0	0	15
6-Feb-24	13	0.5	0	25	0	0	0	0	24	0	0
7-Feb-24	14	79.3	0	10	0	0	25	0	35	0	35
8-Feb-24	15	0.0	0	0	0	0	0	0	0	0	0
9-Feb-24	16	1.5	15	3	0	0	0	0	20	0	0
10-Feb-24	17	0.0	0	0	0	0	0	0	0	0	0
11-Feb-24	18	0.0	0	0	0	0	0	0	0	0	0
12-Feb-24	19	8.0	0	0	0	0	46	0	0	0	0
13-Feb-24	20	0.0	0	0	0	0	25	0	0	0	0
14-Feb-24	21	0.2	0	0	0	0	0	0	0	0	0
15-Feb-24	22	0.0	0	0	0	0	0	0	0	0	0
16-Feb-24	23	2.2	0	0	0	0	28	0	0	0	0
17-Feb-24	24	0.6	0	10	0	0	0	0	0	0	0
18-Feb-24	25	0.5	0	0	0	0	0	0	0	0	0
19-Feb-24	26	0.5	0	35	0	0	0	0	0	0	16
20-Feb-24	27	0.2	0	37	0	0	0	0	0	0	25
21-Feb-24	28	142.8	15	18	0	0	37	0	30	0	0
22-Feb-24	29	1.6	210	15	0	0	0	0	37	0	0
23-Feb-24	30	0.0	0	0	0	0	15	0	11	0	0
24-Feb-24	31	101.3	0	0	0	0	0	0	0	0	0
25-Feb-24	32	1.3	0	0	0	0	0	0	11	0	0
26-Feb-24	33	0.0	0	0	0	0	15	0	0	0	0
27-Feb-24	34	0.0	0	0	0	0	0	0	0	0	0
28-Feb-24	35	241.9	0	0	0	0	0	0	0	0	0
29-Feb-24	36	0.0	0	17	0	0	6	0	10	0	0
1-Mar-24	37	0.0	0	0	0	0	10	0		0	0
2-Mar-24	38	0.0	0	0	0	0	0	0	0	0	0
3-Mar-24	39	0.0	23	0	0	0	0	0	100	0	0
4-Mar-24	40	0.0	63	0	0	0	15	0	0	0	0
5-Mar-24	41	0.0	0	0	0	0	11	0	0	0	0
6-Mar-24	42	0.0	0	0	0	0	0	0	0	0	0
7-Mar-24	43	0.0	0	0	0	0	10	0	0	0	0

8-Mar-24	44	0.0	0	0	0	0	0	0	0	0	0
9-Mar-24	45	0.0	0	0	0	0	0	0	0	0	0
Total (bbl)		Volume of Built	Dump/Contaminate	Shakers	Evaporation	Centrifuge	Trip	Mud cleaner	Pit Cleaning	Formation	Other
		1,458.4	326	296	0	0	258	111	278	0	167.0

Table 10-5: AZNS-85 Loss Distribution

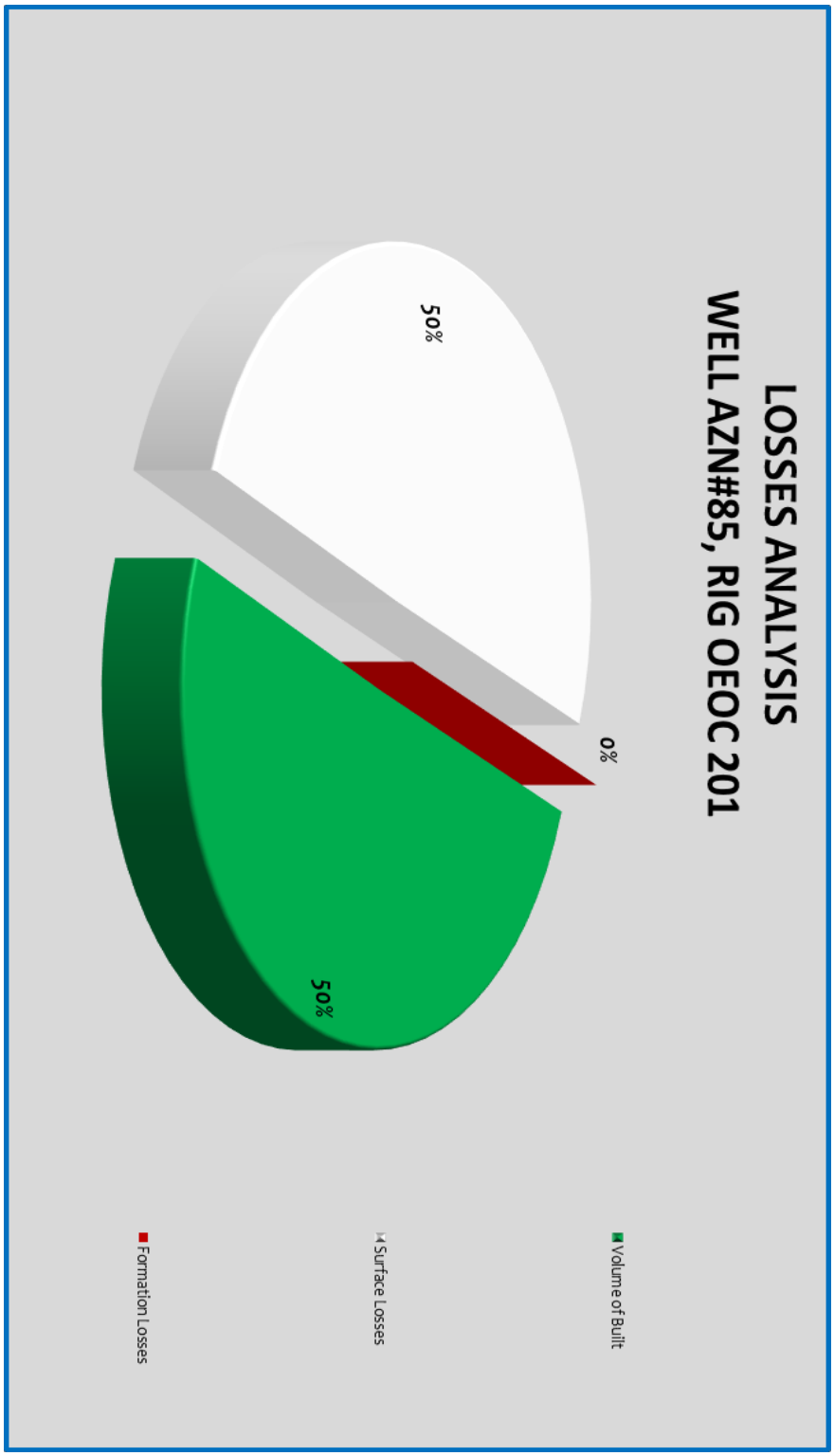


Figure 10-5: AZNS-85 Loss Analysis

10.7 Interval Fluid Properties Recap

25-Jan-24												
26-Jan-24												
27-Jan-24												
28-Jan-24												
29-Jan-24												
30-Jan-24	4090	89	40	27	8	N.C	9.7	30	105,000	3600	3720	Salt Poly
31-Jan-24	4090	81	31	11	3	N.C	9.5	23	73,000	3400	3560	Salt Poly
1-Feb-24	4090	90	36	17	4	N.C	9.2	26	100,000	4400	4520	Salt Poly
2-Feb-24	4090	94	36	16	4	N.C	10	27	100,000	4800	4920	Salt Poly
3-Feb-24	4090	103	36	18	7	N.C	10.8	30	100,000	4000	4120	Salt Poly
4-Feb-24	4090	103	38	9	20	N.C	10.8	30	100,000	3400	4400	Salt Poly
5-Feb-24	4090	103	38	8	19	N.C	10.8	30	100,000	3600	4800	Salt Poly
6-Feb-24	4090	103	37	20	12	N.C	10.9	30	100,000	4000	6000	Salt Poly
7-Feb-24	4090	103	42	20	12	N.C	10.9	30	100,000	4000	6000	Salt Poly
8-Feb-24	4090	103	42	20	12	N.C	10.9	30	100,000	4000	6000	Salt Poly
9-Feb-24	4090	103	42	20	12	N.C	10.9	30	100,000	4000	6000	Salt Poly
10-Feb-24	4090	103	42	20	12	N.C	10.9	30	100,000	4000	6000	Salt Poly
11-Feb-24	4090	103	42	20	12	N.C	10.9	30	100,000	4000	6000	Salt Poly
12-Feb-24	4090	103	42	20	12	N.C	10.9	30	100,000	4000	6000	Salt Poly
13-Feb-24	4090	103	40	18	10	N.C	10.9	30	106,000	3800	6000	Salt Poly
14-Feb-24	4090	103	40	18	10	N.C	10.9	30	106,000	3800	6000	Salt Poly
15-Feb-24	4090	103	40	18	10	N.C	10.9	30	106,000	3800	6000	Salt Poly
16-Feb-24	4090	103	40	18	10	N.C	10.9	30	106,000	3800	6000	Salt Poly
17-Feb-24	4090	103	40	18	10	N.C	10.7	30	106,000	4000	6000	Salt Poly
18-Feb-24	4090	103	40	18	10	N.C	10.7	30	106,000	4000	6000	Salt Poly
19-Feb-24	4090	103	41	18	10	N.C	10.9	30	106,000	4000	6000	Salt Poly
20-Feb-24	4090	103	41	18	10	N.C	10.9	30	106,000	4000	6000	Salt Poly
21-Feb-24	4090	103	41	18	10	N.C	10.8	30	106,000	4000	6000	Salt Poly
22-Feb-24	4090	103	41	18	10	N.C	10.8	28	106,000	4000	6000	Salt Poly
23-Feb-24	4090	103	41	18	10	N.C	10.8	28	106,000	3600	5300	Salt Poly
24-Feb-24	4090	103	41	18	10	N.C	10.8	28	106,000	3600	5300	Salt Poly
25-Feb-24	4090	103	41	22	14	N.C	10.5	28	106,000	3600	5600	Salt Poly
26-Feb-24	4090	103	48	19	12	N.C	10.5	28	106,000	3600	5600	Salt Poly
27-Feb-24	4090	103	44	14	11	N.C	10.5	28	106,000	3600	5600	Salt Poly
28-Feb-24	4090	103	48	19	12	N.C	10.4	28	106,000	3600	5600	Salt Poly
29-Feb-24	4090	103	45	19	12	N.C	10.4	28	106,000	3600	5600	Salt Poly
1-Mar-24	4090	103	44	19	12	N.C	10.4	28	106,000	3600	5600	Salt Poly
2-Mar-24	4090	103	42	14	11	N.C	10.3	28	106,000	3600	5600	Salt Poly
3-Mar-24	4090	103	42	14	11	N.C	10.3	28	106,000	3600	5600	Salt Poly
4-Mar-24	4090	103	42	14	11	N.C	10.3	28	106,000	3600	5600	Salt Poly
5-Mar-24	4090	103	42	14	11	N.C	10.3	28	106,000	3600	5600	Salt Poly
6-Mar-24	4090	103	42	14	11	N.C	10.3	28	106,000	3600	5600	Salt Poly
7-Mar-24	4090	103	42	14	11	N.C	10.3	28	106,000	3600	5600	Salt Poly
8-Mar-24	4090	103	42	14	11	N.C	10.3	28	106,000	3600	5600	Salt Poly

Drilling Start
Drilling End

Table 10-6: AZNS-85 Fluid Properties Recap

10.8 Solid Control Disscusion

Rig OEOC 201 has 3 shale shakers were dressed with:

Shaker#1: 120 Mesh Screen

Shaker#2: 120 – 170 Mesh Screen

Shaker#3: 120 Mesh Screen

WELL AZN#85, RIG OEOC 201	
Solid Control Equipment Working Hr.s	
Shaker # 01	285
Shaker # 02	346
Shaker # 03	91

Table 10-7: AZNS-85 Solid Control Desscusion

10.9 Inhibited $CaBr_2$ Brine

The completion fluid for this section will be $CaCl_2$ brine mixed with $CaBr_2$ brine treated with a scavenger, inhibitor.

Completion						
Products	Units Used	Unit wt lbs	Total wt		Effective Concentration	
695 bbl Completion fluid (Inhibited CaBr2 brine) has been made						
CaBr2	625	BBL	568	gals	0.77	gpb
CaCl2	17	1 MT	15	lbs	164.75	ppb
Corrision Inhibitor	5	55 Gal	5	gals	0.40	gpb
O2 Scavenger	2	55 Gal	3	gals	0.24	gpb
H2S Scavenger	2	55 Gal	3	gals	0.24	gpb
Cacl2 Brine			*200.3*			
MW :			*104.0*			*pcf*
Initial Volume :			*0.0*			*bbl*
Total Volume Built (Completion Fluid) :			*695.0*			*bbl*

Note: Preapred 200^{bbl} $CaCl_2$ brine (Prepared 50^{bbl} for spot @ 7" Liner and prepared 150^{bbl} for completion fluid that is 70^{bbl} for $CaBr_2$, 30^{bbl} as spacer and the rest dead Vol.)

Table 10-8: AZNS-85 Inhibited $CaBr_2$ Brine

11. List of Contractors

Item	Service Item	Contractor
1	Drilling Rig, Mobilization and Demobilization the Rig, Rig Move Between Well Pads	OEOC
2	Drilling Fluids and Completion Fluids	VIRA
3	Waste Management	Pars Haffaran Arvand
4	Electrical Logging and Perforation	PANIZ
5	Mud Logging	MAPSA KISH
6	Directional Drilling (MWD)	VIRA
7	H2S Monitoring and Control	MAPSA KISH BONYAN IMEN
8	Casing and Tubing Running	OEOC
9	Cementing	SPAD ENERGY & Mehran
10	Dry Test, DST and RTTS	OEOC
11	Slick Line	OEOC
12	Stimulation (Coiled Tubing and Acidizing and Nitrogen Lifting)	OEOC
13	Surface Well Testing	OEOC
14	Liner Hanger	PARS POOLAD
15	Well Head	OEOC

Table 11-1: AZNS-85 Contractors Involved

12. Appendix

12.1 DDR Summary

Item	Date	Remark
1	25 – January	P/U & M/U 5" DPs jt by jt (10Stds)
2	26 – January	P/U & M/U 5" DPs jt by jt (totally 53Stds), meanwhile prepare 600^{bbl} WBM (90^{pcf}), change the WETCO T.H.S valves w/ new ones then Press. test the same, Ok
3	27 – January	Cont. P/U 81 Jts 5" DPs jt by jt & M/U 27 Stds (totally 80 Stds), R/B Same, then try to retrieve TBG HGR several times, No Succ.
4	28 – January	Retrieve TBG HGR, start install BOP
5	29 – January	Cont. install BOP & perform BOP test, while prepare to perform test on Blind/ Shear Rams Obs. 11" P.Tester is stock, work on it till getting free, P/U & M/U 8 – 1/2" D/O BHA and RIH to 201^{m}
6	30 – January	Cont. RIH w/ 8 – 1/2" D/O BHA f/201 to 1024^{m}, CBU, resume W/D w/ 8 – 1/2" D/O BHA f/ 1024 to 2086^{m}.
7	31 – January	Cont. Ream Dn w/ 8 – 1/2" D/O BHA f/2086 to 2374^{m}, pump 30^{bbl} Hi-Vis Pill before each Conn.
8	01 – February	Cont. Ream Dn w/ 8 – 1/2" D/O BHA f/2374 to 2629^{m}, pump 30^{bbl} Hi-Vis Pill before each Conn.
9	02 – February	Cont. Ream Dn w/ 8 – 1/2" D/O BHA f/2629 to 2763^{m}, pump 30^{bbl} Hi-Vis Pill before each Conn. Then Start D/O CMT f/2763 to 2820^{m}, while pump 30^{bbl} Hi-Vis Pill & reciprocate last Std before each Conn.
10	03 – February	Prepared Hi-vis pill and pumped each stand. W/Up Mw to 103^{pcf} by Co.Man's order. Used sodium Bic due to improved mud properties. washed & clean divider.

11	04 – February	POOH 8 – 1/2" D/O BHA f/2258^{m} to surface, L/D BHA, P/U & M/U new 5 – 7/8" BHA followed by 5" DPs and RIH to 1861^{m}.
12	05 – February	Cont RIH w/ 5 – 7/8" BHA f/1861 to 2904^{m}, Cont. Rean Dn f/2904 to 2909^{m}, while smooth inside 7" LNR in this interval & D/O Plug, Cont. RIH f/2909 to 2965^{m} & D/O plug, Cont. Rean Dn f/2966 to 3002^{m}.
13	06 – February	Cont. D/O CMT hard w/5 – 7/8" Clean out BHA f/3001 to 3069^{m},work on TDS ,Cont. soft Ream Dn w/5 – 7/8" Clean out BHA f/3069 to 3417^{m},Cont. med to hard Ream Dn w/5 – 7/8" Clean out BHA f/3417 to 3444^{m},Cont. soft Ream Dn f/3444 to 3495^{m} (mud settle & contaminate mud).Cont. med Ream Dn Clean out BHA f/3495 to 3560^{m} (mud settle & contaminate mud).Cont. Ream Dn w/5 – 7/8" Clean out BHA f/3560 to 3643^{m} w/NL-NG.
14	07 – February	Cont. Ream Dn w/5 – 7/8" Clean out BHA to 3665^{m},took weight,tag CMT plug 2. pump 40^{bbl} hi-vis pill, C.B.U, Pooh to surf, L/D 5 – 7/8" B.BIT & BHA.RIH 15 std 5" DPs to 430^{m} as RTTS PKR tail, OBS "static", P/U & M/U 9 – 5/8" RTTS PKR , RIH & set RTTS PKR @40^{m}, N/D BOP stack 13 – 5/8" 10k in progress.
15	08 – February	N/D & HANG UP 13 – 5/8" BOP stack W/BOP CARRIER, disconnect & N/D OLD T.H.S & DSA, AFTER SEVERAL TRYING. cut, DRESS & BEVEL 31^{cm} OF 9 – 5/8" csg NECK. install DSA, TEST BETWEEN 2 FSEALS OF DSA W/4000^{psi}, OK. TEST BETWEEN BTM F-SEAL OF DSA & RING GASKET W/4000^{psi}, OK. THEN INSTALL new 13 – 5/8", 5K * 11", 10K T.H.S (MSP,S/N:TH1806FN180005), TEST PSEAL 7" RING GASKET W/4000^{psi}, OK. N/U 13 – 5/8" BOP, PREFORM BOP TEST AS NIOC POLICY "OK", P/U 1std 5" DPs, RIH to 40^{m}, unset RTTS PKR then pooh & L/D RTTS PKR.
16	09 – February	CONT. POOH TO SURF W/15 STD 5" DPs. P/U & M/U & RIH W/8 – 1/2" b.BIT (SN:2003235, K.D,217, OPEN) + 9 – 5/8" SCRAPER FOLLOW BY 5" DPs TO 2900^{m} WHILE

		SCRAP SEVERAL TIME F/2750 TO 2900^{m} MEANWHILE F/U STRG EACH 15 STD, C.B.S, PUMP 50^{bbl} HI-VIS PILL, POOH TO1600^{m}, TILL MIDNIGHT, W/NL-NG .
17	10 – February	CONT. POOH TO 115^{m} ,SLIP & CUT 99^{ft} DRLG LINE, CONT. POOH TO SURF, L/D 9 – 5/8" SCRAPER & 8 – 1/2" B.BIT (BIT COND.:1-1-WT-A-E-IG-NO-DST), P/U & M/U 9 – 5/8" DRY TEST TOOLS (OEOC) TO 1220^{m}, WHILE F/U STRG W/WTR CUSHION (F.WATER 63^{pcf}), MEANWHILE PASS DRIFT THROUGH EACH STD ,W/NL-NG.
18	11 – February	CONT. RIH 9 – 5/8" DRY TEST TOOLS (OEOC) W/5" DPs TO 1844^{m}, WHILE F/U STRG W/WTR CUSHION (63^{pcf}) MEANWHILE PASS DRIFT THROUGH DPs, CONT. RIH W/9 – 5/8" DRY TEST TOOLS (OEOC) FOLLOWED BY 5" DPs F/1844 TO 1935^{m}, WHILE PASS DRIFT THROUGH DPS, STOP OPERATION DUE TO OBS DROP ANNL. HEAD & THEN WTR CUSHION U-TUBING F/DRILL PIPES, POOH TO SURFWHILE STRG FULL OF MUD & WTR. AT SURF. OBS CRACKED RETAINER ABOVE DRAG BLOCKS OF PKR & SEPARATION 5NO DRAG BLOCKS WHILE CIRC. C.H.H .(1NO DRAG BLOCKS WAS LOST), OBS 9 – 5/8" DRY TEST PKR BACK UP WAS FAILDE WHILE WATING TO REC. NEW 9 – 5/8" DRY TEST PKR .
19	12 – February	NPT for OEOC: M/U & RIH 8 – 1/2" B.bit + bit sub (float in) folow by 5" DPs to 500^{m}, while slow Circ. to Rec. new 9 – 5/8" dry test PKR. POOH to Surf & L/D 8 – 1/2" B.bit & BHA, P/U & RIH fishing magnet BHA as: 5 – 3/16" Rev Circ. magnet + 5" junk sub + B.sub(w/o float valve) +1std 4 – 3/4" DC follow by 3 – 1/2" DPs to 952^{m}, Cont. RIH w/5" DPs to 2240^{m},w/NL-NG.
20	13 – February	NPT FOR OEOC: CONT. RIH FISHING MAGNET W/5" DPs TO 3662^{m}, TOOK WEIGHT, RIH WITH W/D TO 3665^{m}, WORK ON JUNK SUB SEVERAL TIME, DROP BALL START PUMPING, O.B.S INC. PRESS. WORK ON MAGNET. POOH FISHING MAGNET & BHA TO 2560^{m},

		STOP OPERATION DUE TO FAILURE IN DRAW WORKS CHAIN SYS, WOK ON IT "SUCC.", CONT. POOH FISHING MAGNET & BHA F/2560 TO 1130^{m} WHILE F/U STRG EACH 10STD, W/NL-NG.
21	14 – February	CONT. POOH FISHING MAGNET & BHA F/1130^{m} TO SURF, CHECK & L/D FISHING MAGNET & BHA, NOT RETRIEVED THE FISH (DRAG BLOCK), TO 1844^{m}. WHILE F/U STRG W/WTR CUSHION (F. WATER, 63^{pcf}) MEANWHILE PASS DRIFT THROUGH EACH STD ,W/NL-NG.
22	15 – February	CONT RIH W/9 – 5/8" DRY TEST TOOLS TO 2886^{m}, SET PKR @2884^{m}, OPEN THE TOOLS FOR 3HRS, NO OBS BUBBLE ON SURF & OBS THE ANN: STATIC, CLOSE THE TOOLS FOR 6HRS, DROP BAR & SHEAR REV SUB, REV CIRC & GET WTR CUSHION, CONT CIRC TO BALANCE MW, UNSET PKR, OBS THE WELL: STATIC, POOH F/2883 TO 2540^{m} W/NLNG.
23	16 – February	CONT. POOH W/9 – 5/8" DRY TEST TOOLS TO SURF, L/D 9 – 5/8" DRY TEST TOOLS WHILE CIRC ACROSS WELL HEAD, Obs. took 340 – 350^{psi} pressure in closed postion, R/U slick line & run magnet (size: 2.3" & 2.8") @ two stage as catch the fish "no succ". Due to Obs. took weight about 220^{m} above TOC plug while try to pass same point "no succ". POOH to surf & L/D slickline, set 11" wear bushing, M/U & RIH w/5 – 7/8" j.mill (sn:618, new, 5no. 14/32" port nz) & BHA to 280^{m} while F/U string each 10 std w/NL-NG.
24	17 – February	Cont. RIH w/5 – 7/8" j.mill to 3586^{m}, Cont. RIH to 3664.5^{m} (0.5^{m} Above TOC) With P/D, took weight @3664.5^{m} try to W/D "no succ". Then W/R to 3665^{m} (TOC), start milling the fish & cont. D/O soft to med CMT plug to 3709^{m}, while pump hi-vis pill periodically, w/NL-NG.
25	18 – February	Cont. D/O hard. CMT plug f/3709 to 3804^{m}, while B/R each std meanwhile pump hi-vis pill periodically, w/NL-NG.
26	19 – February	Cont. D/O hard cement plug with 5 – 7/8" j.mill f/3804 to 3879^{m}, while B/R each std meanwhile pump hi-vis pill

		periodically, w/NL-NG. Cont. hard W/R DOWN to 3913^{m} while B/R each std meanwhile pump hi-vis pill periodically, w/NL-NG.
27	20 – February	Cont. Hard Ream Dn w/5 – 7/8" Milling BHA f/3913 to 4066^{m}, taking SCR in two Steps.
28	21 – February	Cont. Hard Ream Dn f/4066 to 4078^{m}, pump 50^{bbl} Hi-Vis Pill, CBU till Sh.Shakers clean, POOH 5 – 7/8" Milling BHA to Surface, L/D BHA, RIH w/9 – 5/8" * 7" Tandem Scraper to 255^{m}, w/NL-NG.
29	22 – February	Cont. RIH w/9 – 5/8" * 7" Tandem Scraping Assy. to 4076^{m} S.N PKR & Comp. Strg PKR several times, CBU till Sh.Shakers got cleaned, Displace in-hole mud w/fresh mud, Spot 30^{bbl} $CaCl_2$ in 7" LNR as non solid fluid.
30	23 – February	POOH 9 – 5/8" * 7" Tandem Scraping Assy. to surface, Prepare for Logging job, RIH W/L tools to 4045^{m}, Obs. holding up W/L cable tension, try to pass the same point, No success, Log up to 7" L.L, POOH to surface.
31	24 – February	RIH W/L Tools as ReRun#2 (SBT-VDL-GR-CCL) to 2904^{m}, Log Up to surface and Repeat ReRun#2 f/1400 to 1280^{m}, Ok, POOH W/L Tools to surface, L/D W/L Tools, P/U & M/U new 5 – 7/8" BHA to RIH to 2987^{m}.
32	25 – February	Cont. RIH f/2987 to 4087^{m}, Work on Junk Sub, Pump up to 7" L.Lap, Spot 60^{bbl} Hi-Vis Pill, CBU till Sh.Shakers get cleaned, RIH f/2907 to 3533^{m}.
33	26 – February	RIH back to 4078^{m}, Pump 30^{bbl} non solid fluid as balance in 7" L.Lap , POOH to surface, Obs. no considerable debris in junk subs, RIH w/ SBT, Obs. holding up cable tension, POOH to Surf. RIH w/CBIL
34	27 – February	RIH w/CBIL to evaluate 9 – 5/8" CSG, POOH to surface, L/D W/L Tools in progress @ M.N
35	28 – February	Cont. L/D W/L Tools, RIH W/L Tools as Dummy Run and try to passs the choke points, No Success, POOH & L/D W/L

		Tools, R/D W/L Tools, RIH w/ New BHA to 3649^{m}.
36	29 – February	Cont. RIH w/ New BHA to 4075^{m}, CBU to condition the Mud, Spot 200^{bbl} Fresh Mud in 7" LNR, POOH to 1235^{m}. (During POOH operation Drawworks got failed)
37	01 – March	Cont. POOH to surface, L/D BHA & 5 – 7/8" J.Mill, R/U W/L Tools, P/U & M/U SBT, RIH and perform SBT, POOH to surface, L/D SBT, R/U CBIL and RIH to perform it, POOH, L/D the same.
38	02 – March	Change BOP Top Rams, Test BOP & Choke manifold, P/U & M/U Perforation W/L and perform Perforation @ target intervals: 4057 – 4061^{m}, 4045 – 4041^{m}, all Bullets were shot successfully, w/ NL-NG, wait on Day light.
39	03 – March	Cont. Perforation w/ W/L and perform Perforation @ target intervals: 4036 – 4041^{m}, 4031 – 4046^{m}, all Bullets were shot successfully, w/ NL-NG, L/D W/L Tools & Eqp. and P/U & M/U 7 – 5/16" Polish BHA and RIH to 215^{m}.
40	04 – March	Cont. RIH w/ 7 – 5/16" Polish BHA and Polish inside 7" Tie-Back Sleeve and make mark on Polish nose, POOH to surface, Obs. LG mark on P.Mill nose, P/U & M/U S.N PKR & RIH to 397^{m}.
41	05 – March	Cont. RIH w/ S.N PKR Assy. to 2907^{m}, set the same, POOH to surface, L/D 5" BHA jt by jt, Change Top Rams, test the same, RIH w/ 5 – 7/8" Milling BHA.
42	06 – March	Change BOP Top Ram & Test Same. RIH w/ 5 – 7/8" Clean Out BHA to 4065^{m} (Below Perf. Intervals), CBU.
43	07 – March	Cont. CBU, POOH 5 – 7/8" to surface, P/U & M/U 6 – 1/8" BHA and RIH to 885^{m}.
44	08 – March	Cont. RIH w/ 6 – 1/8" BHA f/885 to 3100^{m}, Scrap inside 7" LNR f/2094 to 3065^{m}.
45	09 – March	Slow circ @3100^{m} W/NL-NG meanwhile prepared completion fluid.

46	10 – March	While Wait on Prepare Completion Fluid Slow Circ. Thru Strg at 3100^{m} w/ NL-NG, Disp. the Well w/ Comp. fluid, Obs. the Well, Static, POOH to 1716^{m}.
47	11 – March	Cont. POOH f/1716^{m} to surface, Prepare for Running Comp. Strg.
48	12 – March	Cont. POOH f/1716^{m} to surface, Running Comp. Strg to 75.5^{m}, connect CILs to CIVs, check same, Ok, Cont. Running Comp. Strg f/75.5 to 256.7^{m}, try to Press. Test Comp. Strg several times, No Succ.
49	13 – March	Cont. Running Comp. Strg f/256.7 to 741^{m}, try to Press. Test Comp. Strg several times, No Succ., Cont. Running Comp. Strg f/741 to 1035^{m}.
50	14 – March	Cont. Running Comp. Strg f/256.7 to 741^{m}, try to Press. Test Comp. Strg several times, No Succ. due to failure of Press. Recorder, Cont. Press. Test using Analog Gauge, Ok, Cont. Running Comp. Strg f/1035 to 2126^{m}.
51	15 – March	Cont. Running Comp. Strg f/2126 to 2949^{m}, Perform Press. Test on TRSSV and its functionality, Ok.
52	16 – March	Running Comp. Strg as per program, Set PKR, Ok, Shear Traveling Joint, Ok, Perform Press. Test against Comp. Strg & PKR and HGR Seals, Ok.
53	17 – March	R/U Slick Line Tools and RIH to 62^{m} and retrieve 3.437" Protection Sleeve of TRSSSV, POOH to surface, R/D Slick Line Tools, Back off landing joint on TBG HGR & L/D same, M/U 2WCV and RIH w/2WCV & installthe same in TBG HGR, Close Shear Ram and Perform Press. Test against 2WCV w/7000^{psi} – 15min, Ok. N/D BOP. Install MM-Seal, N/U BTM Section of Wellhead including DSA Pack-off Head Adaptor & 11" * 4 – 1/16" 10k B.M.V MSP, while pass 1/4" CL & 3/8" CIL Thru. DSA Pack-off Head Adaptor, Connect 1/4" CL to CL Exit Cap, L/U CMT Unit to B.M.V & Perform Press.

54	18 – March	Open all Valves of X – Mass Tree, Close Bottom Master Valve, L/U CMT unit to Kill Valve Companion Flange, Perform Press. Test X – Mass tree body and Connection between X – Mass Tree and BTM Section Flange, 8000^{psi} – 15min, Ok. Open X – Mass Tree Cap and Retrieve 2WCV, Install X – Mass Tree Cap and close all Valves except BTM Master Valve, Release the OEOC Rig 201 @18 – Mar – 2024 @4:00 a.m

Table 12-1: AZNS-85 Daily Drilling Report Summary

13. Comments & Revision

Drilling Engineering							
2	• RTE اصلاح گردد.	9	CM	• اصلاح شد.			
	حفره "26، 151 متری می TDبا توجه به اینکه میبایست تصحیح گردد.Shoeباشد لذا عمق	9-10	CM	• اصلاح شد.			
3	• ترتیب قرارگیری رزوه های ”9 5/8 اصلاح گردد.	9-40-10	CM	• اصلاح شد.			
4	• پونداژ ”13 3/8، گرید NKT میبایست اصلاح گردد.	10-40	CM	• اصلاح شد.			
6	• رزوه Tieback#1 اصلاح گردد.	-9-40-10	CM	• اصلاح شد.			
7	• NPT با توجه به گزارش روزانه کارفرما مغایرت دارد، لذا میبایست اصلاح گردد.	24,27,28	CM	• اصلاح شد.			
Completion Engineering							
1	• The pressure test graphs are not totally clear.		CM	• اصلاح شد.			
2	• با توجه به Tally موجود، تمامی اعماق ابزار رشته مغایرت Completion sketch & tally تکمیلی با	82-83	CM	• اصلاح شد.			

	دارد، لذا اعماق و اخرین Tally رشته تکمیلی بررسی و یکسان سازی گردد.						
Drilling Fluid							
1	• در بخش Drilling Fluid Remark وزن سیال حفاری از 106 PCF به 103 PCF اصلاح شود.	106	CM	• اصلاح شد.			
Production							
1	• قسمت Well FLOWING اضافه گردد.		CM	• گزارش زنده‌سازی چاه به صورت جداگانه ارسال می‌گردد.			
Geology							
1	• Target Coordinate در جدول اطلاعات چاه اصلاح گردد.	9	CM	• اصلاح شد.			
2	• تاپ سازند ها باید بر اساس scope of work تغییر کند.	21	CM	• اصلاح شد.			